Gated Action Dissipation for Bayesian Inverse Problems

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Abstract

In a nonlinear Bayesian inverse problem, the observations can constrain some directions in parameter space much more strongly than others, and these directions can change from one region of the posterior to another. We propose Gated Action Dissipation (GAD) to sample such posteriors. From samples of a current law and derivatives of the unnormalized target density at those samples, GAD estimates the score of the noise-perturbed target as the score of the noise-perturbed current law plus a correction weighted by a matrix formed from target curvature. Reverse diffusion with this score produces new unweighted samples, and the step is repeated. For a Gaussian target, using exact target derivatives gives the exact noisy score from any nonempty set of reference samples. For strongly log-concave perturbations of Gaussian targets whose derivatives vary sufficiently little, we bound the forward Kullback–Leibler divergence after each round with exact conditional expectations and show that the contribution from the initial error decays geometrically. Synthetic tests show how the choice of gate affects sampling accuracy and how separation slows the correction of mixture weights. In acoustic trajectory inversion and joint earthquake location, GAD with Gauss–Newton curvature and a gradually introduced likelihood follows curved structure in the reference posterior projections and attains the lowest mean error among the tested samplers on each problem's primary sliced Wasserstein metric at the final query allowance.

Keywords. Bayesian inverse problems, diffusion sampling, score estimation, matrix gates, Gauss–Newton approximation.

1 Introduction

An observation can tell us a great deal about one combination of parameters and very little about another. The posterior can then concentrate near a lower-dimensional set, with substantial uncertainty along the set. For a nonlinear forward model, the combinations that the data constrain most strongly can change along the set. To sample the posterior, we need to resolve both the narrow directions and how probability is distributed along the set [4, 36].

Suppose we draw samples from a broad distribution and weight them by a sharply concentrated likelihood. A few samples can account for most of the weight [1]. The same concentration can make the sampling dynamics stiff because the target score $s_0 = \nabla \log p_0$, where p_0 is the posterior density, can change rapidly across the narrow directions. We use its derivative to identify these directions locally through the observed information $H_0 = -\nabla^2 \log p_0$. In a Bayesian inverse

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problem we can approximate H_0 by a Gauss–Newton matrix formed from derivatives of the forward model [4, 27]. Computing s_0 and H_0 requires only the unnormalized target density.

In Gated Action Dissipation (GAD), we begin with samples from a current law q. The *reference samples* for a round are the points at which we evaluate the target derivatives. We build the score field from the samples and their derivatives, then use reverse diffusion to transport fresh draws from the terminal law. The unweighted endpoints serve as the reference samples for the next round. Each round can start from samples of q even when its density is unavailable. The target enters through derivatives of its unnormalized density.

Reverse diffusion requires $s_t = \nabla \log p_t$, where p_t is the target density after adding noise for time t [3, 35]. For the Ornstein–Uhlenbeck (OU) process, fix a noisy state and consider the clean states that could have produced it. Tweedie's identity gives s_t from the conditional mean of the clean state [15]. The target-score identity (TSI) gives the same score from the conditional mean of the clean score [10]. We previously combined the two estimates with a scalar weight that depends on the noisy state and time [13]. The Laplace–Fisher Gate Identity (LFGI) gives a matrix weight determined by the conditional mean of H_0 under the target law [14].

GAD takes the conditional means of the two score signals and of H_0 under the current law q. The q-LFGI gate is formed from the mean of H_0. It weights the difference between the two score means, and we add the weighted difference to the noisy score of q. For a Gaussian target, the gate gives less weight to the correction in directions of high curvature. With exact target derivatives, the gated signal is constant in the clean state. Its average over any nonempty set of reference samples equals the noisy target score (Section 6). When the target curvature varies with position, the estimate can depend on where the reference samples lie.

We ask how many target queries are needed to reach a given sampling accuracy. A query evaluates the target energy, gradient, or curvature at a point. Within a GAD round, the same cached derivatives supply the score field throughout reverse transport.

1. We derive the q-LFGI correction from a conditional Poisson equation for the difference between the target and current scores. The error depends on the conditional covariance between the target curvature and the gradient of the Poisson solution (Sections 4 to 6).
2. We use the correction to construct the iterative sampler in Algorithm 1. For strongly log-concave perturbations of Gaussian targets with sufficiently small variation in the perturbation derivatives (Section 8), we bound the forward Kullback–Leibler (KL) divergence of the iteration with exact conditional expectations. The contribution from the initial error decays geometrically for any initial density with finite forward KL.
3. We test Gaussian score recovery, the correction of mode weights, and the effect of the matrix gate on synthetic targets. We compare GAD with the Metropolis-adjusted Langevin algorithm (MALA) and Hamiltonian Monte Carlo (HMC) on two-dimensional targets. For two nonlinear inverse problems, we compare GAD with Markov chain Monte Carlo (MCMC) methods at common target-query allowances (Sections 11 to 13). At the final allowance for each inverse problem, annealed Gauss–Newton GAD (GN-GAD) has the lowest mean error on the primary sliced Wasserstein metric. GAD corrects the component weights slowly in the most separated mixtures tested.

2 Related work

Tweedie's identity expresses the noisy score through a conditional mean of the clean state [15]. The same conditional mean underlies denoising score matching [38]. The target-score identity uses a conditional mean of the clean score [10]. Our earlier work combines sample estimates

from the two identities with a scalar weight that varies with state and time [13]. Control-variate score matching derives a time-dependent scalar coefficient for the same pair of signals [23]. LFGI gives the matrix that minimizes conditional mean squared score error under the target law [14]. The two signals have the same conditional mean under the target, so the LFGI gate changes the variance of the blended signal. GAD uses the same matrix formula with conditional means under the current law. Those means can differ, and GAD weights their difference with the gate to correct the current score (Sections 4 and 5).

Reverse Diffusion Monte Carlo estimates the score by sampling a conditional distribution at a noisy state [22]. Stochastic Localization via Iterative Posterior Sampling estimates a conditional denoising mean by MCMC [18]. Diffusive Gibbs Sampling alternates Gaussian noising with Metropolis-within-Gibbs denoising to move between separated modes [7]. Conditional Diffusion Sampling uses parallel tempering to sample its initial law and then transports those samples with a closed-form drift [5]. Sampling by Averaging estimates intermediate scores with coupled slow and fast diffusions [8]. In a GAD round, the conditional means at all integration queries use the same reference samples and the target derivatives already evaluated there.

Iterated Denoising Energy Matching alternates sampling from a current model with further score training [2]. Path Integral Samplers, controlled diffusion methods, and Adjoint Sampling learn controls or reverse dynamics [20, 32, 37, 41]. The Particle Denoising Diffusion Sampler uses a learned score to guide a sequential Monte Carlo (SMC) sampler [31]. Diffusion Path Samplers via Sequential Monte Carlo (DPSMC) use SMC to evolve auxiliary variables for score estimation and derive control-variate schedules [40]. GAD builds the score field directly from the current samples and passes equally weighted endpoints to the next round. The gate experiment substitutes coefficients adapted from earlier work into the same GAD iteration (Section 11.3).

In Bayesian inverse problems, Riemann-manifold Langevin and Hamiltonian methods use local geometry to construct MCMC proposals [4, 17]. Stochastic Newton MCMC uses gradients and Hessians to form local Gaussian proposals [27]. A variant reuses a low-rank Hessian approximation at the maximum a posteriori (MAP) estimate [30]. Dimension-independent likelihood-informed MCMC uses Hessian information to identify a subspace for proposal adaptation [9]. In GAD, conditional averages of target curvature determine the matrix that weights the correction to the diffusion score. Our inverse-problem comparisons include MCMC methods with a fixed MAP Hessian or local Gauss–Newton matrices.

Stein variational gradient descent moves particles along a kernel-based functional gradient of reverse KL [26]. Stein variational Newton (SVN) uses the second variation of the same objective to construct a Newton update and also describes curvature-informed kernels [12]. Stochastic SVN adds noise and a drift correction to obtain a continuous-time system with the product target law as its invariant law [25]. Affine-invariant interacting Langevin dynamics preconditions particle updates by their ensemble covariance [16]. A GAD round uses its current ensemble to estimate a diffusion score and transports fresh draws with that field. Our forward KL bound applies to the law of the transported draws at the end of a round.

Importance weights can concentrate when the proposal and target differ, with the severity related to intrinsic dimension [1]. GAD gives equal weight to the endpoints passed between rounds. Within a round, the OU kernel can assign most of the weight in a conditional average to a few reference samples. Gaussian score recovery holds even when those weights concentrate (Section 6). For separated mixtures, the clean score can change little when the component weights change [39]. We vary component separation and measure the clean-score discrepancy and the reduction in forward KL after one round (Section 11.2).

Girsanov comparisons relate reverse-diffusion error to integrated squared score error and the mismatch at the terminal law [6, 24]. For the perturbed Gaussian targets covered by our theorem,

we control the integrated error of the corrected score by the KL dissipated along the forward OU diffusion. Combining the two bounds gives the one-round result with exact conditional expectations (Section 8).

3 Relative scores and reverse diffusion

The OU process smooths the target and current laws with the same Gaussian kernel. Their score difference controls how quickly the KL divergence between the two laws decreases along the forward diffusion. Adding the score difference to the current score recovers the target score. Table 1 collects the notation used for both steps.

Table 1: Notation for the target and current OU paths and the GAD iteration. Scores use s, observed information matrices use H, and matrix gates use G. Hats indicate quantities computed from reference samples.

Notation	Meaning
d, I_d	Parameter dimension and $d \times d$ identity matrix.
X_0, Y_t, x, y	Clean and noisy random states and their realizations.
α_t, γ_t	OU mean factor e^{-t} and noise variance $1 - e^{-2t}$.
$K_t^{\mathrm{OU}}(y \mid x)$	OU transition density.
$\widetilde{p}_0, p_0, q$	Unnormalized target density, normalized target density, and current law.
p_t, q_t	Target and current OU marginal densities.
$\rho_{y,t}, \rho_{y,t}^q$	Laws of $X_0 \mid Y_t = y$ when X_0 has law p_0 or q.
s_0, s_t, s_t^q	Clean target score and noisy scores for the target and current laws.
H_0, H^q	Target observed information and its mean under $\rho_{y,t}^q$.
$H_{0,\beta}^{\mathrm{GN}}$	Gauss–Newton approximation to the observed information at likelihood power β.
R_t, d_t	Density ratio p_t/q_t and score difference $\nabla \log R_t$.
$m_t^q, G_{\mathrm{LFGI}}^q$	Conditional mean of the target-score signal minus the Tweedie signal under $\rho_{y,t}^q$, and the matrix gate.
a_t^q, e_t^q	Correction $G_{\mathrm{LFGI}}^q m_t^q$ and score error $d_t - a_t^q$.
$s_t^{a,q}, \widehat{s}_t^{a,q}$	Corrected score with exact expectations and its sample-based version.
$D(q), E(q)$	Path norms of d_t and e_t^q weighted by p_t on $[0, T]$.
ϕ_d, Φ_T, Φ	Standard Gaussian density, exact-expectation GAD map at horizon T, and its infinite-horizon limit.
N, K	Number of reference samples per round and number of rounds.
U_{like}, β	Data misfit and likelihood power in Section 9.

3.1 Target and current OU paths

Let $\widetilde{p}_0$ be a positive unnormalized target density, and let $p_0 = \widetilde{p}_0/Z_{\mathrm{tar}}$ where $Z_{\mathrm{tar}} = \int_{\mathbb{R}^d} \widetilde{p}_0(x)\,\mathrm{d}x$. The target derivatives are

$$s_0(x) = \nabla_x \log \widetilde{p}_0(x), \qquad H_0(x) = -\nabla_x^2 \log \widetilde{p}_0(x).$$

Here s_0 is the clean score and H_0 is the observed information. Both are independent of Z_{tar}. The convergence theorem uses H_0, and the inverse-problem implementation uses the Gauss–Newton matrix in (9.1).

Let X_0 have law q on $\mathbb{R}^d$. For $t > 0$, the OU state has the conditional representation below, with $\xi \sim \mathcal{N}(0, I_d)$ independent of X_0.

$$Y_t = \alpha_t X_0 + \sqrt{\gamma_t}\,\xi, \qquad \alpha_t = e^{-t}, \qquad \gamma_t = 1 - e^{-2t}.$$

Write $K_t^{\mathrm{OU}}(y \mid x)$ for the $\mathcal{N}(\alpha_t x, \gamma_t I_d)$ transition density evaluated at y. The OU marginal density q_t is the integral of this kernel against q. The law of X_0 given $Y_t = y$ is

$$\rho_{y,t}^q(\mathrm{d}x) := \frac{K_t^{\mathrm{OU}}(y \mid x) q(\mathrm{d}x)}{q_t(y)}.$$

When q has a density we also write it as $q(x)$. We use p_0 for the target density and, when used as a law, the measure $p_0(x)\,\mathrm{d}x$. Replacing q by p_0 gives the target OU marginal p_t and the target OU posterior

$$\rho_{y,t}(\mathrm{d}x) = \frac{K_t^{\mathrm{OU}}(y \mid x) p_0(x) \mathrm{d}x}{p_t(y)}.$$

Write $s_t(y) = \nabla_y \log p_t(y)$ and $s_t^q(y) = \nabla_y \log q_t(y)$. Given a specified field $s(y,t)$, the reverse OU stochastic differential equation (SDE) is [3, 35]

$$\mathrm{d}Z_t = \{-Z_t - 2s(Z_t, t)\}\mathrm{d}t + \sqrt{2}\,\mathrm{d}\overline{B}_t, \qquad t : t_{\max} \downarrow t_{\min}. \tag{3.1}$$

Here $\overline{B}_t$ is standard Brownian motion in the reverse time direction. The initial condition for (3.1) is specified at $t_{\max}$. GAD uses ϕ_d, the $\mathcal{N}(0, I_d)$ density, as the terminal density. In the analysis, the interval is $[0, T]$, and the difference between ϕ_d and the exact terminal density p_T contributes the additive error $\mathrm{KL}(p_T \| \phi_d)$.

3.2 Relative density, relative score, and KL dissipation

For a fixed current law q, suppress its superscript in the density ratio and relative score. Define

$$R_t(y) = \frac{p_t(y)}{q_t(y)}, \qquad d_t(y) = \nabla \log R_t(y) = s_t(y) - s_t^q(y).$$

For the two laws evolved by the same OU diffusion, the relative de Bruijn identity gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{KL}(p_t \| q_t) = -\mathbb{E}_{p_t}[\|d_t\|^2].$$

The expectation $\mathbb{E}_{p_t}$ is over $Y_t \sim p_t$, and the norm inside it is Euclidean. For a vector field $u_t(y)$ on a horizon $T > 0$, write

$$\|u\|_p^2 := \int_0^T \mathbb{E}_{p_t}[\|u_t\|^2]\mathrm{d}t.$$

Write $D(q) := \|d\|_p$, with the horizon T implicit in $D(q)$ and in the error norm $E(q)$ below. Integrating the derivative gives

$$\boxed{D(q)^2 = \mathrm{KL}(p_0 \| q) - \mathrm{KL}(p_T \| q_T).} \tag{3.2}$$

The Girsanov bound (8.7) in Section 8 uses the same p_t weighting. Under Assumption 8.1, Appendix B.3 justifies (3.2) for every current density with $\mathrm{KL}(p_0 \| q) < \infty$.

3.3 Corrected score and its error

Let a_t^q be a correction that may depend on the current law q. Adding it to s_t^q gives the corrected score, whose error relative to s_t is

$$s_t^{a,q} := s_t^q + a_t^q, \qquad e_t^q := d_t - a_t^q, \qquad s_t^{a,q} = s_t - e_t^q. \tag{3.3}$$

Let $E(q) := \|e^q\|_p$. The two reverse SDEs driven by $s^{a,q}$ and s_t have the same noise coefficient and differ in drift by $2e_t^q$. The Girsanov bound (8.7) gives an endpoint KL bound of $E(q)^2$ plus the terminal error. By (3.2), if $E(q)^2 \le \theta^2 D(q)^2$ with $0 \le \theta < 1$, the forward KL after one round is at most θ^2 times its starting value plus the terminal error. The correction a_t^q uses conditional means that we can estimate from the reference samples.

4 Conditional score identities

Estimating $d_t = s_t - s_t^q$ from the current samples requires information about the target. The clean target score s_0 supplies that information in a conditional mean. A Poisson equation relates this mean to d_t.

4.1 Tweedie and target-score signals

At a noisy state $y \in \mathbb{R}^d$ and time $t > 0$, define the score signals, the integrands of the two conditional means, by

$$b(x; y, t) = \frac{\alpha_t x - y}{\gamma_t}, \qquad c(x; t) = \frac{s_0(x)}{\alpha_t}, \qquad \delta(x; y, t) = c(x; t) - b(x; y, t).$$

Tweedie's identity gives $s_t = \mathbb{E}_{\rho_{y,t}}[b]$, and the target-score identity gives $s_t = \mathbb{E}_{\rho_{y,t}}[c]$ [13, 15]. These expectations are over X_0 conditional on $Y_t = y$, with X_0 drawn initially from the target. Thus the disagreement signal δ has zero mean under $\rho_{y,t}$. Under the current OU posterior $\rho_{y,t}^q$, the corresponding means are

$$\mathbb{E}_{\rho_{y,t}^q}[b] = s_t^q(y), \qquad m_t^q(y) := \mathbb{E}_{\rho_{y,t}^q}[\delta]. \tag{4.1}$$

When q has a differentiable positive density and integration by parts is valid, the conditional mean m_t^q in (4.1) satisfies

$$m_t^q(y) = \frac{1}{\alpha_t} \mathbb{E}_{\rho_{y,t}^q}[\nabla_x \log(p_0/q)(X_0)]\,.$$

Appendix A.2 derives the density-ratio formula. The conditional mean m_t^q in (4.1) uses q as a law for averaging and applies to an empirical measure.

4.2 Conditional Poisson equation and its moments

Fix y and $t > 0$, and assume $\rho_{y,t} \ll \rho_{y,t}^q$. The centered density ratio

$$g = \frac{\mathrm{d}\rho_{y,t}}{\mathrm{d}\rho_{y,t}^q} - 1, \qquad \mathbb{E}_{\rho_{y,t}^q}[g] = 0,$$

represents the difference between the two conditional laws. Let $u = u_{y,t}$ solve the conditional Stein–Poisson equation in weak form

$$\mathbb{E}_{\rho_{y,t}^q}[\nabla u \cdot \nabla \varphi] = \mathbb{E}_{\rho_{y,t}^q}[g\varphi] \quad \text{for all centered } \varphi \in H^1(\rho_{y,t}^q). \tag{4.2}$$

We assume a centered weak solution exists and that the components of x and $\nabla_x \log \rho_{y,t}(x)$, the test functions used for (4.3), belong to this weighted Sobolev space after centering. A finite Poincaré constant for $\rho^q_{y,t}$ and $g \in L^2(\rho^q_{y,t})$ suffice for solvability (Appendix A.1). The convergence proof uses the residual identity (8.2), which holds independently of Poisson solvability.

The right-hand side of (4.2) is the difference of conditional means

$$\mathbb{E}_{\rho^q_{y,t}}[g\varphi] = \mathbb{E}_{\rho_{y,t}}[\varphi] - \mathbb{E}_{\rho^q_{y,t}}[\varphi].$$

The weak equation (4.2) expresses this difference as an average involving ∇u, the Poisson flux. Write $A_0(x;t) = \alpha_t^2 I_d + \gamma_t H_0(x)$ for the shifted observed-information matrix. Testing (4.2) with the components of x and $\nabla_x \log \rho_{y,t}(x)$ gives

$$\mathbb{E}_{\rho^q_{y,t}}[\nabla u] = \frac{\gamma_t}{\alpha_t} d_t, \qquad \mathbb{E}_{\rho^q_{y,t}}[A_0 \nabla u] = \alpha_t \gamma_t \, m_t^q. \tag{4.3}$$

The mean flux determines d_t. Weighting the same flux by A_0 gives a mean proportional to m_t^q. The derivation is in Appendix A.3.

5 Reconstructing the relative score

The mean of the matrix A_0 in (4.3) is

$$H^q(y,t) := \mathbb{E}_{\rho^q_{y,t}}[H_0(X_0)], \qquad A^q(y,t) := \alpha_t^2 I_d + \gamma_t H^q(y,t) = \mathbb{E}_{\rho^q_{y,t}}[A_0(X_0;t)].$$

We assume $A^q(y,t)$ is positive definite wherever it is inverted. If $H_0(x)$ is positive semidefinite for all x, then $A^q \succeq \alpha_t^2 I_d$.

Subtracting the mean flux gives a centered vector $v_\perp = \nabla u - \mathbb{E}_{\rho^q_{y,t}}[\nabla u]$. Define $r_H(y,t) = \mathbb{E}_{\rho^q_{y,t}}[(H_0 - H^q)v_\perp]$. The vector $r_H \in \mathbb{R}^d$ is the mean product of the centered curvature matrix and the centered flux evaluated at the same state. Since $A_0 - A^q = \gamma_t(H_0 - H^q)$, expanding the product yields

$$\mathbb{E}_{\rho^q_{y,t}}[A_0 \nabla u] = A^q \, \mathbb{E}_{\rho^q_{y,t}}[\nabla u] + \gamma_t r_H. \tag{5.1}$$

Inserting the two means from (4.3) into (5.1) gives

$$\alpha_t^2 m_t^q = A^q d_t + \alpha_t r_H. \tag{5.2}$$

We can estimate m_t^q and A^q from the reference samples using target derivatives. The term r_H requires the conditional Poisson solution. Solving (5.2) with r_H omitted gives a correction a_t^q and, by (3.3), its error e_t^q:

$$a_t^q = \alpha_t^2 (A^q)^{-1} m_t^q, \qquad e_t^q = -\alpha_t (A^q)^{-1} r_H. \tag{5.3}$$

If H_0 is constant, r_H vanishes and $a_t^q = d_t$. The matrix multiplying m_t^q in (5.3) is the q-LFGI gate.

6 The q-LFGI gate

For a symmetric $d \times d$ matrix P such that $\alpha_t^2 I_d + \gamma_t P$ is positive definite, define

$$\Psi_t(P) = \alpha_t^2 (\alpha_t^2 I_d + \gamma_t P)^{-1}.$$

The LFGI matrix function Ψ_t from [14] evaluated at H^q gives the q-LFGI gate

$$G^q_{\mathrm{LFGI}}(y,t) = \Psi_t(H^q(y,t)) = \alpha_t^2 (A^q(y,t))^{-1}. \tag{6.1}$$

For a matrix $G(y,t)$ fixed at a given y,t, the gated signal is $z_G(x;y,t) = b(x;y,t) + G(y,t)\delta(x;y,t)$. By (4.1) its mean under $\rho^q_{y,t}$ is $s^q_t + Gm^q_t$. Using (6.1) in (5.3) gives

$$a^q_t = G^q_{\mathrm{LFGI}} m^q_t, \qquad s^{a,q}_t = \mathbb{E}_{\rho^q_{y,t}}[z_{G^q_{\mathrm{LFGI}}}]. \tag{6.2}$$

Equation (6.2) defines the corrected score without a Poisson solve.

In the Gaussian case of LFGI [14], the curvature is constant. For $p_0 = \mathcal{N}(m, P^{-1})$ with $P \succ 0$, $H_0 = P$, so averaging it under any current OU posterior gives the same matrix. The gate coefficient in an eigendirection of P with eigenvalue λ_j is

$$g_j(t) = \frac{\alpha_t^2}{\alpha_t^2 + \gamma_t \lambda_j}.$$

At a given $t > 0$, $g_j(t)$ decreases with λ_j. In this Gaussian case the gate makes the derivative of z_G with respect to x vanish by (A.3), so z_G is constant in x. Since the constant signal has target OU posterior mean s_t, its average over any nonempty set of reference samples is s_t for any normalized weights shared by the two signals. Exact Gaussian sampling in one round additionally requires the exact terminal law p_T and exact reverse integration.

When $q = p_0$, the conditional mean is $H^{p_0}(y,t) = \mathbb{E}_{\rho_{y,t}}[H_0(X_0)]$. The gate $G^q_{\mathrm{LFGI}} = \Psi_t(H^{p_0})$ then coincides with $G_\star$, the LFGI minimizer of conditional mean squared score error over $G \in \mathbb{R}^{d\times d}$ at fixed y,t [14]. With the finite moments and positive-definite conditional covariance in Appendix A.2, the minimizer is given by

$$G_\star(y,t) = \arg\min_{G\in\mathbb{R}^{d\times d}} \mathcal{R}(G;y,t), \qquad \mathcal{R}(G;y,t) := \mathbb{E}_{\rho_{y,t}}[\|z_G(X_0;y,t) - s_t(y)\|^2]. \tag{6.3}$$

For $q \neq p_0$ we use the reconstruction (6.2). Its error is given by (5.3) under the solvability assumptions stated after (4.2). For the convergence analysis, Lemma 8.2 gives a separate error identity that uses only conditional expectations.

7 Gated Action Dissipation

With exact conditional expectations, a GAD round fixes the score $s^{a,q}_t$ from (6.2) and integrates (3.1) from $Z_T \sim \phi_d$ at $t = T$ to $t = 0$. Write $\Phi_T(q)$ for the law of Z_0. The next round starts from this law according to

$$q_{k+1} = \Phi_T(q_k). \tag{7.1}$$

The *gated action* $a^q_t = G^q_{\mathrm{LFGI}} m^q_t$ is the correction added to s^q_t. It vanishes at $q = p_0$ because the two signals have the same conditional mean.

Algorithm 1 approximates each conditional expectation by an average over the reference samples with the normalized OU kernel weights in (I.1). We keep the reference samples and their target derivatives fixed during a round. At each integration query (y,t), we recompute the weights and conditional means. The synthetic experiments in Sections 11 and 12 use the reverse SDE. The inverse problems use the probability-flow ordinary differential equation (ODE) in (10.1).

The conditional weights in Algorithm 1 are used only to evaluate the score field. All endpoints have the same weight when they become the next round's reference samples. For the inverse problems, (I.2) gives the GN score field and Appendix I.3 gives the likelihood schedules and request accounting.

1. **Input.** Provide N unweighted reference samples $X_{1:N}^{(0)} = (X_1^{(0)}, \dots, X_N^{(0)})$, a grid $0 < t_{\min} = t_0 < \cdots < t_{L_t} = t_{\max}$, a terminal distribution, a reverse SDE integrator (3.1) or probability-flow ODE integrator (10.1), evaluators for the clean target score and curvature, and a round count K.
2. **For $k = 0, \dots, K-1$, perform the following steps.**
   (a) Cache $s_0(X_i^{(k)})$ and the observed information $H_i = H_0(X_i^{(k)})$, or its Gauss–Newton approximation (9.1), at each reference state and hold the cache fixed for the round. For annealed GN-GAD, use the derivatives at power β_{k+1} in (9.4).
   (b) At each integration query (y, t), compute the normalized weights $w_i(y, t)$ from (I.1). For the current empirical law q_k, average the signals b, c from Section 4.1 and the cached curvature:
   $$\widehat{b}^{q_k}(y,t) = \sum_{i=1}^{N} w_i(y,t) b(X_i^{(k)}; y, t),$$
   $$\widehat{c}^{q_k}(y,t) = \sum_{i=1}^{N} w_i(y,t) c(X_i^{(k)}; t),$$
   $$\widehat{H}^{q_k}(y,t) = \sum_{i=1}^{N} w_i(y,t) H_i.$$
   Use the score field
   $$\widehat{m}_t^{q_k}(y) = \widehat{c}^{q_k} - \widehat{b}^{q_k}, \qquad \widehat{s}_t^{a,q_k}(y) = \widehat{b}^{q_k} + \Psi_t(\widehat{H}^{q_k}) \widehat{m}_t^{q_k}(y).$$
   Apply Ψ_t as in (6.1) when the empirical shifted matrix is positive definite. For the toy targets with indefinite observed information, use the eigenvalue safeguards in Appendix F.3.
   (c) Draw N fresh states at $t_{\max}$ from the terminal distribution. Integrate with $\widehat{s}_t^{a,q_k}$ down to $t_{\min}$ and set $X_{1:N}^{(k+1)}$ to the unweighted endpoints.
3. **Output.** Return the unweighted samples $X_{1:N}^{(K)}$.

Algorithm 1: Gated Action Dissipation with the q-LFGI gate.

7.1 A one-dimensional illustration

We illustrate the iteration on a one-dimensional target with three Gaussian components. Each round starts from a fitted Gaussian mixture, uses quadrature for the conditional means, and fits another mixture to the transported endpoints. The sampler in Algorithm 1 instead uses the endpoints directly. The illustration starts from a broad, nearly Gaussian mixture and uses exact target derivatives with a gate safeguard for negative curvature.

In Figure 1, the first round moves most of the missing mass to the separated left mode, and later rounds make smaller corrections. Estimated forward KL falls from 0.572 initially to 0.076 after one round, and later rounds bring it down close to the error from the finite diffusion horizon. The separated mixture serves as a numerical illustration beyond the strongly log-concave target class of Assumption 8.1. Appendix G.1 gives the target and initial law, the mixture fit, the gate safeguard, and the construction of the displayed densities.

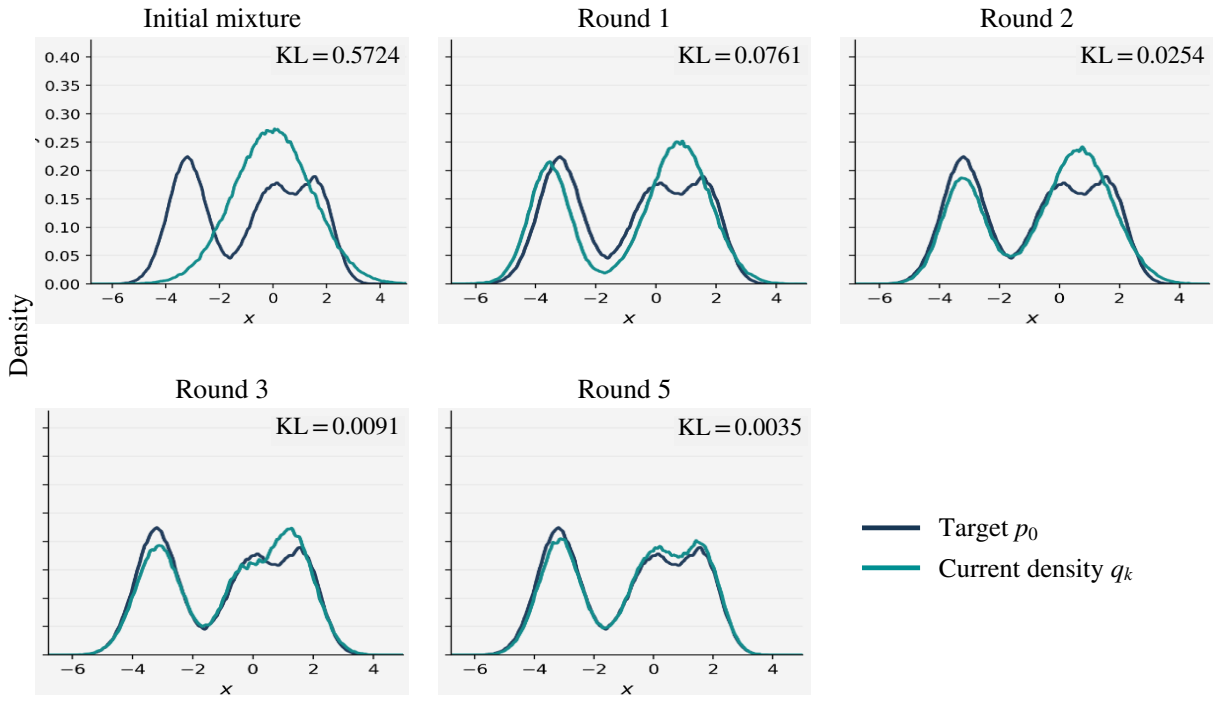


Figure 1: **GAD rounds on a one-dimensional Gaussian mixture.** The target p_0 (dark blue) and the fitted current density q_k (teal) are shown initially and after rounds 1, 2, 3, and 5. The horizontal axis is the state x, and the vertical axis is density. The curves are smoothed histograms of independent draws from the target and fitted mixtures. Each panel reports an estimate of $\mathrm{KL}(p_0\|q_k)$, and lower values indicate closer agreement. Appendix G.1 gives the display construction.

8 Convergence with exact expectations for perturbed Gaussian targets

For strongly log-concave perturbations of Gaussian targets, we bound the integrated squared score error $E(q)^2$ using the curvature and gradient bounds on the perturbation. The score error then bounds forward KL after one round of (7.1). Appendix B gives the estimates used in the proof.

8.1 Target class and score error

The difference between the target Hessian and its constant quadratic part determines one term of the score error. The following assumptions bound that difference and the variation of the perturbation gradient. We use $\succeq$ for the Loewner order on symmetric matrices and $\|\cdot\|_{\mathrm{op}}$ for the Euclidean operator norm.

Assumption 8.1 (Strongly log-concave perturbations of Gaussian targets)**.** Up to an additive constant,

$$-\log \widetilde{p}_0(x) = \tfrac{1}{2}x^\top \Lambda x + V(x), \qquad \Lambda = \Lambda^\top \succ 0, \qquad V \in C^2(\mathbb{R}^d),$$

and there are constants $\kappa > 0$, $L_V \geq 0$, and $\Delta_V < \infty$ such that, for all $x, x' \in \mathbb{R}^d$,

$$H_0(x) \succeq \kappa I_d, \qquad \|\nabla^2 V(x)\|_{\mathrm{op}} \leq L_V, \qquad \|\nabla V(x) - \nabla V(x')\| \leq \Delta_V. \tag{8.1}$$

Under (8.1), $\kappa I_d \preceq H_0 \preceq (\|\Lambda\|_{\mathrm{op}} + L_V) I_d$. Hence p_0 is normalizable with Gaussian tails and has moments of all orders. The gradient ∇V is bounded because its variation is bounded, and a linear term in $-\log \widetilde{p}_0$ can be absorbed into V without changing L_V or Δ_V. For every current law, $H^q = \mathbb{E}_{\rho^q_{y,t}}[H_0] \succeq \kappa I_d$, so $A^q \succeq (\alpha_t^2 + \kappa\gamma_t) I_d$ is invertible and the q-LFGI gate is defined. The current law enters only through the domain

$$\mathcal{Q} := \left\{ q : \ q \text{ is a probability density on } \mathbb{R}^d \text{ and } \mathrm{KL}(p_0 \| q) < \infty \right\}.$$

A density in $\mathcal{Q}$ is positive almost everywhere, but it need not be smooth, have moments, or have a bounded ratio to p_0.

The correction $a_t^q = G^q_{\mathrm{LFGI}} m_t^q$ in (6.2) yields a score error that depends on $H^q - \Lambda$ and the difference between the two conditional means of ∇V.

Lemma 8.2 (q-LFGI residual identity)**.** *Under Assumption 8.1, let q be a probability law, $t > 0$, and $y \in \mathbb{R}^d$. Then*

$$e_t^q(y) = \gamma_t (A^q)^{-1} (H^q - \Lambda)\, d_t(y) + \alpha_t (A^q)^{-1} \big(\mathbb{E}_{\rho^q_{y,t}}[\nabla V] - \mathbb{E}_{\rho_{y,t}}[\nabla V] \big). \tag{8.2}$$

Consequently,

$$\|e_t^q(y)\| \le \frac{\gamma_t}{\alpha_t^2 + \kappa\gamma_t} \left\{ L_V \|d_t(y)\| + \frac{\alpha_t \Delta_V}{\gamma_t} \sqrt{\tfrac{1}{2}\mathrm{KL}(\rho_{y,t} \| \rho^q_{y,t})} \right\}. \tag{8.3}$$

Proof. All averages are finite because $\rho^q_{y,t}$ has moments of all orders for $t > 0$ (Appendix B.1). The Gaussian tails of the strongly log-concave target OU posterior justify taking the mean of its score in (A.1) to obtain $\mathbb{E}_{\rho_{y,t}}[\delta] = 0$. Tweedie's identity gives $d_t = (\alpha_t/\gamma_t)(\mathbb{E}_{\rho_{y,t}}[X_0] - \mathbb{E}_{\rho^q_{y,t}}[X_0])$. Since $-s_0(x) = \Lambda x + \nabla V(x)$, the signal $-\alpha_t \delta(x) = (\Lambda + \alpha_t^2 \gamma_t^{-1} I_d) x + \nabla V(x) - \alpha_t \gamma_t^{-1} y$ is affine in x apart from ∇V. Therefore

$$\alpha_t m_t^q = \alpha_t (\mathbb{E}_{\rho^q_{y,t}}[\delta] - \mathbb{E}_{\rho_{y,t}}[\delta]) = \frac{1}{\alpha_t}(\alpha_t^2 I_d + \gamma_t \Lambda) d_t - (\mathbb{E}_{\rho^q_{y,t}}[\nabla V] - \mathbb{E}_{\rho_{y,t}}[\nabla V]).$$

Multiplying by $\alpha_t (A^q)^{-1}$ gives a_t^q. Subtracting a_t^q from d_t and using $A^q - \alpha_t^2 I_d - \gamma_t \Lambda = \gamma_t (H^q - \Lambda)$ proves (8.2). For (8.3), use $A^q \succeq (\alpha_t^2 + \kappa\gamma_t) I_d$ and $\|H^q - \Lambda\|_{\mathrm{op}} = \|\mathbb{E}_{\rho^q_{y,t}}[\nabla^2 V]\|_{\mathrm{op}} \le L_V$. For a unit vector v, the function $v \cdot \nabla V$ has oscillation at most Δ_V. Its two posterior means therefore differ by at most Δ_V times the total-variation distance between the posteriors, which Pinsker's inequality bounds by $\{\frac{1}{2}\mathrm{KL}(\rho_{y,t} \| \rho^q_{y,t})\}^{1/2}$. □

For a Gaussian target, V is affine, both terms in (8.2) vanish, and the corrected score equals the target score for every current law. In general, the posterior KL in (8.3) is controlled by the forward KL already dissipated along the OU path. For $q \in \mathcal{Q}$ and $t > 0$,

$$\mathbb{E}_{p_t}\Big[\mathrm{KL}(\rho_{Y_t,t} \| \rho^q_{Y_t,t}) \Big] = \mathrm{KL}(p_0 \| q) - \mathrm{KL}(p_t \| q_t) = \int_0^t \mathbb{E}_{p_u}[\|d_u\|^2] \mathrm{d}u. \tag{8.4}$$

The first equality is the chain rule for relative entropy applied to the joint laws of (X_0, Y_t), which share the OU kernel. The second is (3.2) on $[0, t]$ (Appendix B.3).

8.2 Uniform score-error bound and contraction

The KL argument needs $E(q)^2 \le \theta^2 D(q)^2$ uniformly over $q \in \mathcal{Q}$ and $T > 0$, with $\theta^2 < 1$ for contraction. Under Assumption 8.1, a closed-form choice for θ^2 uses the curvature and gradient variation bounds

$$a := \frac{L_V^2}{\kappa^2}, \qquad b := \frac{\Delta_V^2}{4\kappa}, \qquad C_{\mathrm{closed}} := \frac{a + b + \sqrt{a^2 + b^2}}{2}. \tag{8.5}$$

The bound in (8.5) is at most $a+b$, the bound from a fixed Young inequality. Lemma B.5 gives another bound $S(1)$ by retaining H^q inside the inverse matrix and using a scalar function S to bound the accumulated error. When $\Lambda = \lambda I_d$, (B.8) tests the two endpoints κ and M of the allowed curvature interval. In that case (B.9) also gives a closed-form bound. Each choice depends only on the target.

Proposition 8.3 (Uniform bound for the integrated score error)**.** *Under Assumption 8.1, let θ^2 be C_{closed}, the value $S(1)$ of an admissible function S from Lemma B.5, or, when $\Lambda = \lambda I_d$, the joint bound (B.9) with $\kappa I_d \preceq H_0 \preceq M I_d$. Then every $q \in \mathcal{Q}$ and every horizon $T > 0$ satisfy*

$$E(q)^2 \le \theta^2 D(q)^2 = \theta^2 \{\mathrm{KL}(p_0\|q) - \mathrm{KL}(p_T\|q_T)\}. \tag{8.6}$$

By (8.4), the mean conditional KL at time t is the integral of $\mathbb{E}_{p_u}\|d_u\|^2$ from 0 to t. The proof in Appendix B.4 multiplies the mean conditional KL by a function of time chosen from the target bounds (8.1). In the transformed time, the derivative of that product bounds the integrand of $E(q)^2$. Integrating gives (8.6).

Under Assumption 8.1, the apparent singular terms in the corrected score cancel in (B.2). Lemma B.2 bounds the score by $C_q(1+\|y\|)$ uniformly for $t > 0$ for each current probability law q. Lemma B.3 then gives a unique nonexplosive strong solution of (3.1) for any terminal law, up to and including $t = 0$. If the terminal law has a density φ with $\mathrm{KL}(p_T\|\varphi) < \infty$, applying the localized Girsanov comparison of Lemma B.4 on $[0,T]$ gives an endpoint density r satisfying

$$\mathrm{KL}(p_0\|r) \le \mathrm{KL}(p_T\|\varphi) + E(q)^2. \tag{8.7}$$

The proof uses a stopped change of drift in place of a Novikov condition.

Theorem 8.4 (Geometric forward-KL contraction)**.** *Assume Assumption 8.1, choose any bound $\theta^2 < 1$ from Proposition 8.3, and set $\varepsilon_T := \mathrm{KL}(p_T\|\phi_d)$.*

(i) *For every $q \in \mathcal{Q}$ and $T > 0$,*

$$\mathrm{KL}(p_0\|\Phi_T(q)) \le \varepsilon_T + \theta^2 \mathrm{KL}(p_0\|q). \tag{8.8}$$

Hence $\Phi_T(\mathcal{Q}) \subseteq \mathcal{Q}$, and the iteration (7.1) started from $q_0 \in \mathcal{Q}$ satisfies

$$\mathrm{KL}(p_0\|q_k) \le \theta^{2k}\mathrm{KL}(p_0\|q_0) + \frac{1-\theta^{2k}}{1-\theta^2}\,\varepsilon_T. \tag{8.9}$$

(ii) *For every $q \in \mathcal{Q}$, the limit $\Phi(q) := \lim_{T\to\infty}\Phi_T(q)$ exists in total variation and has a density. Moreover,*

$$\Phi(\mathcal{Q}) \subseteq \mathcal{Q}, \qquad \mathrm{KL}(p_0\|\Phi(q)) \le \theta^2 \mathrm{KL}(p_0\|q). \tag{8.10}$$

Consequently, $q_{k+1} = \Phi(q_k)$ satisfies $\mathrm{KL}(p_0\|q_k) \le \theta^{2k}\mathrm{KL}(p_0\|q_0)$ for every $q_0 \in \mathcal{Q}$, and p_0 is the unique fixed point of Φ in $\mathcal{Q}$.

Proof. For part (i), apply (8.7) with $\varphi = \phi_d$, then (8.6) and $D(q)^2 \le \mathrm{KL}(p_0\|q)$. This gives $\mathrm{KL}(p_0\|\Phi_T(q)) \le \varepsilon_T + \theta^2 D(q)^2$, hence (8.8). The right side is finite and $\Phi_T(q)$ has a density, so $\Phi_T(q) \in \mathcal{Q}$. Induction gives (8.9). Part (ii) is proved in Appendix B.6. □

The condition $\theta < 1$ in Theorem 8.4 involves only the target derivatives. Any initial density with $\mathrm{KL}(p_0\|q_0) < \infty$ is admissible, including a broad Gaussian whose density ratio to p_0 is unbounded. The terminal error vanishes with increasing horizon by (B.10). For a Gaussian target, the infinite-horizon map reaches the target in one round. At finite horizon, only the terminal error remains in the bound of Theorem 8.4.

For the non-Gaussian target with $-\log \widetilde{p}_0(x) = \frac{1}{2}\|x\|^2 + \epsilon \cos x_1$ and $0 \le \epsilon < 1$, we have $\Lambda = I_d$ and $V(x) = \epsilon \cos x_1$. In Assumption 8.1, one may take $\kappa = 1 - \epsilon$, $L_V = \epsilon$, and $\Delta_V = 2\epsilon$. Then the closed-form choice (8.5) gives a bound below one for $\epsilon < 0.4639$, and $\theta^2 < 0.2683$ at $\epsilon = 0.3$. Appendix B.5 gives examples with coordinate coupling and with a sharper bound. Theorem 8.4 assumes a strongly log-concave target with bounded perturbation gradient. The iteration in the theorem uses exact observed information and conditional expectations in continuous-time reverse diffusion.

9 Computational extensions for inverse problems

In the acoustic and earthquake experiments, we form the gate from a Gauss–Newton approximation to the observed information. We compare two likelihood schedules numerically. One uses the full likelihood from the start. The other begins at a fraction of the log likelihood and increases it to full power. Theorem 8.4 applies to exact observed information and a fixed target.

9.1 Gauss–Newton information

Let $\alpha \in \mathbb{R}^d$ have standard Gaussian prior, and let $F\colon \mathbb{R}^d \to \mathbb{R}^{n_{\mathrm{obs}}}$ be the forward map to the observations. The data model is $y_{\mathrm{obs}} = F(\alpha_\star) + \eta$, where $\alpha_\star$ is the data-generating parameter and $\eta \sim \mathcal{N}(0, \Gamma)$ is independent Gaussian noise with $\Gamma \succ 0$. Write $r(\alpha) = \Gamma^{-1/2}(F(\alpha) - y_{\mathrm{obs}})$, $U_{\mathrm{like}}(\alpha) = \|r(\alpha)\|^2/2$, and let $J(\alpha) = \nabla r(\alpha) \in \mathbb{R}^{n_{\mathrm{obs}} \times d}$ be the Jacobian of the whitened residual. The posterior has unnormalized density $\widetilde{p}_0(\alpha) = \exp\{-\|\alpha\|^2/2 - U_{\mathrm{like}}(\alpha)\}$. We retain its exact clean score and use the standard Gauss–Newton approximation to the observed information [4, 27]

$$H_0^{\mathrm{GN}}(\alpha) = I_d + J(\alpha)^\top J(\alpha). \tag{9.1}$$

The matrix in (9.1) is positive definite, while the exact observed information is

$$H_0(\alpha) = H_0^{\mathrm{GN}}(\alpha) + R_{\mathrm{nl}}(\alpha), \qquad R_{\mathrm{nl}}(\alpha) = \sum_j r_j(\alpha) \nabla^2 r_j(\alpha). \tag{9.2}$$

The residual R_{nl} vanishes for an affine forward map. For a nonlinear map, accurate clean scores and a positive-definite proxy do not by themselves make the curvature substitution accurate.

In the gate (6.1) we replace the conditional mean of H_0 by that of H_0^{GN}. The shifted matrix stays positive definite. We assess the effect of the added curvature term in (I.3) numerically.

9.2 Likelihood annealing

Intermediate target distributions are a standard device in annealed importance sampling and sequential Monte Carlo [11, 28]. We use the likelihood-power family

$$p_{0,\beta}(\alpha) = Z_\beta^{-1} \exp\{-\tfrac{1}{2}\|\alpha\|^2 - \beta U_{\mathrm{like}}(\alpha)\}, \qquad 0 \le \beta \le 1, \tag{9.3}$$

where Z_β normalizes the density. Thus $p_{0,0}$ is the prior and $p_{0,1} = p_0$. The likelihood power β in (9.3) is distinct from OU time t. At each fixed power the clean score and GN proxy are

$$s_{0,\beta}(\alpha) = -\alpha - \beta J(\alpha)^\top r(\alpha), \qquad H_{0,\beta}^{\mathrm{GN}}(\alpha) = I_d + \beta J(\alpha)^\top J(\alpha). \tag{9.4}$$

At full power, $H_{0,1}^{\mathrm{GN}} = H_0^{\mathrm{GN}}$. The likelihood power multiplies the likelihood terms and leaves the prior terms fixed.

Choose a nondecreasing sequence $\beta_1, \dots, \beta_K = 1$. Round k in Algorithm 1 evaluates (9.4) at power β_{k+1}, holds the reference samples and their derivatives fixed during transport, and passes

the unweighted endpoints to the next round. We call this implementation annealed GN-GAD. Direct GN-GAD uses $\beta = 1$ throughout. Once $\beta = 1$, any additional rounds keep the posterior as their target.

The acoustic and earthquake experiments compare specified likelihood powers with direct $\beta_k = 1$ rounds at the same allowances for target evaluations. The annealed MCMC methods use the same tempered targets (9.3).

10 Samplers and comparison settings

In the experiments, each GAD round computes the score field $\widehat{s}_t^{a,q}$ of Algorithm 1 from the current N unweighted reference samples, holds the field fixed while it transports fresh Gaussian terminal samples, and uses the endpoints as the next reference samples. No density fit or acceptance step is applied to these endpoints. The mixture-fitted illustration in Section 7.1 is a separate diagnostic. The synthetic experiments in Sections 11 and 12 use the reverse OU SDE (3.1), exact target scores, and exact observed information. Both inverse problems in Section 13 use the Gauss–Newton score estimate (I.2) with the score and curvature at the likelihood power in (9.4). The probability-flow ODE in decreasing OU time is

$$\frac{\mathrm{d}Z_t}{\mathrm{d}t} = -Z_t - \widehat{s}_t^{a,q}(Z_t), \qquad t : T \downarrow t_{\min}, \qquad Z_T \sim \phi_d. \tag{10.1}$$

With the exact score and the same exact terminal law, the reverse SDE and probability flow have the same marginal laws [35]. Theorem 8.4 analyzes the reverse SDE (3.1). Transport grids, initial reference samples, and safeguards are given in Appendices C to G.

MALA applies the Hastings correction to a Langevin proposal using the unnormalized target and its clean score [33, 34]. For the inverse problems, let $U = -\log \widetilde{p}_0$, let α_{MAP} be its minimizer, and let $C_{\mathrm{MAP}} = [\nabla^2 U(\alpha_{\mathrm{MAP}})]^{-1}$ when the Hessian is positive definite. We use MAP–Laplace for the Gaussian $\mathcal{N}(\alpha_{\mathrm{MAP}}, C_{\mathrm{MAP}})$. MAP-preconditioned MALA (MAP-pMALA) uses the proposal

$$\alpha' = \alpha + hC_{\mathrm{MAP}} s_0(\alpha) + \sqrt{2h}\, C_{\mathrm{MAP}}^{1/2}\xi, \qquad \xi \sim \mathcal{N}(0, I_d),$$

with step size $h > 0$ and fixed covariance preconditioning [30]. The MAP Hessians in the two main applications are positive definite. Each MAP-pMALA chain starts from MAP–Laplace, adapts h during warmup, and fixes it for production. Additional inverse-problem methods use fixed-metric MALA, simplified manifold MALA with a local Gauss–Newton matrix (GN-smMALA) [17], and fixed-mass HMC. Appendix I.4 gives their likelihood powers, initialization, configuration selection, and the chain states retained for evaluation.

The two-dimensional comparison uses isotropic MALA and unit-mass HMC, each with four chains started from the same broad initial law as GAD (Section 12). HMC uses Gaussian momentum and leapfrog trajectories with a Metropolis correction [29]. Appendix F.4 gives the step-size adaptation and the other settings.

All methods within a study access the same unnormalized target. GAD also uses exact observed information or the stated Gauss–Newton approximation. The two-dimensional and inverse-problem comparisons match the allowances for target evaluations using Appendix F.4 and section 13.1. We evaluate against independent target samples when available and samples from the No-U-Turn Sampler (NUTS) for the inverse problems [21]. In the inverse problems, the NUTS samples also define the metric used to select configurations on the calibration seed. No sampler uses NUTS draws for its initialization, preconditioning, or adaptation.

11 Numerical tests of the score estimator and gate

We start with a Gaussian target and change the law of the reference samples to check the cancellation in (A.3). We then increase the separation of two mixture components while holding the initial weight mismatch fixed. We measure how much the clean scores differ and how much KL falls in one round. Finally, we hold the sample and round counts fixed and vary the gate on an eight-dimensional target.

The Gaussian, gate, and two-dimensional sampler tests compare equal-size sample sets by the empirical sliced Wasserstein-2 distance (SW_2). The empirical distance (E.1) uses one-dimensional projections. We report it as a percentage of the target's root mean squared coordinate spread. Lower values indicate closer agreement. Distances between two independent target samples give a scale for comparison. We use SD for the sample standard deviation of the metric over the indicated runs. The mixture study uses forward KL because the evolving densities are available.

11.1 Gaussian score recovery with finitely many reference samples

For an anisotropic Gaussian target, we change the reference law from a broad Gaussian to a bimodal mixture. We compare the estimated score with the exact noisy score and compare the endpoints after one round. The two reference laws yield the exact score and the same paired endpoints to numerical precision. The endpoint distance to independent target draws is comparable to the distance between two independent target samples (Table 2). This distance includes numerical integration and finite-sample error, even when the score estimate is exact. Appendix E.2 specifies the target, reference laws, and numerical settings.

Table 2: **Gaussian cancellation and one-round sample agreement.** Both reference laws use 4096 reference samples and 4096 transported endpoints. Score residuals are maximum absolute errors before field interpolation. Normalized SW_2 is in percent, mean ± sample SD over three seeds, against an independent 4096-point target sample. Lower values are better. The final column compares two independent target samples. Appendix E.2 gives the paired-run design.

Reference law	Max score residual	One-round SW_2 (%)	Target/target (%)
Gaussian	2.80×10^{-11}	3.69 ± 0.49	4.05 ± 0.63
Bimodal	1.27×10^{-10}	3.69 ± 0.49	4.05 ± 0.63

11.2 Sensitivity of the clean score to mixture weights

Suppose the current law has the right mixture components but gives them the wrong weights. Where one component dominates, changing its weight has little effect on the clean score. Increasing the separation leaves appreciable differences between the clean scores mainly where the components overlap, a region of low probability [39]. For a smooth current density q, define

$$K_{\mathrm{mix}}(q) := \mathrm{KL}(p_0\|q), \qquad F_{\mathrm{mix}}(q) := \mathbb{E}_{p_0}\|s_0 - \nabla \log q\|^2, \qquad \chi(q) := \frac{\sigma^2 F_{\mathrm{mix}}(q)}{K_{\mathrm{mix}}(q)}. \tag{11.1}$$

The two Gaussian components have the same standard deviation σ. We vary their separation, measured in units of σ, and hold the initial weight mismatch fixed. The target weights are equal and the initial law assigns weights $0.7, 0.3$ to the same components (Appendix E.3). For each fixed target, we measure fractional KL progress by

$$g_k := 1 - \frac{K_{\mathrm{mix}}(q_{k+1})}{K_{\mathrm{mix}}(q_k)}, \qquad q_{k+1} = \Phi_T(q_k). \tag{11.2}$$

Larger g_k in (11.2) means greater progress. The clean-score discrepancy $F_{\mathrm{mix}}(q)$ in (11.1) compares scores at time zero. The integral $D(q)^2$ in (3.2) measures their squared difference over the OU path.

We evolve the densities with exact clean target derivatives and conditional expectations computed by quadrature, obtaining each next density from a numerical solution of the Fokker–Planck equation. The calculation uses neither reference samples nor a Gaussian mixture fit. Between separations 5 and 9, the initial sensitivity χ and the first-round gain each fall by about three orders of magnitude. After 30 rounds, GAD has almost recovered the weights of the less separated mixture. Forward KL remains appreciable for the more separated mixture. In Figure 2, the less the clean score changes with the component weights, the more slowly the iteration corrects the imbalance for these mixtures. Appendix E.3 gives the values at each separation and the numerical checks.

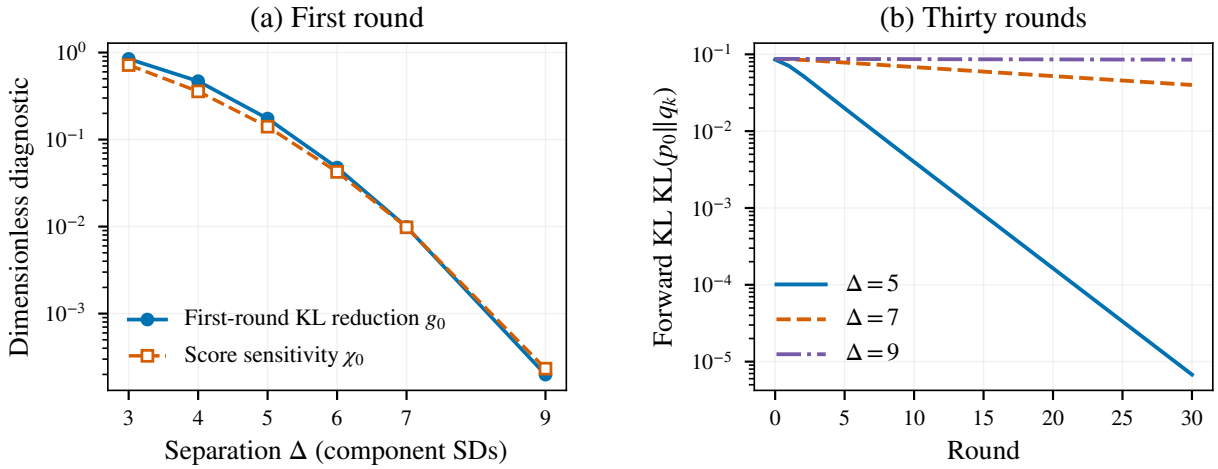


Figure 2: **Clean-score sensitivity and KL progress for separated mixtures.** The target has equal component weights and the initial law has weights $0.7, 0.3$ on the same components. (a) First-round fractional forward-KL reduction g_0 and clean-score sensitivity χ_0 versus separation in component standard deviations. Larger g_0 means a greater KL reduction. (b) Forward KL over 30 exact-expectation rounds, with no mixture fit, for separations 5, 7, and 9. Lower KL indicates closer agreement. These are deterministic density calculations, so there are no sampling bands. Appendix E.3 gives the grid resolution checks.

11.3 Effect of the gate over several rounds

We vary the gate in Algorithm 1 on a smooth eight-dimensional target with positive, position-dependent curvature. Appendix E.4 defines the density and the coordinates used in the figure. The gates share the same initial reference samples, sample counts, and reverse-SDE random draws within each seed. Tweedie and TSI use the gates 0 and I_d, respectively. The spatially uniform scalar and matrix gates depend only on t. The local scalar and local Matrix Blend gates also depend on y through conditional moments under q. Their formulas and numerical safeguards are given in Appendix E.5. The uniform scalar q-CV gate adapts a control-variate (CV) coefficient to the current law. The local scalar gate q-CVSI adapts the control-variate score identity (CVSI) [23]. The q-DPSMC gate adapts a uniform matrix coefficient from DPSMC [40]. The local Matrix Blend uses a conditional covariance quotient [14]. All of these coefficients are evaluated within Algorithm 1.

With equal sample and round counts, q-LFGI has the lowest mean SW_2 at both rounds three and four (Tables 3 and 16). At round three its error is 8.56 ± 0.40, near the target/target scale 7.87 ± 0.29. The local scalar gate has the next lowest round-three error. None of the TSI runs completes three rounds under the stopping rule in Appendix E.5. Figure 3 shows the first two transformed coordinates, and the metric uses all eight physical coordinates.

Table 3: **Gate comparison at equal sample and round counts.** Normalized full eight-dimensional SW_2 in percent, mean ± sample SD over five seeds, using 1024 GAD endpoints and 1024 independent target samples per run. Lower is better. Every gate uses 1024 reference samples per round. The target/target scale is 7.87 ± 0.29. "Runs" counts completed round-three runs. No TSI run completes round three under the rule in Appendix E.5.

Gate	Round 1	Round 2	Round 3	Runs
Tweedie	201.81 ± 2.56	200.61 ± 2.65	199.41 ± 2.18	5/5
TSI	31.22 ± 2.21	$(1.2 \pm 1.1) \times 10^6$	failed	0/5
Uniform scalar (q-CV)	160.54 ± 2.17	116.82 ± 2.05	71.06 ± 2.13	5/5
Uniform matrix (q-DPSMC)	144.32 ± 2.75	99.54 ± 1.36	63.46 ± 2.53	5/5
Local scalar (q-CVSI)	74.39 ± 2.13	30.98 ± 1.85	12.12 ± 0.75	5/5
Local Matrix Blend	57.13 ± 3.67	40.50 ± 2.94	30.05 ± 3.47	5/5
q-LFGI	22.57 ± 1.50	9.45 ± 1.60	8.56 ± 0.40	5/5

12 Low-dimensional sampler benchmarks

We compare GAD with isotropic MALA and unit-mass HMC on an anisotropic Gaussian, unequal overlapping lobes, a ring, and an S-shaped chain. The densities and parameters are given in Appendix F.1 in the order shown in Figure 4. All methods start from the same broad Gaussian law. The comparison uses a common allowance for energy, gradient, and Hessian requests, including MCMC initialization and warmup. The request counts and settings are given in Appendices F.3 to F.5.

Table 4 compares 1536 samples from each method with the same 1536 independent target samples within each run. The MCMC samples retain chain dependence and states repeated after rejections. GAD has the lowest mean error on the Gaussian, lobes, and S-chain. HMC has the lowest mean error on the ring. The between-seed SDs are comparable to or larger than many of the differences in mean error.

Table 4: **Two-dimensional comparison with MALA and HMC.** Normalized empirical SW_2 in percent, mean ± sample SD over three seeds. Lower values are better. Each method contributes 1536 samples, compared with 1536 independent target samples shared by all methods. GAD uses 4096 reference samples per round and three rounds. Every method receives 24,576 energy, gradient, and Hessian requests in total, including MCMC initialization and warmup. Neither MCMC method uses preconditioning. Target/target compares two independent target samples of the same size. The normalization is given in (E.1).

Target	GAD	MALA	HMC	Target/target
Gaussian	6.09 ± 1.17	6.93 ± 2.79	7.29 ± 2.79	6.12 ± 0.13
Unequal lobes	5.41 ± 0.29	6.82 ± 1.38	5.60 ± 1.98	4.55 ± 0.26
Ring	5.42 ± 1.02	6.27 ± 1.54	4.94 ± 1.70	4.23 ± 0.83
S-shaped chain	5.29 ± 2.45	7.65 ± 3.27	5.40 ± 3.22	5.29 ± 1.82

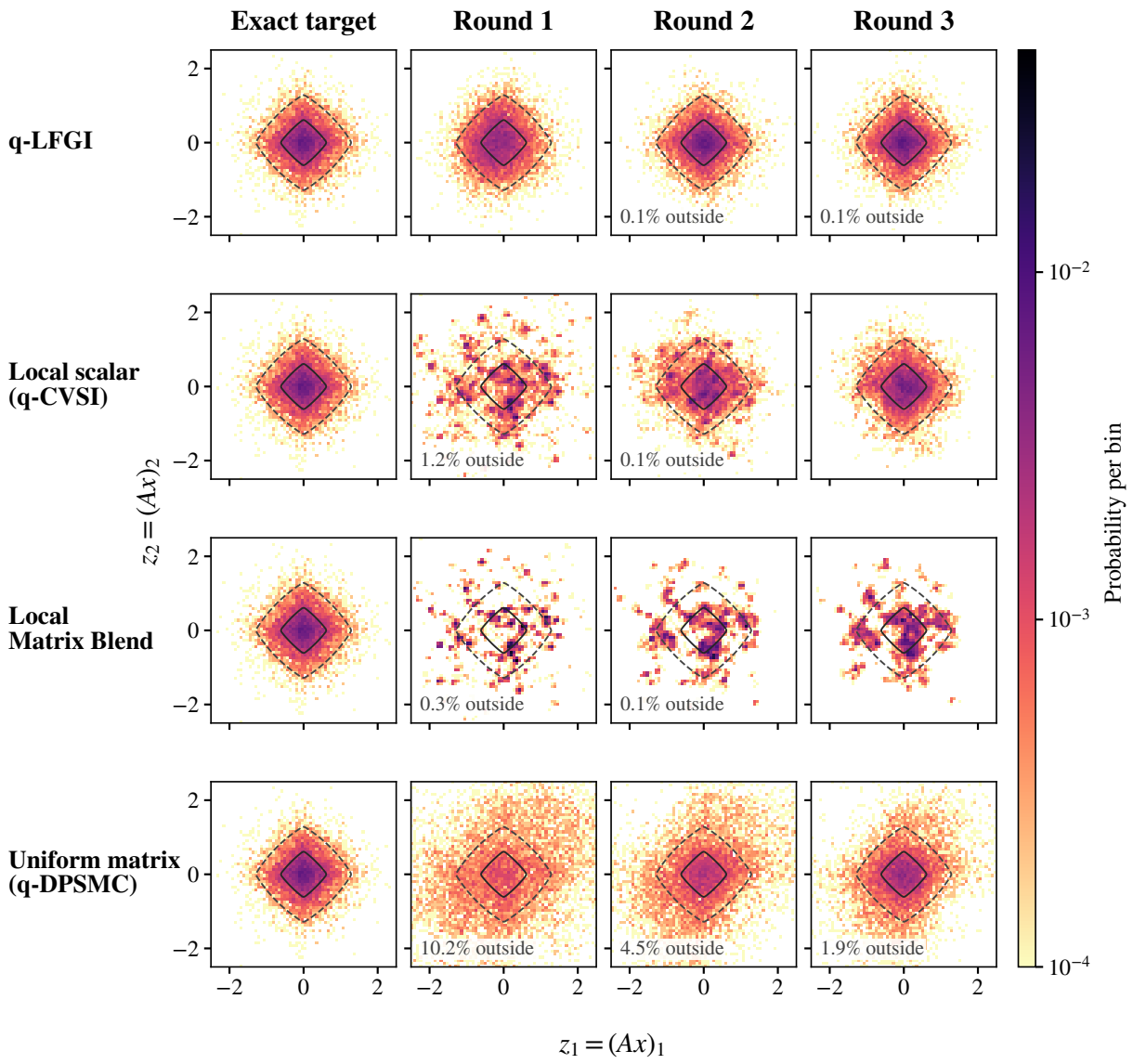


Figure 3: **Gate comparison over three rounds on an eight-dimensional target.** Rows show q-LFGI, local scalar (q-CVSI), local Matrix Blend, and uniform matrix (q-DPSMC) gates. Columns show the exact target and GAD rounds 1–3. Axes are the first two transformed coordinates $z = Ax$ defined in Appendix E.4. Panels share axes and a logarithmic probability-per-bin scale normalized by the full sample count. Contours enclose 50% and 90% of the exact projected target. The histograms use display draws from a single run. The errors in Table 3 use all eight physical coordinates and five seeds. Appendix E.5 gives the display construction.

Once a GAD score field is computed, fresh draws can be transported without further target evaluations. In Figure 4, the GAD display uses those fresh draws, while the MCMC display resamples the budget-limited chains. The number of independent draws affects the smoothness of the histograms. Figures 12 and 13 repeat the display with equal base samples and show the samples used in Table 4.

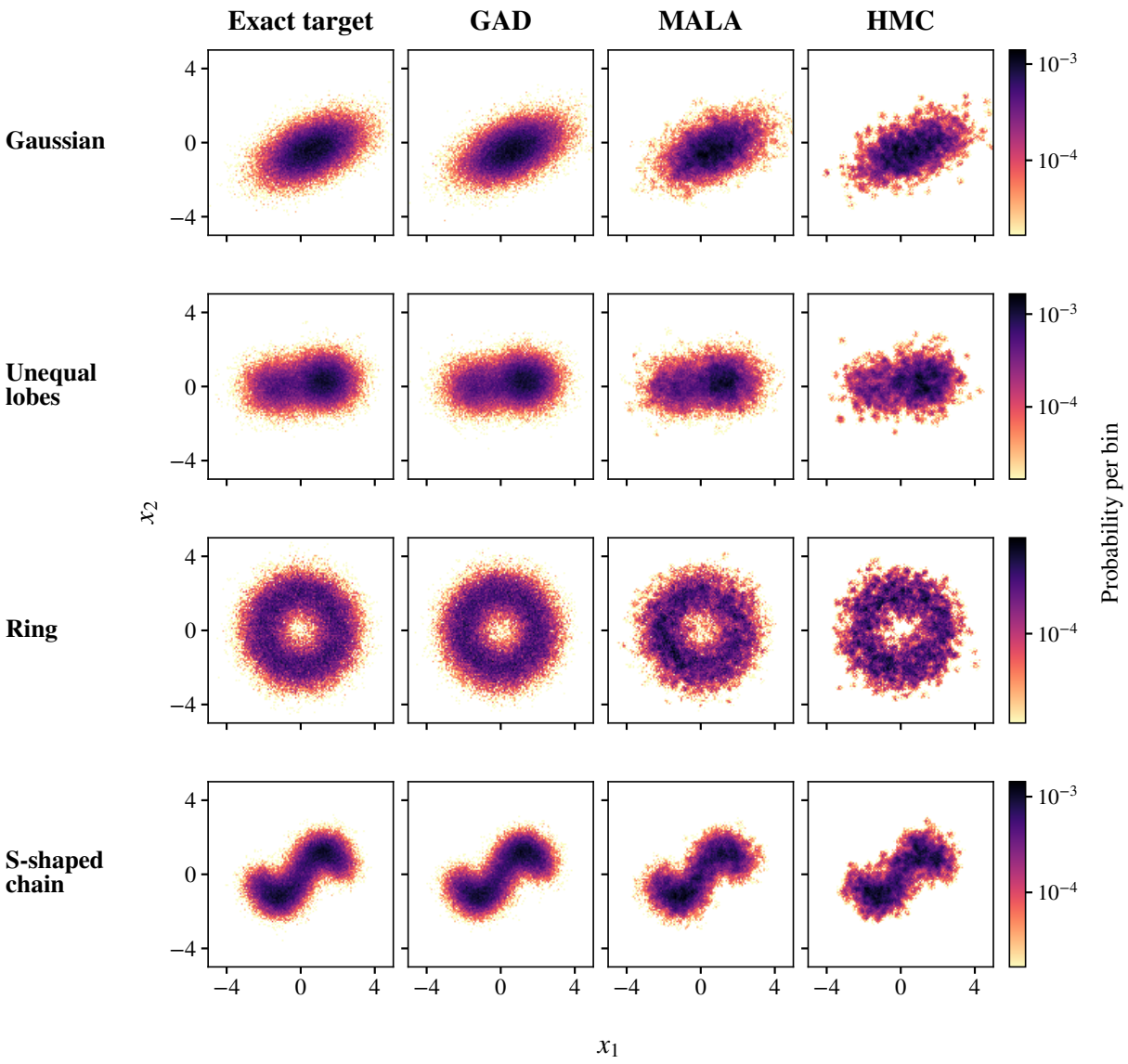


Figure 4: **Two-dimensional distributions at the selected request budget.** Rows show the Gaussian, unequal-lobe, ring, and S-shaped targets. Columns show the target and samples from three-round GAD, MALA, and HMC in the original coordinates for one seed. The target and GAD panels start from independent draws, with the GAD draws transported by the third-round score field. The MCMC panels start from the budget-limited production states. All panels use the same resampling and Gaussian jitter for display. Colors show probability per bin on a shared log scale within each target. Appendix F.2 gives the display construction and identifies the samples used for the metrics.

13 Posterior sampling in nonlinear inverse problems

We compare direct and annealed GN-GAD at common allowances for target evaluations on two nonlinear inverse problems. The acoustic example infers a trajectory from 16 coefficients. The earthquake example infers five event locations and origin times together with three shared velocity parameters, for a total of 23 unknowns. Appendix H gives two initial-field inversions whose reported posterior summaries are well described by MAP–Laplace.

13.1 Experimental setup and evaluation

Both examples use prior-whitened coefficients $\alpha \sim \mathcal{N}(0, I_d)$ and observations $y_{\text{obs}} = F(\alpha) + \eta$, with $\eta \sim \mathcal{N}(0, \Gamma)$. The posterior target is

$$\widetilde{p}_0(\alpha) = \exp\left\{-\frac{1}{2}\|\alpha\|^2 - \frac{1}{2}\|\Gamma^{-1/2}(F(\alpha) - y_{\text{obs}})\|^2\right\}, \qquad p_0(\alpha) = \frac{\widetilde{p}_0(\alpha)}{Z_{\text{tar}}}. \tag{13.1}$$

The forward map F and noise covariance Γ are specified in Appendices C.1 and D.1. We differentiate the log posterior in (13.1) for the clean score and use the Gauss–Newton approximation (9.1) to the observed information in the gate. The powered versions of the score and observed-information approximation are in (9.4).

Each problem uses one fixed synthetic dataset. We select the MCMC settings by the final primary error on a calibration seed, then hold the settings fixed for five evaluation seeds (Appendix I.4). Within each problem the seeds share the dataset and the NUTS evaluation samples.

Direct and annealed GN-GAD share their initial samples and terminal Gaussian draws within each seed. Both use the same sample count per round, round count, and request allowance. The comparison varies the likelihood schedule. The MCMC methods in the main tables start from the same initial law as GN-GAD, except for MAP-pMALA, which starts from MAP–Laplace and uses the inverse exact posterior Hessian at the MAP. The MAP-pMALA comparison includes the effect of starting from MAP–Laplace. Appendix I.3 gives the starting laws, schedules, and allowances.

We measure how sampling accuracy changes with cumulative access to the target. The query count is

$$W = N_E + N_G + N_H, \tag{13.2}$$

where the three terms count pointwise energy, gradient, and curvature requests. Each curvature request counts once whether it returns an exact Hessian or a Gauss–Newton matrix. The count includes sampler setup and warmup. Constructing the GAD score field and integrating the reverse dynamics reuse the cached derivatives from the start of the round. Appendix I.3 gives the accounting for each method.

The primary acoustic metric is joint sliced Wasserstein distance in prior-whitened coefficients, normalized by the root mean squared coefficient spread of the NUTS samples. For earthquakes, the primary metric averages the normalized distances for the location and origin-time blocks of the four sparsely observed events. Maximum event error is the largest absolute difference from the NUTS event probabilities, in percentage points. The data-fit ratio compares the median noise-normalized squared residual with the NUTS median, so the target is one. Appendices C and D give the exact metrics and events. Tables and uncertainty bands report the mean and one sample SD over the evaluation seeds. The physical histograms illustrate a single seed.

13.2 Acoustic trajectory inversion

We infer the planar path of a moving source from differences in pulse arrival times at two fixed receivers. The path is represented by eight modes of a zero-mean Gaussian process in each coordinate. At each known pulse time, the forward map takes the difference in range to the receivers and divides by the sound speed. The likelihood uses Gaussian timing noise. Appendix C.1 specifies the path basis, forward map, and observation covariance.

Each pulse constrains the difference in range to the two receivers. Many source positions can give the same range difference. The temporal prior links these positions into paths, whose

uncertainty can follow a curve that a local Gaussian may describe poorly. Figure 5 shows the generating path, receivers, and arrival differences. The data are generated from a path with a nonzero mean and a larger basis than the common inference prior (Appendix C.1).

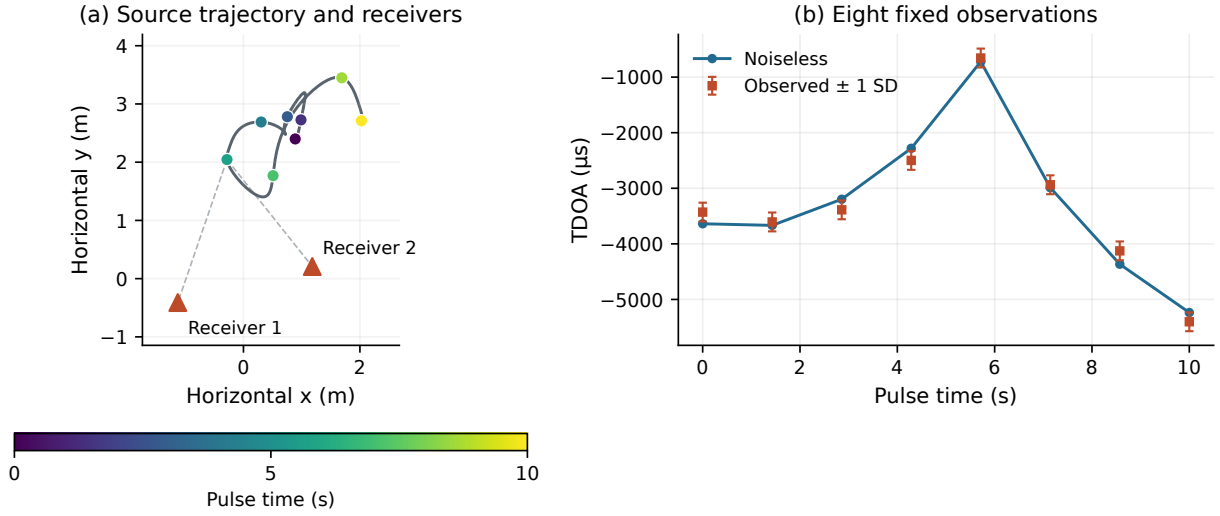


Figure 5: **Acoustic trajectory acquisition and fixed observations.** (a) Generating path in meters, with eight pulses (circles) and two receivers (triangles). Pulse color gives emission time. Dashed segments show the two ranges for one pulse. (b) Noiseless and observed time differences of arrival in microseconds at the same pulse times. Error bars show one noise SD. Appendix C.1 gives the forward map and the fixed observations.

Both GAD variants follow the broad curve in the NUTS midpoint samples at the final budget. The annealed HMC samples are less evenly spread along it in this seed (Figure 6). Appendix C gives the errors by coefficient block and summaries of the paths.

At the final allowance, annealed GN-GAD attains a joint SW_2 of 0.134 ± 0.012, 37.1% below annealed HMC (0.213 ± 0.004), the MCMC method with the lowest error on this measure (Table 5). Annealing reduces GAD's joint error in every paired seed. Annealed GN-smMALA has the lowest mean maximum event error, and MAP-pMALA comes closest to NUTS on median data misfit.

Table 5: **Acoustic endpoint comparison.** Five-seed means ± sample SD under an allowance of 27517 E+G+H requests. Event error is the largest absolute probability error in percentage points (pp). Lower SW_2 and event errors are better, and a fit ratio of one matches NUTS. Appendix C gives the metric definitions, actual request counts, and full comparison.

Method	Joint $SW_2 \downarrow$	Event error (pp) $\downarrow$	Fit ratio $\to 1$
Annealed GN-GAD	0.134 ± 0.012	15.2 ± 2.3	2.03 ± 0.12
Direct GN-GAD	0.154 ± 0.009	19.3 ± 2.3	2.15 ± 0.07
MAP-pMALA	0.630 ± 0.020	47.3 ± 7.7	0.98 ± 0.07
Annealed MALA	0.218 ± 0.001	16.7 ± 3.5	2.33 ± 0.25
GN-smMALA	0.312 ± 0.113	16.4 ± 4.5	7.36 ± 1.61
Annealed GN-smMALA	0.239 ± 0.026	13.2 ± 3.6	6.71 ± 1.27
Annealed HMC	0.213 ± 0.004	14.6 ± 4.3	2.16 ± 0.12

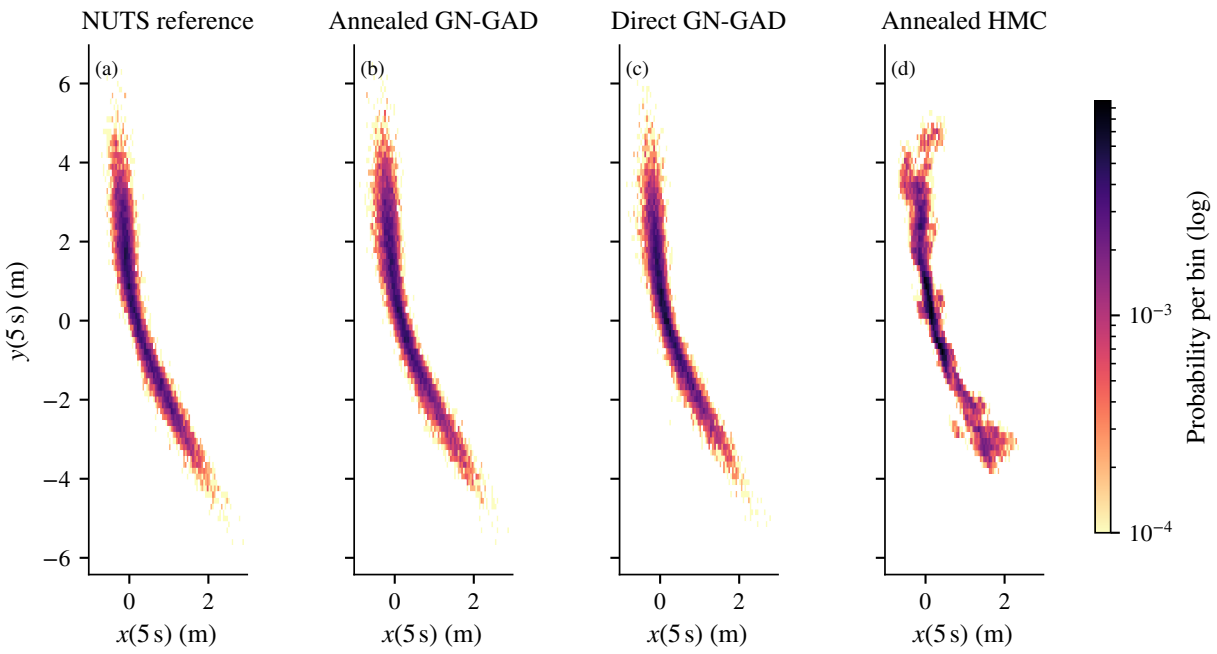


Figure 6: **Acoustic midpoint geometry at the final allowance.** The axes give the two coordinates of the path at its temporal midpoint, 5 s, in meters. Panels show NUTS samples and a single seed for annealed GN-GAD, direct GN-GAD, and annealed HMC at the final request allowance. Panels use common bins and a shared logarithmic probability-per-bin scale, with counts divided by the full sample size. Appendix C.4 gives the display construction.

13.3 Joint earthquake location and velocity inference

We jointly infer the locations and origin times of five earthquakes and three shared velocity parameters from sparse arrival-time observations. Each event contributes east, north, depth, and origin-time coordinates. The velocity model describes how the primary (P) and secondary (S) wave speeds change with depth. The unknowns are the primary wave speed at the surface, its vertical gradient, and the ratio of P- to S-wave speeds. A pick is modeled as the event origin time plus the travel time to the station, with phase-dependent Gaussian noise. Appendix D.1 gives the priors, the analytic travel-time formula, and its domain.

E1 serves as an anchor because we observe both P and S arrivals at every station. For each of E2–E5 we have just two P arrivals and one S arrival, for a total of 24 observations with fixed station associations (Figure 7). Differences between S and P arrivals and between pairs of P arrivals constrain distances to a pair of stations. Many locations along a curved set can explain those differences. The likelihood depends on each event's origin time and on velocities shared across events. The curvature persists under the affine change from event to whitened coordinates. Appendix D.1 lists the generating parameters and picks.

Figure 8 compares the location–depth tradeoffs for events E2 and E4 at the final allowance.

At the final allowance, annealed GN-GAD attains an event-block SW_2 of 0.154 ± 0.019, 42.5% below direct GN-smMALA, the MCMC method with the lowest block error (Table 6). Annealing roughly halves GAD's mean block error and reduces it in all five paired seeds. Annealed GN-GAD also has the lowest mean event-probability error, and MAP-pMALA comes closest to the NUTS median data misfit. The same likelihood schedule increases the event-block error of GN-smMALA at this allowance.

Figure 9 plots primary error against the common request allowance at each checkpoint. Both

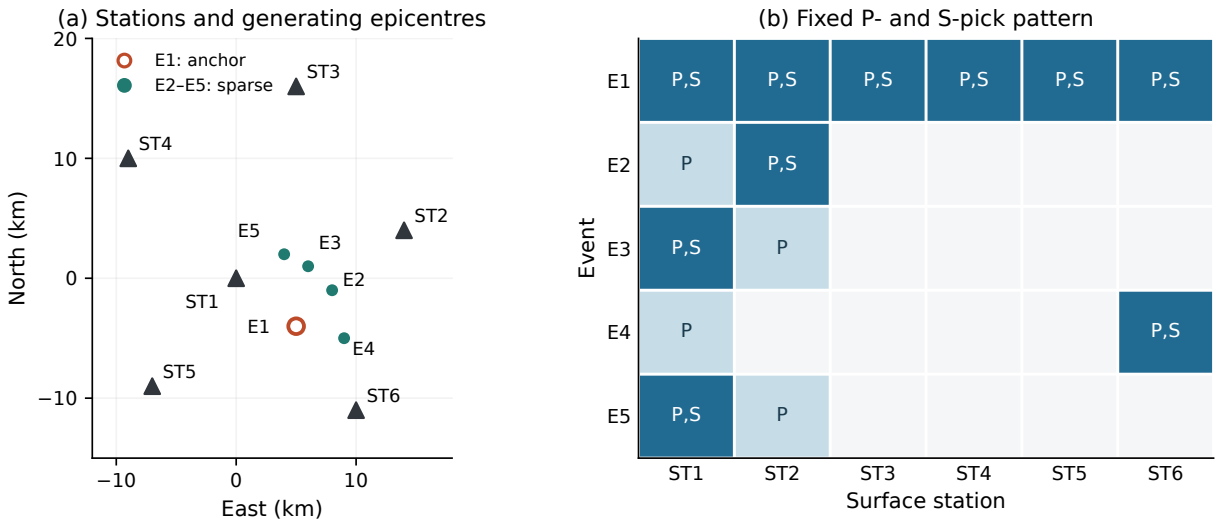


Figure 7: **Earthquake acquisition geometry and observed phases.** (a) Surface stations (triangles) and generating epicenters (circles) in east and north coordinates in km. (b) P- and S-wave arrivals observed for each event and station. The anchor event E1 has both phases at all six stations. A blank cell indicates no pick. Appendix D.1 gives the generating parameters and fixed arrival times.

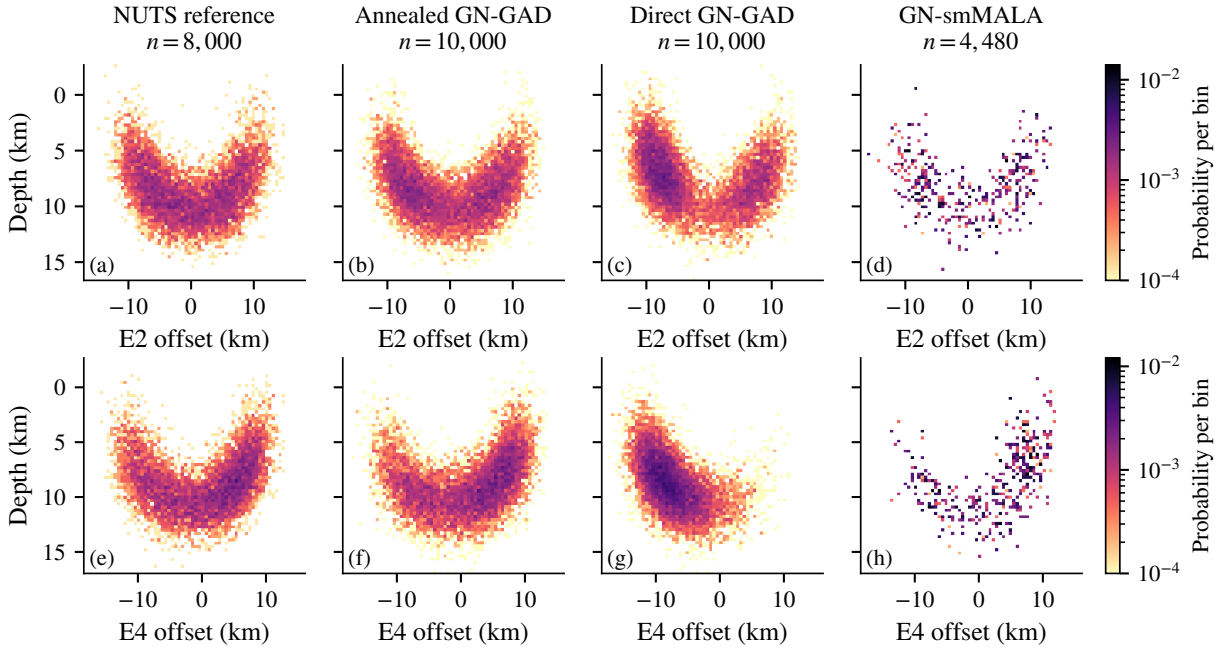


Figure 8: **Earthquake location and depth at the final allowance.** For events E2 and E4, horizontal offset is measured along the leading horizontal principal axis of the NUTS samples and centered at their mean. Offset and depth are in km, with depth positive downward. Columns show NUTS, annealed GN-GAD, direct GN-GAD, and GN-smMALA. Sampler panels use one seed. Each row shares bins and a logarithmic probability-per-bin scale normalized by the full sample count. Appendix D.4 gives the display construction.

annealed GAD runs reach full likelihood power for their final two rounds and continue to improve

Table 6: **Earthquake-location endpoint comparison.** Five-seed means ± sample SD under an allowance of 27000 E+G+H requests. The block metric averages the normalized SW_2 distances for E2–E5, and event error is the largest absolute probability error in percentage points (pp). Lower errors are better, and a fit ratio of one matches NUTS. Appendix D gives the metric definitions, actual request counts, and full comparison.

Method	Event-block $SW_2 \downarrow$	Event error (pp) $\downarrow$	Fit ratio $\to 1$
Annealed GN-GAD	0.154 ± 0.019	7.7 ± 1.2	1.45 ± 0.05
Direct GN-GAD	0.311 ± 0.104	8.9 ± 3.3	1.36 ± 0.04
MAP-pMALA	0.491 ± 0.025	15.3 ± 4.3	1.00 ± 0.02
Annealed MALA	0.653 ± 0.020	41.7 ± 4.5	2.73 ± 0.07
GN-smMALA	0.267 ± 0.028	16.5 ± 2.0	3.76 ± 0.12
Annealed GN-smMALA	0.374 ± 0.021	29.0 ± 2.7	4.92 ± 0.25
Annealed HMC	0.359 ± 0.028	28.6 ± 5.3	1.28 ± 0.06

at full power. Intermediate-power points show distance to the final posterior.

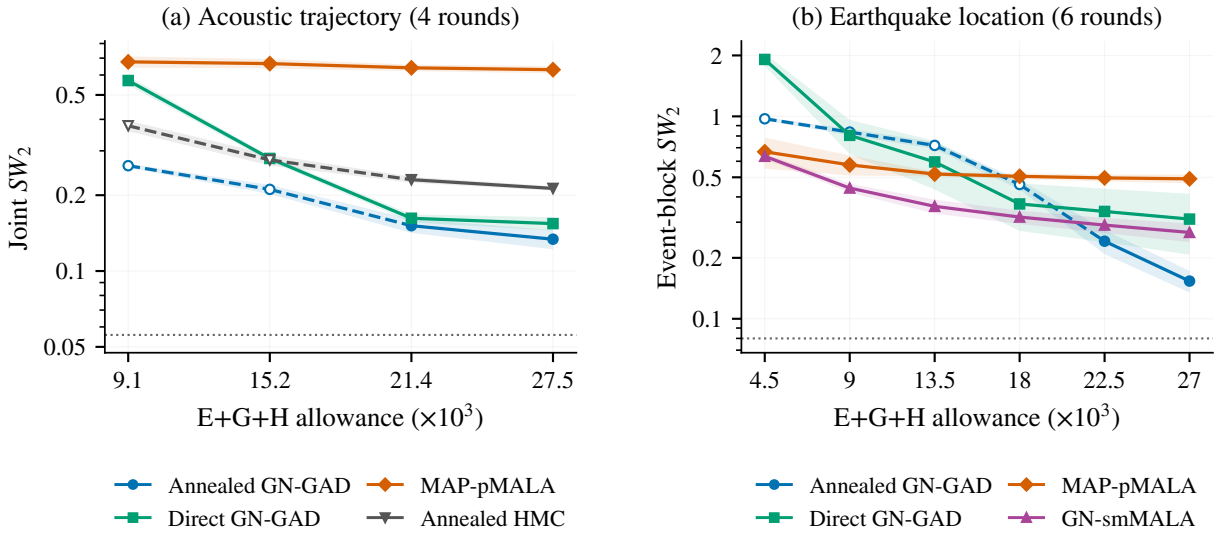


Figure 9: **Geometric accuracy under common request allowances.** The left panel shows joint normalized SW_2 for acoustic inversion, and the right panel shows the mean normalized SW_2 over event blocks E2–E5. Lower values indicate closer agreement with NUTS. Markers give means over five sampler seeds and shading gives one sample SD. The horizontal axis gives each checkpoint's common allowance in thousands of energy, gradient, and curvature requests (Appendix I.3). Hollow markers denote checkpoints without full-power samples, and dashed segments touch those checkpoints. Dotted lines show discrepancies between disjoint groups of NUTS chains. The vertical axes are logarithmic.

14 Discussion

With conditional means under the target law, LFGI minimizes the conditional mean squared score error. With conditional means under the current law, the gate weights the correction to the current score. The error of the corrected score depends on how target curvature varies across the clean states that could have produced the noisy state (Sections 5 and 6). For the perturbed

Gaussian targets, the contraction argument bounds this error using conditional averages, without requiring a Poisson solve or a smooth initial density (Section 8).

For the Gaussian target with exact target derivatives, constant curvature makes the score estimate exact regardless of where the reference samples lie. On the eight-dimensional target with varying curvature, changing the gate changes the sample error after the same number of rounds (Tables 2 and 3). In the mixture experiment, the more separated components show both a smaller clean-score response to changing their weights and slower improvement in forward KL (Figure 2). Among the two-dimensional benchmarks, GAD has the lowest mean error on three targets and HMC on the ring (Table 4).

Annealed GN-GAD comes closest to NUTS on the primary sliced Wasserstein measure in both inverse problems, while MAP-pMALA comes closest on median data misfit (Tables 5 and 6). Different trajectories or event locations can predict similar arrival times. The median residual measures agreement with the observations. The distributional distances and physical projections assess where the samplers place probability among the trajectories or locations that can produce those observations.

In the two initial-field inversions, MAP–Laplace agrees closely with the NUTS reference on the displayed projections and gives the smallest reported field-moment errors. GN-GAD improves on MAP-pMALA in sliced Wasserstein distance, with a smaller gain over MAP–Laplace (Appendix H). In the acoustic and earthquake examples, the NUTS projections instead show curved range-difference and location–depth tradeoffs. The orientation of uncertainty changes along those projected sets (Figures 6 and 8).

15 Limitations and future work

Theorem 8.4 concerns exact conditional expectations and exact curvature in reverse diffusion for a fixed, strongly log-concave target with sufficiently small variation of the perturbation derivatives. The separated mixtures are outside this class. In the inverse problems, we estimate the conditional means from samples, approximate the curvature by Gauss–Newton, and integrate a deterministic flow numerically. We choose a likelihood schedule for each problem and change the target between rounds as the likelihood power increases. We also test the earthquake schedule with GN-smMALA (Table 6 and appendix I.2).

Dense curvature storage and repeated field evaluations contribute to the cost of reusing the target derivatives during transport (Appendix I.5). Estimating densities along the probability flow may also allow importance weighting to correct errors in component probabilities. The density calculations in [14] use posterior pilot samples. Here the question is how accurately the density can be evaluated using the samples generated by GAD itself.

16 Conclusion

We estimate a diffusion score using the current samples and derivatives of the unnormalized target density. The q-LFGI gate weights the conditional difference between two score signals. We add the result to the current score and use reverse diffusion to produce the next set of unweighted samples. For a Gaussian target with exact target derivatives, the corrected score equals the noisy target score for any nonempty set of reference samples. For strongly log-concave perturbations of Gaussian targets whose derivatives vary sufficiently little, the contribution of the initial error to the forward KL bound decays geometrically across rounds with exact conditional expectations.

On the acoustic and earthquake problems, annealed GN-GAD follows the curved posterior structure seen in the reference and gives the lowest mean primary sliced Wasserstein error among the methods compared at the final query allowance. Each round reuses its target derivatives throughout transport. Structured curvature and faster reverse integration could reduce the cost of evaluating the score field along those trajectories.

A Poisson equation and q-LFGI identities

A.1 Weak Poisson solvability

For the weak problem (4.2), assume $g \in L^2(\rho^q_{y,t})$ and a finite current-posterior Poincaré constant $C_{\mathrm{P}}(\rho^q_{y,t})$. For centered $\varphi \in H^1(\rho^q_{y,t})$,

$$|\mathbb{E}_{\rho^q_{y,t}}[g\varphi]| \le \|g\|_{L^2(\rho^q_{y,t})}\|\varphi\|_{L^2(\rho^q_{y,t})} \le \sqrt{C_{\mathrm{P}}(\rho^q_{y,t})}\|g\|_{L^2(\rho^q_{y,t})}\|\nabla\varphi\|_{L^2(\rho^q_{y,t})}.$$

The bilinear form $\mathbb{E}_{\rho^q_{y,t}}[\nabla u \cdot \nabla\varphi]$ is coercive on mean-zero $H^1(\rho^q_{y,t})$ modulo constants. Lax–Milgram gives a unique mean-zero weak solution under these conditions, which justify the Poisson representation used in Sections 4 and 5. The convergence proof uses the score-error identity (8.2).

A.2 Paired identities and calibrated q-LFGI risk

Because

$$\nabla_x K_t^{\mathrm{OU}}(y \mid x) = -\alpha_t b(x; y, t) K_t^{\mathrm{OU}}(y \mid x),$$

differentiation of the OU marginals gives $\mathbb{E}_{\rho^q_{y,t}}[b] = s^q_t$ and $\mathbb{E}_{\rho_{y,t}}[b] = s_t$. If q has a positive differentiable density and the boundary term vanishes, integration by parts in x gives $\mathbb{E}_{\rho^q_{y,t}}[\nabla \log q(X_0)] = \alpha_t \mathbb{E}_{\rho^q_{y,t}}[b]$. Subtracting the mean of b from the mean of $c = s_0/\alpha_t$ gives the expression for m^q_t in terms of the density ratio following (4.1). The target OU posterior score is

$$\nabla_x \log \rho_{y,t}(x) = s_0(x) - \frac{\alpha_t}{\gamma_t}(\alpha_t x - y) = \alpha_t\{c(x;t) - b(x;y,t)\}. \tag{A.1}$$

Its mean under $\rho_{y,t}$ vanishes under the boundary condition for the target, giving $\mathbb{E}_{\rho_{y,t}}[c] = s_t$. For a matrix gate $G(y,t)$, set $a^G_t = Gm^q_t$ and $e^G_t = d_t - a^G_t$. Both score signals have target-OU-posterior mean $s_t(y)$, hence

$$e^G_t(y) = \mathbb{E}_{\rho_{y,t}}[z_G] - \mathbb{E}_{\rho^q_{y,t}}[z_G] = \mathbb{E}_{\rho^q_{y,t}}\left[g\{z_G - \mathbb{E}_{\rho^q_{y,t}}[z_G]\}\right]. \tag{A.2}$$

Direct differentiation gives

$$J_G(x;y,t) := \nabla_x z_G(x;y,t) = \frac{1}{\alpha_t\gamma_t}\{\alpha_t^2 I_d - G(y,t)A_0(x;t)\}. \tag{A.3}$$

Averaging (A.3) under the current OU posterior gives

$$\mathbb{E}_{\rho^q_{y,t}}[J_G] = \frac{1}{\alpha_t\gamma_t}\{\alpha_t^2 I_d - GA^q(y,t)\}.$$

Hence

$$\mathbb{E}_{\rho^q_{y,t}}[J_G] = 0 \quad\Longleftrightarrow\quad G = \alpha_t^2 (A^q)^{-1} = \Psi_t(H^q) = G^q_{\mathrm{LFGI}}.$$

At $q = p_0$, assume the second moments and integration by parts used in the following identities are valid. Integrating the derivative of the target OU posterior score (A.1) under $\rho_{y,t}$ gives the covariance of the zero-mean disagreement signal

$$\mathbb{E}_{\rho_{y,t}}[(c-b)(c-b)^\top] = \frac{1}{\gamma_t} I_d + \frac{1}{\alpha_t^2} H^{p_0}(y,t),$$

and integration by parts with $b - s_t$ gives

$$\mathbb{E}_{\rho_{y,t}}[(b-s_t)(c-b)^\top] = -\frac{1}{\gamma_t} I_d.$$

Assume the disagreement covariance is positive definite, as holds when $\alpha_t^2 I_d + \gamma_t H^{p_0}$ is positive definite. The conditional squared-error risk in (6.3) has normal equation

$$G\left(\frac{1}{\gamma_t} I_d + \frac{1}{\alpha_t^2} H^{p_0}(y,t)\right) = \frac{1}{\gamma_t} I_d.$$

Its unique solution is

$$G = \Psi_t(H^{p_0}) = G_\star,$$

the LFGI population-optimal gate [14]. This proves (6.3).

A.3 The two Poisson moments

The weak Poisson equation (4.2) implies, componentwise for every sufficiently regular vector-valued f,

$$\mathbb{E}_{\rho_{y,t}}[f] - \mathbb{E}_{\rho^q_{y,t}}[f] = \mathbb{E}_{\rho^q_{y,t}}[(\nabla f)\nabla u].$$

Taking $f = z_G$ and using (A.2) gives

$$e_t^G = \mathbb{E}_{\rho^q_{y,t}}[J_G \nabla u].$$

Taking $f = \nabla\varphi$ yields the differentiated Stein identity

$$\mathbb{E}_{\rho^q_{y,t}}[\nabla^2\varphi\, \nabla u] = \mathbb{E}_{\rho_{y,t}}[\nabla\varphi] - \mathbb{E}_{\rho^q_{y,t}}[\nabla\varphi].$$

For $\varphi(x) = \|x\|^2/2$,

$$\mathbb{E}_{\rho^q_{y,t}}[\nabla u] = \mathbb{E}_{\rho_{y,t}}[X_0] - \mathbb{E}_{\rho^q_{y,t}}[X_0] = \frac{\gamma_t}{\alpha_t} d_t.$$

For $\varphi = \log \rho_{y,t}$, the target OU posterior satisfies

$$\nabla^2 \log \rho_{y,t} = -\frac{1}{\gamma_t} A_0, \qquad \mathbb{E}_{\rho_{y,t}}[\nabla \log \rho_{y,t}] = 0, \qquad \mathbb{E}_{\rho^q_{y,t}}[\nabla \log \rho_{y,t}] = \alpha_t m_t^q.$$

Substitution gives

$$\mathbb{E}_{\rho^q_{y,t}}[A_0 \nabla u] = \alpha_t \gamma_t\, m_t^q,$$

proving (4.3).

B Proofs for the population convergence theorem

Throughout, Assumption 8.1 holds and q is a probability law on $\mathbb{R}^d$. A density and finite $\mathrm{KL}(p_0\|q)$ are assumed only where stated. At a query (y,t) with $t > 0$, write $\mu^q := \mathbb{E}_{\rho^q_{y,t}}[X_0]$. Statements about (3.1) on a window $[t_1, t_2]$ use the reverse clock $u = t_2 - t$, in which the equation driven by a score s reads

$$\mathrm{d}Z_u = \{Z_u + 2s(Z_u, t_2 - u)\}\mathrm{d}u + \sqrt{2}\,\mathrm{d}B_u, \qquad 0 \le u \le t_2 - t_1. \tag{B.1}$$

B.1 Posterior moments and growth of the corrected score

For $t > 0$, the kernel $x \mapsto K_t^{\mathrm{OU}}(y \mid x)$ is positive, and every polynomial in x times it is bounded. Hence $\rho_{y,t}^q$ has moments of all orders even when q has none. Since H_0 is bounded and s_0 has linear growth, all posterior averages in Sections 4 to 6 are finite. Because $\nabla_y \log \rho_{y,t}^q(x) = b(x; y, t) - s_t^q(y)$, differentiation under the integral gives, componentwise, $\nabla_y \mathbb{E}_{\rho_{y,t}^q}[F] = \mathrm{Cov}_{\rho_{y,t}^q}(F, b)$ for every $F(x)$ of polynomial growth. On each compact set $[\varepsilon, T] \times \{\|y\| \le n\}$ with $0 < \varepsilon < T$, the Gaussian kernel bounds the polynomial-weighted numerators uniformly, and the positive continuous density $q_t(y)$ has a positive minimum. The positive minimum and the bounds on the numerators bound the derivatives on each compact set. In particular, $s^{a,q}$ is continuous on $(0, \infty) \times \mathbb{R}^d$ and locally Lipschitz in y, uniformly for t in compact subsets of $(0, \infty)$.

Lemma B.1 (Posterior moment bound)**.** *Choose $R < \infty$ with $\beta_R := q(\{\|x\| \le R\}) > 0$. For all $t > 0$ and $y \in \mathbb{R}^d$,*

$$\mathbb{E}_{\rho_{y,t}^q} \|\alpha_t X_0 - y\|^2 \le (\|y\| + R)^2 + 2 \log(1/\beta_R), \qquad \|\alpha_t \mu^q\| \le 2\|y\| + R + \sqrt{2 \log(1/\beta_R)}.$$

Proof. Let $\ell := \int \exp\{-\|\alpha_t x - y\|^2/(2\gamma_t)\}\, q(\mathrm{d}x)$, so that $\mathrm{d}\rho_{y,t}^q/\mathrm{d}q = \exp\{-\|\alpha_t x - y\|^2/(2\gamma_t)\}/\ell$. Nonnegativity of $\mathrm{KL}(\rho_{y,t}^q \| q)$ gives $\mathbb{E}_{\rho_{y,t}^q} \|\alpha_t X_0 - y\|^2 \le -2\gamma_t \log \ell$. Restricting the integral to $\{\|x\| \le R\}$ gives $\ell \ge \beta_R \exp\{-(\|y\| + \alpha_t R)^2/(2\gamma_t)\}$. The first bound follows from $\alpha_t, \gamma_t \le 1$. The second follows from Jensen's inequality and the triangle inequality. □

Lemma B.2 (Score cancellation and linear growth)**.** *For $t > 0$ and $y \in \mathbb{R}^d$,*

$$s_t^{a,q}(y) = -(A^q)^{-1}\{H^q y + \alpha_t(\mathbb{E}_{\rho_{y,t}^q}[\nabla V] - (H^q - \Lambda)\mu^q)\}. \tag{B.2}$$

With R and β_R as in Lemma B.1, $\|s_t^{a,q}(y)\| \le C_q(1 + \|y\|)$ for all $t > 0$ and y, where

$$C_q := \frac{\|\Lambda\|_{\mathrm{op}} + 3L_V + \sup_x \|\nabla V(x)\| + L_V\{R + \sqrt{2\log(1/\beta_R)}\}}{\min\{1, \kappa\}}.$$

The linear growth bound also holds for the target score s_t, with constant C_{p_0}.

Proof. Tweedie's identity gives $s_t^q = (\alpha_t \mu^q - y)/\gamma_t$. Since $-s_0(x) = \Lambda x + \nabla V(x)$, $m_t^q = -\alpha_t^{-1}(\Lambda \mu^q + \mathbb{E}_{\rho_{y,t}^q}[\nabla V]) - (\alpha_t \mu^q - y)/\gamma_t$. Insert $a_t^q = \alpha_t^2 (A^q)^{-1} m_t^q$ into $s_t^{a,q} = s_t^q + a_t^q$ and use $I_d - \alpha_t^2 (A^q)^{-1} = \gamma_t (A^q)^{-1} H^q$. The factors $1/\gamma_t$ cancel, and (B.2) follows. The growth bound uses $A^q \succeq (\alpha_t^2 + \kappa\gamma_t) I_d \succeq \min\{1, \kappa\} I_d$, $\|H^q\|_{\mathrm{op}} \le \|\Lambda\|_{\mathrm{op}} + L_V$, $\|H^q - \Lambda\|_{\mathrm{op}} \le L_V$, and Lemma B.1. At $q = p_0$, $m_t^q = 0$ by (4.1), so $s_t^{a,p_0} = s_t$. □

B.2 Well-posedness and path-entropy comparison

Lemma B.3 (Well-posedness)**.** *Let $0 \le t_1 < t_2$, and let s be $s^{a,q}$ or the target score. For every law of Z_0, (B.1) has a unique strong solution on $[0, t_2 - t_1]$, including the endpoint, and the solution does not explode.*

Proof. For $0 < \varepsilon < t_2$, the drift is continuous and locally Lipschitz on $[0, t_2 - \max\{t_1, \varepsilon\}]$ (Appendix B.1), so a pathwise unique strong solution exists there up to explosion. For $s = s^{a,q}$, Lemma B.2 and Gronwall's inequality give, pathwise,

$$\sup_u \|Z_u\| \le (\|Z_0\| + 2C_q t_2 + \sqrt{2} \sup_{u \le t_2} \|B_u\|) e^{(1+2C_q)t_2},$$

and the target score satisfies Lemma B.2 with C_{p_0}. Hence the solution does not explode, with no moment condition on Z_0. When $t_1 = 0$, the pathwise bound on $\sup_u \|Z_u\|$ makes the drift integrable along each path up to $u = t_2$. Thus Z_u converges as $u \uparrow t_2$, and uniqueness on every $[0, t_2 - \varepsilon]$ gives uniqueness through the endpoint. □

For the exact target score, apply the diffusion time-reversal theorem [19, Theorem 2.1] on the part of the forward interval with $t \geq \varepsilon > 0$. The OU coefficients are Lipschitz and the marginals there are smooth and bounded, with locally bounded scores. The reversed OU process solves (B.1) weakly [see also 3]. Pathwise uniqueness (Lemma B.3) and the Yamada–Watanabe theorem identify its law with that of the solution started from p_{t_2}. Path continuity extends the identification to $t_1 = 0$. In particular,

$$\begin{aligned} Z_u \sim p_{t_2-u}, \qquad 0 \leq u \leq t_2 - t_1, \\ \mathrm{Law}(Z_{t_2-t_1} \mid Z_0 = z) = \mathrm{Law}(Y_{t_1} \mid Y_{t_2} = z) \qquad \text{for } p_{t_2}\text{-almost every } z. \end{aligned} \tag{B.3}$$

The second relation follows by disintegrating the common law with respect to its initial point in reverse time. The conditional law on the right has a density by Bayes' rule with the OU transition density.

Lemma B.4 (Interval path-entropy comparison)**.** *Let $0 \leq t_1 < t_2$, let ν be a probability density with $\mathrm{KL}(p_{t_2}\|\nu) < \infty$, and let r_{t_1} be the endpoint law of (B.1) with $s = s^{a,q}$ and $Z_0 \sim \nu$. Then r_{t_1} has a density and*

$$\mathrm{KL}(p_{t_1}\|r_{t_1}) \leq \mathrm{KL}(p_{t_2}\|\nu) + \int_{t_1}^{t_2} \mathbb{E}_{p_t}[\|e_t^q\|^2]\mathrm{d}t. \tag{B.4}$$

Proof. Write $\tau := t_2 - t_1$, and let $\mathbf{P}^z$ and $\mathbf{R}^z$ be the path laws on $C([0,\tau];\mathbb{R}^d)$ of the target and corrected solutions started at z. By (3.3), their drifts differ by $2e^q_{t_2-u}$. Let σ_n be the exit time from the open ball of radius n, and let $A_n := \{\sigma_n > \tau\}$. By Lemma B.2, the drift difference is bounded before $\sigma_n \wedge \tau$. Girsanov's theorem with this stopped bounded drift change gives a law $\mathbf{R}^{z,n}$ equivalent to $\mathbf{P}^z$ under which the coordinate process solves the corrected equation up to σ_n. Pathwise uniqueness up to σ_n gives $\mathbf{R}^{z,n} = \mathbf{R}^z$ on measurable subsets of A_n. Thus $\mathbf{P}^z$ and $\mathbf{R}^z$ are equivalent on A_n, with the stopped Girsanov density. Paths are continuous and neither process explodes, so $\mathbf{P}^z(A_n) \to 1$ and $\mathbf{R}^z(A_n) \to 1$. Hence $\mathbf{P}^z$ and $\mathbf{R}^z$ are equivalent and, $\mathbf{P}^z$-almost surely,

$$\log \frac{\mathrm{d}\mathbf{P}^z}{\mathrm{d}\mathbf{R}^z} = \sqrt{2}\int_0^\tau e^q_{t_2-u}(Z_u) \cdot \mathrm{d}B_u + \int_0^\tau \|e^q_{t_2-u}(Z_u)\|^2 \mathrm{d}u,$$

where B is the Brownian motion of (B.1) under $\mathbf{P}^z$. This localization replaces a Novikov condition [6, Section 5.2].

Let $\mathbf{P}$ and $\mathbf{R}$ mix the initial points over p_{t_2} and ν, respectively. By (B.3), $Z_u \sim p_{t_2-u}$ under $\mathbf{P}$. The expected quadratic variation of the stochastic integral is $2\int_{t_1}^{t_2} \mathbb{E}_{p_t}[\|e_t^q\|^2]\mathrm{d}t$, which is finite by Lemma B.2 and the finite second moments of p_t. The stochastic integral thus has mean zero under $\mathbf{P}^z$ for p_{t_2}-almost every z. The chain rule for relative entropy [24] gives

$$\mathrm{KL}(\mathbf{P}\|\mathbf{R}) = \mathrm{KL}(p_{t_2}\|\nu) + \int_{t_1}^{t_2} \mathbb{E}_{p_t}[\|e_t^q\|^2]\mathrm{d}t.$$

The data-processing inequality for the endpoint map proves (B.4). Finally, $\nu \ll p_{t_2}$ because $p_{t_2} > 0$. For p_{t_2}-almost every z, the endpoint law of $\mathbf{P}^z$ has a density by (B.3), and the endpoint law of $\mathbf{R}^z$ is equivalent to it. Mixing over ν shows that r_{t_1} has a density. □

B.3 The relative de Bruijn identity

To justify the relative de Bruijn identity and its integral (3.2) on the stated domain, fix $q \in \mathcal{Q}$, $T > 0$, and $0 < \varepsilon < T$. For $t \in [\varepsilon, T]$, the marginals p_t and q_t are smooth and positive. By Tweedie's identity, $\nabla^2 \log q_t = (\alpha_t^2/\gamma_t^2)\,\mathrm{Cov}_{\rho^q_{y,t}}(X_0) - \gamma_t^{-1} I_d$. With β_R as in Lemma B.1,

$$\beta_R(2\pi\gamma_t)^{-d/2}\exp\{-(\|y\|+R)^2/(2\gamma_t)\} \le q_t(y) \le (2\pi\gamma_t)^{-d/2}.$$

The bounds on q_t and the posterior moment estimate of Lemma B.1 show that $|\log q_t|$, $\|\nabla \log q_t\|$, and $\|\nabla^2 \log q_t\|$ grow at most quadratically in y, uniformly for $t \in [\varepsilon, T]$. The same holds for p_t, which in addition has Gaussian tails. These bounds justify differentiation under the integral and integration by parts without boundary terms in the common Fokker–Planck equation $\partial_t p = \nabla \cdot (yp) + \Delta p$. Hence $\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{KL}(p_t\|q_t) = -\mathbb{E}_{p_t}[\|d_t\|^2]$ on $[\varepsilon, T]$, and

$$\mathrm{KL}(p_\varepsilon\|q_\varepsilon) - \mathrm{KL}(p_T\|q_T) = \int_\varepsilon^T \mathbb{E}_{p_t}[\|d_t\|^2]\mathrm{d}t.$$

Data processing gives $\mathrm{KL}(p_\varepsilon\|q_\varepsilon) \le \mathrm{KL}(p_0\|q)$. Joint lower semicontinuity of relative entropy under the weak convergence $(p_\varepsilon, q_\varepsilon) \to (p_0, q)$ gives the reverse inequality in the limit. Monotone convergence as $\varepsilon \downarrow 0$ proves (3.2). For (8.4), the joint laws of (X_0, Y_t) under p_0 and q share the OU kernel. Disintegrating their relative entropy over X_0 gives $\mathrm{KL}(p_0\|q)$, and disintegrating over Y_t gives $\mathrm{KL}(p_t\|q_t) + \mathbb{E}_{p_t}[\mathrm{KL}(\rho_{Y_t,t}\|\rho^q_{Y_t,t})]$. Combining with (3.2) at horizon t gives (8.4) for $t > 0$. At $t = 0$ the conditional laws given the clean state coincide, so both sides are zero.

B.4 Proof of the uniform score-error bound

Set $\omega_t := \alpha_t^2/\gamma_t$ and $u := \kappa/(\omega_t + \kappa)$, so u increases from zero to one and $\mathrm{d}\omega_t/\mathrm{d}t = -2(\alpha_t/\gamma_t)^2$. For fixed $q \in \mathcal{Q}$, write

$$D_{\mathrm{cond}}(u) := \mathbb{E}_{p_{t(u)}}\left[\mathrm{KL}(\rho_{Y_{t(u)},t(u)}\|\rho^q_{Y_{t(u)},t(u)})\right].$$

By (8.4), $D_{\mathrm{cond}}(0) = 0$, $D'_{\mathrm{cond}} \ge 0$ almost everywhere, and $D_{\mathrm{cond}}(u_T) = D(q)^2$, where $u_T = \kappa/(\omega_T + \kappa)$. The functions S in Lemma B.5 and (B.8) are chosen from the target before q is specified.

Lemma B.5 (Target-only matrix storage bound)**.** *Under Assumption 8.1, define the bounded target sets*

$$\mathcal{H} := \overline{\mathrm{conv}}\{H_0(x) : x \in \mathbb{R}^d\}, \qquad \mathcal{F} := \{\nabla V(x) : x \in \mathbb{R}^d\}.$$

At $\omega = \kappa(1-u)/u$, let

$$R_H := (\omega I_d + H)^{-1}, \qquad C_H := R_H(H - \Lambda).$$

For $w > \sup_{H\in\mathcal{H}} \|C_H\|_{\mathrm{op}}^2$, define $B_H(u,w) := (I_d - C_H C_H^\top/w)^{-1/2} R_H$ and set

$$J(u,w) := \frac{\kappa}{4u^2} \sup_{H\in\mathcal{H}} \mathrm{diam}\{B_H(u,w)\mathcal{F}\}^2. \tag{B.5}$$

Suppose $S : [0,1] \to [0,\infty)$ is nondecreasing, $S(0) = 0$, continuous at zero, and piecewise C^1 with finitely many nonnegative jumps. If, for $0 < u \le 1$,

$$S(u) > \sup_{H\in\mathcal{H}} \|C_H\|_{\mathrm{op}}^2, \qquad S'(u) \ge J(u, S(u)) \tag{B.6}$$

away from the jumps, then every $q \in \mathcal{Q}$ and every $T > 0$ satisfy $E(q)^2 \le S(u_T)D(q)^2 \le S(1)D(q)^2$.

Proof. At each query (y,t), let $P=\rho_{y,t}$, $Q=\rho^q_{y,t}$, $H=H^q=\mathbb{E}_Q H_0\in\mathcal{H}$, and $\delta f=\mathbb{E}_Q\nabla V-\mathbb{E}_P\nabla V$. The residual identity (8.2) becomes

$$e=C_H d+(\alpha_t/\gamma_t)R_H\delta f.$$

For $w>\|C_H\|_{\mathrm{op}}^2$, completion of the square gives

$$\|C_H d+z\|^2\le w\|d\|^2+z^\top(I_d-C_HC_H^\top/w)^{-1}z.$$

Apply this with $z=(\alpha_t/\gamma_t)R_H\delta f$. Bounded-function Pinsker applied to $B_H\nabla V$ yields

$$\|e\|^2\le w\|d\|^2+\tfrac{1}{2}(\alpha_t/\gamma_t)^2\operatorname{diam}(B_H\mathcal{F})^2\mathrm{KL}(P\|Q).$$

The inequality holds for every $H\in\mathcal{H}$, including $H^q(y,t)$ for the current law q. Taking the fixed target envelope in (B.5), averaging over p_t, and using $\mathrm{d}u/\mathrm{d}t=2u^2(\alpha_t/\gamma_t)^2/\kappa$ give

$$E(q)^2\le\int_0^{u_T}\{SD'_{\mathrm{cond}}+J(u,S)D_{\mathrm{cond}}\}\mathrm{d}u\le S(u_T)D_{\mathrm{cond}}(u_T).$$

The last inequality follows from (B.6). Each jump of SD_{cond} is nonnegative. The initial product vanishes since $D_{\mathrm{cond}}(0)=S(0)=0$. There is no optimization over current laws, nor any regularity condition on their density ratios. □

Closed-form part of Proposition 8.3. For a,b from (8.5), the pointwise bound (8.3), Minkowski's inequality in $L^2(p_t)$, and the same change of time give

$$E(q)^2\le\int_0^{u_T}\left(\sqrt{a}\,u\sqrt{D'_{\mathrm{cond}}(u)}+\sqrt{b}\sqrt{D_{\mathrm{cond}}(u)}\right)^2\mathrm{d}u. \tag{B.7}$$

For $a,b>0$, set

$$c:=\frac{a-b+\sqrt{a^2+b^2}}{2},\qquad S(u):=bu+cu^2.$$

Then $0<c<a$, $S>au^2$, $S'\ge b$, and

$$(S-au^2)(S'-b)-abu^2=2c(a-c)u^2(1-u)\ge 0.$$

These are the positivity conditions for the quadratic form showing that the integrand in (B.7) is at most $(SD_{\mathrm{cond}})'$. Thus $E(q)^2\le S(u_T)D(q)^2\le S(1)D(q)^2=C_{\mathrm{closed}}D(q)^2$. The degenerate cases follow by continuity. Together with Lemma B.5, the isotropic criterion (B.8), and the joint bound (B.9), this proves Proposition 8.3 for each choice of θ^2 listed there. □

When the quadratic part is isotropic, the matrix condition (B.6) simplifies. If $\Lambda=\lambda I_d$ and $\kappa I_d\preceq H_0\preceq MI_d$, then H commutes with $H-\lambda I_d$. Consequently the squared singular values of B_H are

$$\frac{1}{(\omega+r)^2-(r-\lambda)^2/S},\qquad r\in[\kappa,M].$$

For $S<1$, the denominator is concave in r, so its minimum is at an endpoint. For $S\ge 1$, it is increasing in $r>0$, with the same conclusion. Since $\operatorname{diam}\mathcal{F}\le\Delta_V$, it suffices that S have the regularity in Lemma B.5 and satisfy

$$S'(u)\ge\frac{\Delta_V^2}{4\kappa}\max_{r\in\{\kappa,M\}}\frac{1}{[1+(r/\kappa-1)u]^2-[(r-\lambda)/\kappa]^2u^2/S(u)}, \tag{B.8}$$

with both denominators positive. The scalar criterion applies in any dimension, including to nonseparable targets. At each endpoint, the denominator uses the inverse at that same curvature value. For general anisotropic Λ, use (B.6) or the closed-form bound.

For a computed piecewise affine S, the criterion (B.8) can be certified exactly. On each interval with slope $v > 0$, put $b = \Delta_V^2/(4\kappa)$ and check the two cubic polynomial inequalities

$$v\left\{[1 + (r/\kappa - 1)u]^2 S(u) - [(r-\lambda)/\kappa]^2 u^2\right\} - bS(u) \geq 0, \qquad r \in \{\kappa, M\},$$

together with positivity of the denominators. Nonnegative Bernstein coefficients give a bound valid on the interval. Rational coefficients and upward rounding allow exact verification. An initial segment reaching $u = 0$ must also be verified. Once the inequalities have been verified on every interval, $S(1)$ gives a uniform bound.

A closed-form joint bound is also available. Define

$$a_r := \frac{(r-\lambda)^2}{r^2}, \qquad b_r := \frac{\Delta_V^2}{4r}, \qquad C_{\text{joint}} := \sum_{r\in\{\kappa,M\}} \frac{a_r + b_r + \sqrt{a_r^2 + b_r^2}}{2}. \tag{B.9}$$

Then $E(q)^2 \leq C_{\text{joint}} D(q)^2$. To see this, put $z = (\alpha_t/\gamma_t)\delta f - (\omega+\lambda)d$. In this notation the residual identity (8.2) reads $e = d + (\omega I_d + H)^{-1} z$. In an eigenbasis of H, each squared component is convex as a function of $(\omega + r)^{-1}$. Bounding by the two endpoint values and summing components gives

$$\|e\|^2 \leq \sum_{r\in\{\kappa,M\}} \left\| \frac{r-\lambda}{\omega+r} d + \frac{\alpha_t/\gamma_t}{\omega+r}\delta f \right\|^2.$$

Apply the scalar storage calculation (B.7) to each endpoint field with time scale r and coefficients a_r, b_r, and add the bounds. Every term uses the conditional-KL identity (8.4). Each bound holds on the same finite-KL domain, so their minimum also gives a uniform bound.

B.5 Further examples of the score-error bound

The coordinate-coupling potential $\frac{1}{2}\|x\|^2 + 0.34\cos x_1 \cos x_2$, whose Hessian perturbation has eigenvalues $-0.34\cos(x_1 \pm x_2)$, gives $\kappa = 0.66$, $L_V = 0.34$, $\Delta_V = 0.68$, and the closed-form bound $\theta^2 < 0.380$. For the one-dimensional potential $U(x) = x^2/2 + 0.12\log\cosh(5x)$, we may take $\kappa = \lambda = 1$, $M = 4$, and $\Delta_V = 1.2$. The bound in (B.9) gives $\theta^2 < 0.972$, whereas $a + b = 9.36$.

B.6 Terminal error and the infinite-horizon map

The OU semigroup leaves ϕ_d invariant, so data processing makes the terminal error $\varepsilon_T = \mathrm{KL}(p_T\|\phi_d)$ nonincreasing in T. To bound it, write p_T as the mixture of $\mathcal{N}(\alpha_T x, \gamma_T I_d)$ for $x \sim p_0$. Joint convexity of relative entropy bounds ε_T by the mean KL divergence from these Gaussians to ϕ_d. The Gaussian KL formula gives

$$\varepsilon_T \leq \tfrac{1}{2}\left\{\alpha_T^2 \mathbb{E}_{p_0}\|X_0\|^2 + d(\gamma_T - 1 - \log\gamma_T)\right\} = O(e^{-2T}). \tag{B.10}$$

Here $\alpha_T^2 = e^{-2T}$ and $\gamma_T - 1 - \log\gamma_T = O(e^{-4T})$.

Proof of Theorem 8.4(ii). Fix $q \in \mathcal{Q}$. The corrected field does not depend on the horizon. For $t_1 < t_2$, let $\mathcal{U}^q_{t_2,t_1}$ be the Markov kernel of the corrected solution from physical time t_2 to t_1, so that $\Phi_T(q) = \phi_d \mathcal{U}^q_{T,0}$. By Proposition 8.3 and monotone convergence, $\int_0^\infty \mathbb{E}_{p_t}[\|e_t^q\|^2]\mathrm{d}t \leq \theta^2 \mathrm{KL}(p_0\|q) < \infty$. Hence $E^2_{>T} := \int_T^\infty \mathbb{E}_{p_t}[\|e_t^q\|^2]\mathrm{d}t \to 0$. For $S > T$, the Markov property gives

$\Phi_S(q) = r\,\mathcal{U}^q_{T,0}$ with $r := \phi_d \mathcal{U}^q_{S,T}$, and (B.4) on $[T, S]$ gives $\mathrm{KL}(p_T\|r) \le \varepsilon_S + E^2_{>T}$. Contraction of total variation under $\mathcal{U}^q_{T,0}$, the triangle inequality, and Pinsker's inequality give

$$\mathrm{TV}(\Phi_S(q), \Phi_T(q)) \le \mathrm{TV}(r, p_T) + \mathrm{TV}(p_T, \phi_d) \le \sqrt{\tfrac{1}{2}(\varepsilon_S + E^2_{>T})} + \sqrt{\tfrac{1}{2}\varepsilon_T}.$$

By (B.10), the right side tends to zero uniformly in $S > T$ as $T \to \infty$. Probability measures are complete in total variation, so $\Phi(q)$ exists. Each $\Phi_T(q)$ has a density, and total-variation limits preserve absolute continuity, so $\Phi(q)$ has a density. Lower semicontinuity of $\mathrm{KL}(p_0\|\cdot)$ under total-variation convergence and (8.8) give

$$\mathrm{KL}(p_0\|\Phi(q)) \le \liminf_{T\to\infty} \{\varepsilon_T + \theta^2 \mathrm{KL}(p_0\|q)\} = \theta^2 \mathrm{KL}(p_0\|q).$$

This proves (8.10), and induction gives the geometric bound. Taking $q = p_0$ gives $\Phi(p_0) = p_0$. A fixed point $q \in \mathcal{Q}$ satisfies $\mathrm{KL}(p_0\|q) \le \theta^2 \mathrm{KL}(p_0\|q)$, hence $q = p_0$. □

C Acoustic trajectory experiment details

C.1 Prior discretization, generating trajectory, and fixed data

A moving acoustic source has position $r_\alpha(\tau) \in \mathbb{R}^2$ during the physical-time interval $0 \le \tau \le 10$ s. The unknown is its complete trajectory, represented by two independent zero-mean Matérn–5/2 Gaussian processes with marginal SD 1.5 m and correlation time 2 s. Let (λ_k, ϕ_k) be the temporal covariance eigenpairs and $b_k = \sqrt{\lambda_k}\phi_k$. The trajectory uses eight Karhunen–Loève modes in each coordinate.

$$r_\alpha(\tau) = \left(\sum_{k=1}^{8} b_k(\tau)\alpha_k, \sum_{k=1}^{8} b_k(\tau)\alpha_{8+k} \right), \qquad \alpha \sim \mathcal{N}(0, I_{16}). \tag{C.1}$$

The retained modes are not rescaled to restore the variance of the full process. Physical time τ is distinct from diffusion time t.

Two stationary receivers $a_1, a_2 \in \mathbb{R}^2$ record eight associated pulses at $\tau_i = 10(i-1)/7$ s. With known sound speed $c = 343$ m/s, the time difference of arrival (TDOA), in microseconds, is

$$F_i(\alpha) = \frac{10^6}{c} \left(\|r_\alpha(\tau_i) - a_2\| - \|r_\alpha(\tau_i) - a_1\| \right), \qquad y_{\mathrm{obs}} = F(\alpha) + \eta. \tag{C.2}$$

Independent receiver timestamp errors with SD 120 μs give $\eta \sim \mathcal{N}(0, \Gamma)$, where $\Gamma = 2(120\,\mu\mathrm{s})^2 I_8$. Pulse association, the emission schedule, receiver locations, and sound speed are fixed and known.

For each coordinate, the temporal covariance is

$$k(\tau, \tau') = 1.5^2 \left(1 + \sqrt{5}\,u + \frac{5u^2}{3} \right) e^{-\sqrt{5}u}, \qquad u = |\tau - \tau'|/2.$$

The discretization uses 401 equally spaced nodes on $[0, 10]$ at times τ_j^{grid} with trapezoidal weights w_j. Interior weights are 0.025 s and endpoint weights are half as large. If v_k is a unit eigenvector of the symmetric matrix with entries $\sqrt{w_i}k(\tau_i^{\mathrm{grid}}, \tau_j^{\mathrm{grid}})\sqrt{w_j}$, with eigenvalue λ_k, the scaled mode in (C.1) is evaluated by

$$b_k(\tau) = \frac{1}{\sqrt{\lambda_k}} \sum_j k(\tau, \tau_j^{\mathrm{grid}})\sqrt{w_j}(v_k)_j.$$

Eigenvalues are ordered from largest to smallest, and the largest-magnitude entry of each eigenvector is positive. The leading eight modes are used for inference without variance rescaling. Thus every coefficient affects the source path, and there are no unused Gaussian coordinates.

Table 7: **Fixed acoustic observations.** Physical pulse times in seconds and observed arrival differences in microseconds, rounded for display. Independent TDOA noise SD is $\sqrt{2} \times 120\,\mu$s. These are the data in (C.2).

Pulse	Time (s)	Observed TDOA (μs)
1	0.000000	-3429.083531
2	1.428571	-3605.751643
3	2.857143	-3387.756589
4	4.285714	-2498.507053
5	5.714286	-656.299655
6	7.142857	-2936.812510
7	8.571429	-4126.443627
8	10.000000	-5402.098042

The exact receiver locations, in meters, are

$$a_1 = (-1.1276311449, -0.4104241720), \qquad a_2 = (1.1817693036, 0.2083778132).$$

Ranges replace $\|v\|$ by $\sqrt{\|v\|^2 + 10^{-18}}$, with distances in meters. The generating path is

$$r_\star(\tau) = c_\star + (2\tau/10 - 1)d_\star + \sum_{k=3}^{16} b_k(\tau)(\xi_{1k}, \xi_{2k}),$$
$$c_\star = (0.65, 2.5), \qquad d_\star = (0.20, -0.125),$$

where ξ is a 2-by-16 array of standard normals drawn in row order by NumPy's default generator with seed 20261101. The first two mode coefficients are set to zero. The linear path is added separately and is not projected into the inference basis. Figure 5 uses this generating path.

Table 7 records the eight observed TDOAs used by every method. The noise realization is generated as standard normals with seed 20261102 on the sorted union of the eight pulse times and the 24 equally spaced points of $[0, 10]$, giving 30 distinct times. The eight pulse entries are multiplied by $\sqrt{2} \times 120\,\mu$s and added to the noiseless range differences. Generating noise directly as a difference of receiver timestamps gives the same Gaussian model. The recorded realization uses this direct TDOA draw. The posterior uses $\Gamma = 28800 I_8$ in squared microseconds, with fixed noise variance. We keep the mismatch between the sixteen-mode, nonzero-mean generating trajectory and the eight-mode zero-mean prior used for inference.

C.2 Numerical transport and initialization

Each round stores exact clean scores and the matching likelihood-power GN matrices from (9.4). Deterministic probability flow uses 600 steps of the classical fourth-order Runge–Kutta method (RK4) on a logarithmic OU-time grid from 10 to $6.352721252 \times 10^{-7}$. The 2048 reference samples and their derivatives stay fixed during integration. Annealed GN-GAD passes the same Gaussian integration checks at its intermediate powers without refining the grid. The checks measure integration error for a Gaussian surrogate. The samplers retain their transported endpoints with equal weights.

The initial law is an equal mixture of the prior and MAP–Laplace. The MAP estimate is computed from eight starts that do not use the data-generating parameter. Its cost, 2941 requests, is counted for each method and seed that uses it. The exact MAP Hessian is positive definite without eigenvalue clipping. Each GAD round adds $3 \times 2048 = 6144$ requests, giving allowances $9085, 15229, 21373, 27517$. The other MCMC methods start from the same initial

Table 8: **Acoustic geometric checks at the final allowance.** Means ± sample SD over five seeds. For every error, smaller values indicate closer agreement. All errors use the metric samples. The coefficient blocks and normalization are defined in Appendix C.3.

Method	Joint SW_2	Low-mode SW_2	High-mode SW_2	Covariance error
Annealed GN-GAD	0.134 ± 0.012	0.179 ± 0.017	0.078 ± 0.006	0.335 ± 0.045
Direct GN-GAD	0.154 ± 0.009	0.215 ± 0.017	0.102 ± 0.009	0.395 ± 0.043
MAP-pMALA	0.630 ± 0.020	0.804 ± 0.012	0.436 ± 0.039	0.885 ± 0.006
Annealed MALA	0.218 ± 0.001	0.245 ± 0.005	0.204 ± 0.019	0.416 ± 0.048
Annealed MAP-pMALA	0.784 ± 0.042	0.984 ± 0.043	0.500 ± 0.068	1.732 ± 0.114
GN-smMALA	0.312 ± 0.113	0.341 ± 0.082	0.297 ± 0.146	0.609 ± 0.310
Annealed GN-smMALA	0.239 ± 0.026	0.259 ± 0.044	0.234 ± 0.037	0.486 ± 0.095
Annealed HMC	0.213 ± 0.004	0.216 ± 0.013	0.227 ± 0.018	0.365 ± 0.062

mixture. MAP-pMALA uses eight chains initialized from MAP–Laplace, each with 256 adaptive transitions targeting acceptance 0.574, followed by a fixed step. At the final allowance it has 1279 production states per chain, including rejection repeats. Setup and warmup count toward the allowance. Additional kernels and calibration are specified in Appendix I.4.

C.3 Reference and metric definitions

The shared reference uses four chains with 1500 warmup and 10000 retained states per chain, target acceptance 0.995, and maximum depth 12. NUTS operates on the exact posterior after an affine change to MAP coordinates. Lazy coefficient-reflection proposals use a Metropolis–Hastings (MH) correction at full likelihood power. The reflection proposals are used only for the NUTS run. The last two chains define the evaluation reference, and the first two define its split comparison. The four-chain diagnostics give maximum rank $\widehat{R} = 1.000253$, minimum bulk and tail effective sample sizes (ESS) 15518/11895, zero divergences and depth saturations, and minimum chain energy Bayesian fraction of missing information 0.800. The maximum event Monte Carlo standard error (MCSE) over the full reference is 0.00501285, slightly above the precision target of 0.005 set for this run. The held-out evaluation group's crossing-probability MCSE is 0.00706. Diagnostics and split discrepancies do not certify global coverage.

For a coefficient block of dimension d_b, normalized sliced distance is

$$S_b = \left[\frac{d_b}{\operatorname{tr}(C_{\mathrm{ref},b})} \frac{1}{1024} \sum_{\ell=1}^{1024} W_2^2\Big((v_\ell^\top)_{\#}\widehat{q}_b, (v_\ell^\top)_{\#}\widehat{p}_{\mathrm{ref},b} \Big) \right]^{1/2}. \tag{C.3}$$

Directions are fixed random unit vectors, with equal-size sample sets randomly subsampled to at most 2048 states. The direction and subsampling seeds are 23351 and 9531. Equation (C.3) uses scalar trace normalization in prior coefficient coordinates, not covariance whitening. The split reference discrepancy is 0.05576. Low and high coefficient blocks comprise modes 1–4 and 5–8 per coordinate. Each block has dimension eight. Relative covariance error is $\|C_q - C_{\mathrm{ref}}\|_F / \|C_{\mathrm{ref}}\|_F$, using metric samples. Table 8 reports these complementary checks for every selected acoustic comparator. At the final allowance, the mean joint SW_2 of annealed GN-GAD is 37.1% lower than that of annealed HMC, the best MCMC method on this measure, and 13.3% lower than that of direct GN-GAD. Annealing reduces the joint, covariance, and both block errors of GN-GAD in every paired seed.

Physical events are midpoint height below zero, entry into the disk centered at $(0, 2)$ m with radius 1 m, and crossing of $y = 0$ during $[0, 10]$ s. Disk entry and crossing use a 1601-point trajectory grid. A path crosses if the below-zero residence time lies strictly between zero and

Table 9: **Acoustic physical-event probabilities at the final allowance.** Sampler entries are five-seed means ± sample SD. Agreement is measured against the NUTS value in each column. The SD reflects variation across sampler seeds.

Method	Midpoint below zero	Enters disk	Crosses zero
NUTS	0.442	0.752	0.690
Annealed GN-GAD	0.450 ± 0.028	0.781 ± 0.019	0.842 ± 0.023
Direct GN-GAD	0.448 ± 0.019	0.781 ± 0.009	0.883 ± 0.023
MAP-pMALA	0.462 ± 0.116	0.279 ± 0.077	0.981 ± 0.007
Annealed MALA	0.479 ± 0.019	0.708 ± 0.034	0.857 ± 0.035
Annealed MAP-pMALA	0.470 ± 0.041	0.522 ± 0.046	0.878 ± 0.039
GN-smMALA	0.477 ± 0.078	0.697 ± 0.033	0.849 ± 0.048
Annealed GN-smMALA	0.435 ± 0.053	0.720 ± 0.028	0.821 ± 0.036
Annealed HMC	0.474 ± 0.040	0.742 ± 0.032	0.836 ± 0.043

Table 10: **Acoustic MCMC output-window sensitivity.** Joint normalized SW_2 (lower is better), five-seed mean ± sample SD at each method's actual final expenditure. All-production and latter-half windows use the same cumulative run. Annealed GAD is 0.134 ± 0.012 at this allowance.

Method	All $\beta = 1$ production	Latter half
MAP-pMALA	0.630 ± 0.020	0.620 ± 0.026
Annealed MALA	0.223 ± 0.005	0.218 ± 0.001
Annealed MAP-pMALA	0.701 ± 0.044	0.784 ± 0.042
GN-smMALA	0.352 ± 0.122	0.312 ± 0.113
Annealed GN-smMALA	0.237 ± 0.023	0.239 ± 0.026
Annealed HMC	0.201 ± 0.007	0.213 ± 0.004

ten seconds. Maximum event error is the largest absolute error over these three probabilities, multiplied by 100. The data-fit statistic is the median, over a fixed subsample of at most 512 metric states, of $\|r(\alpha)\|^2/8$, divided by the reference median 0.93964. A ratio of one matches the NUTS median.

Table 9 separates the three physical events. Annealed GN-GAD estimates the midpoint and disk-entry probabilities as 0.450 and 0.781, versus 0.442 and 0.752 in the reference. All reported samplers estimate a higher whole-path crossing probability than the reference value 0.690. For example, annealed GN-GAD, annealed HMC, and annealed MALA give 0.842, 0.836, and 0.857, respectively. A small transverse displacement of a path near the crossing boundary can change its whole-path indicator. Transverse spread in the samples could also increase the arrival residuals.

C.4 Output-window sensitivity and display construction

The MCMC methods other than MAP-pMALA retain the final half of their $\beta = 1$ production states. MAP-pMALA retains all of its production states. We use the saved states from both windows to recompute the errors in Table 10. Annealed GAD has lower joint SW_2 at the final allowance under either window. Annealed HMC spends 25725 requests because one more trajectory for all chains would exceed 27517. All other acoustic endpoint rows spend the full allowance.

Figure 10 checks coordinate uncertainty using the seed-43 samples from the metrics. Pointwise posterior intervals describe physical spread, not uncertainty over sampler seeds. The main histogram instead uses 10000 display draws per panel from that same fixed seed. GAD transports

fresh Gaussian draws with the score field from the final reference samples. MCMC display branches start from the saved beginning of the retained window, choose a parent and production age uniformly, and reuse the frozen kernel. MAP-pMALA uses the saved post-warmup states and ages $1, \ldots, 1279$. Branches do not receive longer trajectories or a new warmup. They illustrate the conditional retained-window law, without creating new initial coverage. The NUTS histogram has 2500 evenly selected draws from each of its four chains.

All histogram panels use 89 bins per axis, common reference-based physical limits, and bin count divided by the full sample size. Empty bins remain empty, off-panel mass remains in the denominator, and no smoothing or in-panel renormalization is applied. Different experiment seeds are never pooled. Display requests are separately recorded and excluded from metrics, coordinate bands, and the comparison allowance.

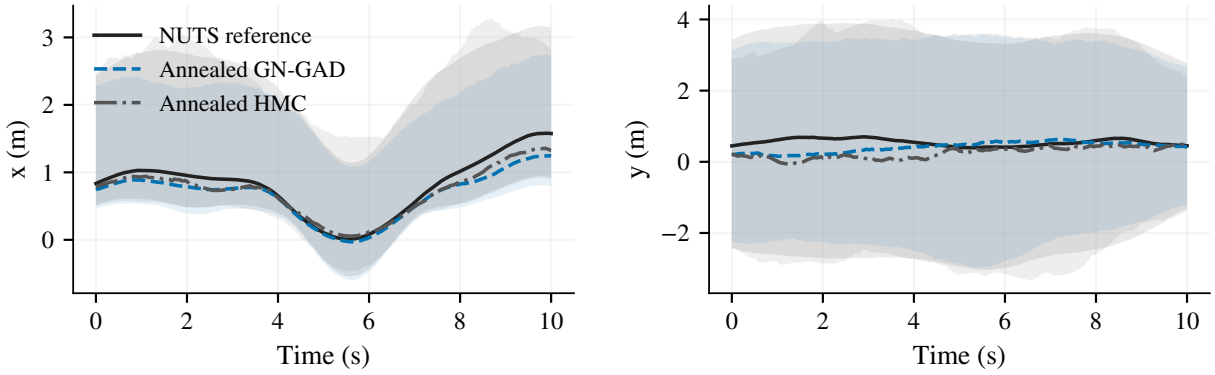


Figure 10: **Acoustic coordinate uncertainty for experiment seed 43.** Medians and pointwise 90% posterior intervals for the two spatial coordinates in meters over $[0, 10]$ s. The NUTS bands use the 20000-state evaluation group. Annealed GN-GAD and annealed HMC use their final metric samples of 2048 and 384 states, respectively. No display-only draws or other experiment seeds enter the intervals. The intervals describe uncertainty at each time along the path. Whole-path event probabilities are reported in Table 9.

D Joint earthquake-location experiment details

D.1 Physical model, priors, and fixed observations

Units are km, seconds, and km/s. The velocity parameters are the surface P-wave speed v_0, the vertical gradient g, and the P- to S-wave speed ratio κ_{PS}. The event coordinates have independent Gaussian priors, and v_0, g, and $\kappa_{\mathrm{PS}} - 1$ have independent lognormal priors. The six surface stations have horizontal coordinates

$$(0, 0),\ (14, 4),\ (5, 16),\ (-9, 10),\ (-7, -9),\ (10, -11),$$

in station-number order. For $\alpha \sim \mathcal{N}(0, I_{23})$, event $k = 1, \ldots, 5$ has

$$e_k = 3 + 9\alpha_{4k-3}, \quad n_k = 2 + 9\alpha_{4k-2},$$
$$z_k = 8 + 3\alpha_{4k-1}, \quad t_{0,k} = \alpha_{4k},$$

$$v_0 = 5.2e^{0.05\alpha_{21}},$$
$$g = 0.06e^{0.3\alpha_{22}},$$
$$\kappa_{\mathrm{PS}} = 1 + 0.73e^{0.1\alpha_{23}}.$$

Depth is positive downward. The Gaussian prior on depth extends above the surface.

Table 11: **Generating earthquake parameters.** Positions are in km, depth is positive downward, and origin times are in seconds. E1 is the anchor.

Event	East	North	Depth	Origin time
E1	5.0	-4.0	6.0	0.00
E2	8.0	-1.0	8.5	0.31
E3	6.0	1.0	7.0	-0.22
E4	9.0	-5.0	10.0	0.12
E5	4.0	2.0	5.5	-0.41

The medium has depth-dependent speeds $v_P(z) = v_0 + gz$ and $v_S(z) = v_P(z)/\kappa_{\mathrm{PS}}$. For a source at depth z_s, receiver at depth $z_r = 0$, and Euclidean source–receiver separation R, the analytic travel times are

$$T_P = \frac{1}{g}\operatorname{arcosh}\left(1 + \frac{g^2R^2}{2(v_0 + gz_s)(v_0 + gz_r)}\right), \qquad T_S = \kappa_{\mathrm{PS}}T_P. \tag{D.1}$$

The formula in (D.1) accounts for rays curved by the velocity gradient. A pick from event k at station j has prediction $t_{0,k} + T_P$ or $t_{0,k} + T_S$. Independent Gaussian pick errors have SD 0.08 s for P and 0.15 s for S, defining the diagonal covariance in (13.1). Derivatives are obtained from this analytic forward map.

For positive likelihood power, the posterior assigns zero likelihood where a source has nonpositive wave speed. The travel-time calculation is nonfinite outside its domain, and the MCMC implementations reject nonfinite proposals. Travel times use (D.1), evaluated in its stable $\log(1 + a + \sqrt{a(a+2)})$ form, where $a = g^2R^2/[2(v_0 + gz_s)(v_0 + gz_r)]$. Automatic differentiation of this analytic map supplies Jacobians, GN matrices, exact scores, and the MAP Hessian. No discretized ray solver or implicit root differentiation is used.

Table 11 lists the generating event parameters. The shared velocity parameters are $(v_0, g, \kappa_{\mathrm{PS}}) = (5.35, 0.07, 1.76)$, with g in s^{-1}. The station numbering is the same in Figure 7 and table 12. Sparse-event picks were assigned to the nearest and second-nearest stations in horizontal distance at the generating locations, with S observed only at the nearest. The list is held fixed. Missing phases and stations do not contribute detection or censoring terms to the likelihood.

Table 12 gives all 24 recorded arrival times. To reproduce the realization, independent standard normals from NumPy's default generator with data seed 71 are drawn first for E1, P then S at stations $1, 6, 2, 5, 4, 3$. For E2–E5, the order is P then S at the nearest station, followed by P at the second-nearest station. Each draw is multiplied by its phase SD, 0.08 s or 0.15 s, and added to the generating origin time plus the travel time in (D.1). The likelihood uses these recorded values and independent errors. The model omits station corrections, unknown phase associations, and additional pick uncertainty.

The acquisition setting followed a reference-based screen for departure from MAP–Laplace and a moment-matched Gaussian. An exploratory sampler pilot compared two likelihood schedules before the six-round path was frozen. Subsequent seed-42 comparator selection and five sampler repeats do not turn this selected problem into a test across unseen datasets.

D.2 Transport controls and reference

Direct and annealed GN-GAD start from the prior and use 1500 reference samples in each of six rounds. Each round costs 4500 requests. Neither variant uses the MAP estimate, so no MAP

Table 12: **Fixed earthquake arrival picks.** Recorded arrival times in seconds, rounded for display. A dash denotes an unobserved phase. Independent noise SDs are 0.08 s (P) and 0.15 s (S).

Event	Station	P arrival (s)	S arrival (s)
E1	ST1	1.582452	2.855981
E1	ST2	2.458755	4.333243
E1	ST3	3.806598	6.602302
E1	ST4	3.667507	6.489049
E1	ST5	2.601111	4.316244
E1	ST6	1.877515	3.306574
E2	ST1	2.411565	–
E2	ST2	2.323569	3.965978
E3	ST1	1.346819	2.779552
E3	ST2	1.720495	–
E4	ST1	2.562834	–
E4	ST6	2.192577	3.807015
E5	ST1	0.965825	1.690106
E5	ST2	1.748654	–

cost is counted. Annealed powers are $(0.02, 0.07, 0.2, 0.45, 1, 1)$, versus six ones for direct GAD. Within each seed, the two variants share initial samples and terminal Gaussian draws. Round scores and GN matrices scale only the likelihood as in (9.4).

The integration grid starts with OU terminal time 5 and 600 RK4 steps on a logarithmic grid. For the Gaussian surrogate associated with the current reference samples, the terminal time is extended and the grid is refined as required to meet a 1% Gaussian integration tolerance. The endpoint is at most 10^{-5}, also constrained by a 1% variance-ratio criterion. Refinement permits up to 64 times the initial step count. The shifted resolvent eigenvalues are guarded below by $\max(10^{-3}, \alpha_t^2/2)$, bounding gate gain by two. The checks measure how the terminal time, endpoint, and time grid affect transport of the Gaussian surrogate. The comparison with NUTS assesses the overall sampling error for the nonlinear target. The saved control settings specify the grid selection rule. Exact realized grids are available for only some of the earthquake runs.

MAP-pMALA uses four MAP–Laplace-initialized chains, the inverse exact MAP Hessian, and 200 warmup transitions per chain targeting 0.574 acceptance. The MAP estimate uses four starts that do not use the data-generating parameter, and its cost is counted for each method that uses it. The MAP Hessian is used without clipping. The other MCMC methods start from the prior and run populations of chains. Their selected settings are given in Appendix I.4.

The NUTS run has four chains with a dense mass matrix, each with 1500 warmup and 2000 retained states, target acceptance 0.95, and maximum depth ten. Starts are independently dispersed by half a MAP–Laplace SD. Maximum rank $\widehat{R}$ is 1.001316, minimum ESS 1951, mean acceptance 0.9551, with zero divergences and depth saturations. Comparing the first two with the last two chains gives event-block and full-space SW_2 discrepancies of 0.07997 and 0.07873. The primary evaluation uses all 8000 NUTS states. These diagnostics come from finitely many NUTS states. They do not prove that the chains explored the whole posterior, and they do not give a certified error floor.

D.3 Metrics and broad endpoint comparisons

SW_2 uses prior-whitened coefficients and the scalar block-trace normalization of (C.3). Here the full-space implementation uses 256 fixed directions, sample sets capped at 1000 evenly indexed

Table 13: **Complete selected earthquake geometric comparison.** Five-seed means ± sample SD, with actual spent E+G+H requests under allowance 27000. Lower errors indicate closer agreement. Block and full SW_2 have the scalar reference normalization in (C.3). GAD continuations include inherited transport work. Settings are frozen on seed 42.

Method	Spent W	Block SW_2	Full SW_2	Covariance error
Annealed GN-GAD	27000	0.154 ± 0.019	0.130 ± 0.019	0.150 ± 0.008
Direct GN-GAD	27000	0.311 ± 0.104	0.273 ± 0.101	0.274 ± 0.037
MAP-pMALA	26999	0.491 ± 0.025	0.424 ± 0.011	0.508 ± 0.072
Annealed MALA	26880	0.653 ± 0.020	0.574 ± 0.016	0.778 ± 0.027
Annealed MAP-pMALA	26863	1.581 ± 0.065	1.583 ± 0.088	3.521 ± 0.354
GN-smMALA	26880	0.267 ± 0.028	0.247 ± 0.016	0.423 ± 0.052
Annealed GN-smMALA	26880	0.374 ± 0.021	0.322 ± 0.016	0.462 ± 0.032
Annealed HMC	26752	0.359 ± 0.028	0.310 ± 0.022	0.454 ± 0.051
Direct MALA	26880	0.583 ± 0.024	0.540 ± 0.031	0.681 ± 0.046
Direct HMC	24832	0.687 ± 0.024	0.850 ± 0.035	1.076 ± 0.086
GAD then MAP-pMALA	26943	0.463 ± 0.090	0.439 ± 0.061	0.443 ± 0.049
GAD then GN-smMALA	26832	0.455 ± 0.032	0.410 ± 0.025	0.466 ± 0.034
GAD then HMC	24272	0.287 ± 0.035	0.268 ± 0.024	0.417 ± 0.008
MAP-start HMC	26623	0.539 ± 0.182	0.475 ± 0.152	1.050 ± 0.609
Replica-exchange HMC	25903	0.775 ± 0.216	1.002 ± 0.356	3.784 ± 1.473
Replica-exchange GN-smMALA	26880	0.618 ± 0.117	0.594 ± 0.065	1.151 ± 0.233
SMC–MALA	26912	1.369 ± 0.237	1.360 ± 0.171	0.952 ± 0.010
SMC–GN-smMALA	26880	1.051 ± 0.170	0.992 ± 0.191	0.887 ± 0.274

states, and 1000 midpoint quantiles per projection. We approximate the quantile integral using the midpoint quantiles. Each event block uses 64 directions in its four east/north/depth/time coordinates. The primary error is the mean of the four normalized block distances for E2–E5. Projection seed 42 is fixed for evaluation. Full-space SW_2 covers all 23 coordinates. Relative covariance error uses all retained metric states. The different block dimensions and implementations preclude comparing error magnitudes between applications. Table 13 retains full-space and covariance results alongside the primary block error. Annealing lowers the mean block SW_2 of GN-GAD by 50.6%, to 42.5% below direct GN-smMALA, the MCMC method with the lowest block error. Annealed GN-GAD has the lowest mean full-space SW_2 among the tested methods. It also improves on direct GN-GAD in mean event and depth-interval errors and has the lowest mean errors on both physical measures (Table 14).

Physical events are $z_k < 5$ km and $z_k > 10$ km for each of the five events. Maximum event error is the largest of the ten absolute probability errors, in percentage points. Depth-interval error is the root mean squared (RMS) discrepancy of the 5% and 95% depth quantiles for E2–E5, in km. The data-fit statistic uses at most 512 metric states and is the median of $\|r(\alpha)\|^2/24$, divided by its reference value. Table 14 retains the physical checks for the same frozen comparator settings.

The broad comparison includes direct and annealed fixed-metric methods, transport-to-MCMC continuations, replica exchange, and SMC. We select the settings on seed 42 and use them for all reported seeds. MAP-pMALA has higher block error with the likelihood schedule of annealed GN-GAD than it does at full power throughout (Tables 13 and 17). The prior-started GN-smMALA kernel has lower block error than MAP-pMALA. Its own block error increases with annealing at the tested allowance. The selected SMC–MALA and SMC–GN-smMALA runs reach $\beta = 1$. None of the 24 tested SMC–HMC settings reaches $\beta = 1$ within 27000 requests on any of seeds 42–47. The endpoint table reports errors for runs that reach full likelihood power.

Table 14: **Earthquake physical checks for the complete selected set.** Same five-seed endpoints as Table 13. Event and depth-interval errors are smaller when agreement is closer. The data-fit ratio targets one. Between-seed SD excludes reference uncertainty.

Method	Event error (pp)	Depth interval RMSE (km)	Fit ratio $\to 1$
Annealed GN-GAD	7.7 ± 1.2	0.39 ± 0.10	1.45 ± 0.05
Direct GN-GAD	8.9 ± 3.3	0.49 ± 0.12	1.36 ± 0.04
MAP-pMALA	15.3 ± 4.3	1.08 ± 0.16	1.00 ± 0.02
Annealed MALA	41.7 ± 4.5	1.68 ± 0.04	2.73 ± 0.07
Annealed MAP-pMALA	49.6 ± 5.5	3.21 ± 0.18	341.69 ± 14.91
GN-smMALA	16.5 ± 2.0	0.69 ± 0.11	3.76 ± 0.12
Annealed GN-smMALA	29.0 ± 2.7	0.96 ± 0.12	4.92 ± 0.25
Annealed HMC	28.6 ± 5.3	0.96 ± 0.08	1.28 ± 0.06
Direct MALA	37.8 ± 3.7	1.62 ± 0.15	2.26 ± 0.04
Direct HMC	43.8 ± 3.0	1.82 ± 0.10	36.97 ± 1.80
GAD then MAP-pMALA	35.1 ± 7.3	1.30 ± 0.41	2.16 ± 0.09
GAD then GN-smMALA	33.2 ± 5.8	1.25 ± 0.17	2.60 ± 0.16
GAD then HMC	17.3 ± 2.6	0.77 ± 0.15	1.71 ± 0.14
MAP-start HMC	16.7 ± 7.1	1.59 ± 0.96	4.78 ± 8.16
Replica-exchange HMC	31.6 ± 19.5	2.58 ± 1.49	1.61 ± 0.40
Replica-exchange GN-smMALA	23.1 ± 3.5	1.54 ± 0.36	1.60 ± 0.09
SMC–MALA	68.9 ± 21.7	3.77 ± 0.50	2.26 ± 0.65
SMC–GN-smMALA	60.4 ± 18.3	2.51 ± 0.71	1.70 ± 0.40

D.4 Physical displays and seed robustness

Figure 8 uses experiment seed 43 throughout. For each event, the horizontal axis projects east/north displacement onto the leading eigenvector of its NUTS reference covariance, after subtracting that reference mean. Depth remains its physical coordinate. The two GAD panels transport 10000 fresh Gaussian draws with the score field from their own final 1500 reference samples. The NUTS reference has 8000 states, and GN-smMALA uses its 4480 saved metric states. Each event row uses common 66-by-66 bins and a shared logarithmic probability-per-bin scale. Bin counts are divided by the full sample count without smoothing. The additional GAD draws are used only in the display.

Figure 11 complements that fixed-seed display with the archived all-seed comparison. The density panels retain the original E4 histograms of 1000 states per seed at their original physical coordinates, without rebinning or pooling. The paired block distances come from the archived metric rows.

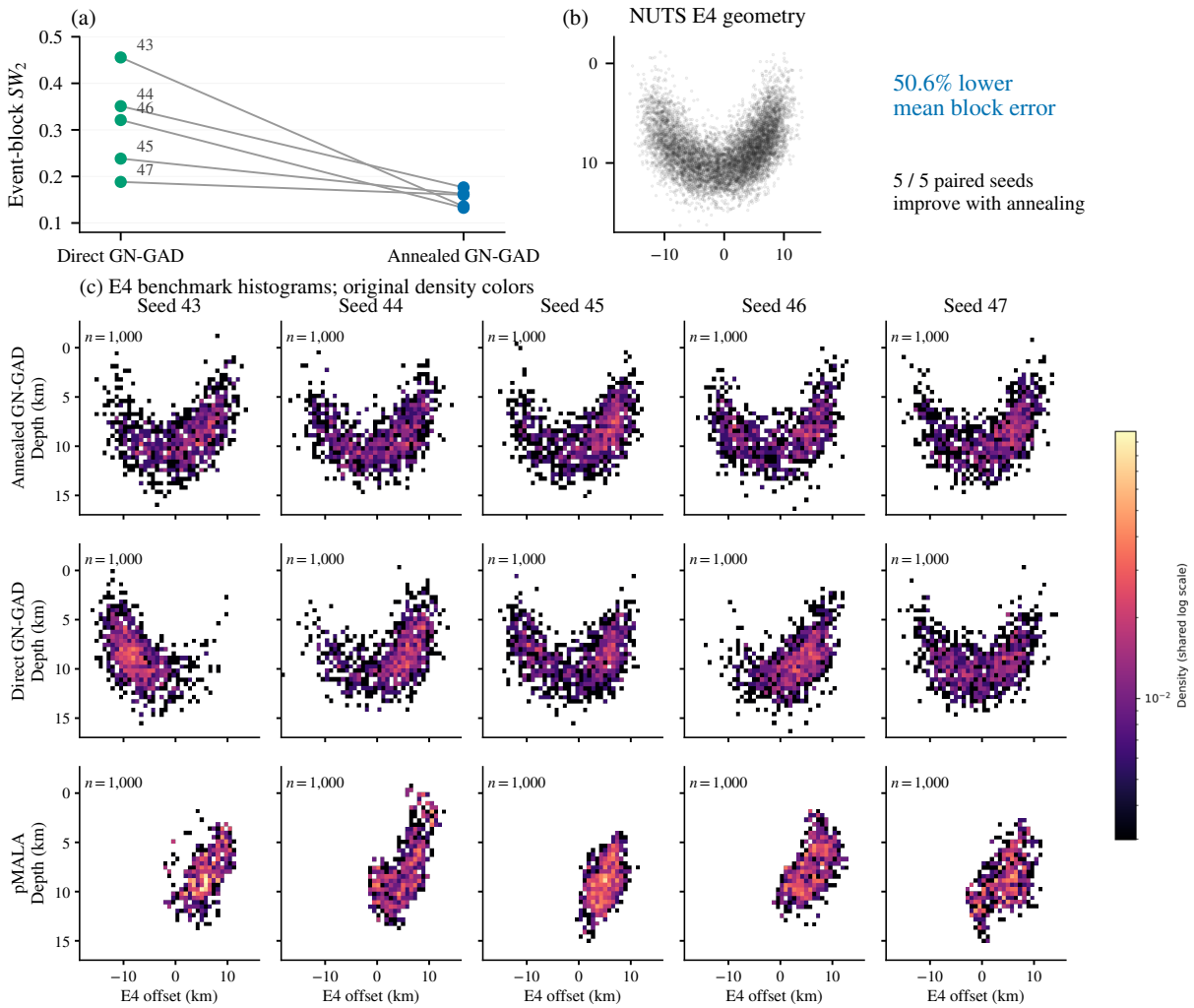


Figure 11: **Earthquake annealing comparison across the five seeds.** (a) Final E2–E5 block SW_2 for direct and annealed GN-GAD in each paired seed. Lower values are better. (b) NUTS reference for E4. (c) Archived E4 densities for annealed GN-GAD, direct GN-GAD, and MAP-pMALA (labeled pMALA), using 1000 states per seed. Panels (b,c) show horizontal offset and depth in km, with depth positive downward. The histograms in (c) share a logarithmic density scale. Appendix D.4 describes how the archived panels and the probability-per-bin panels in Figure 8 were made.

E Gaussian, mixture-weight, and gate experiments

The Gaussian and gate studies estimate conditional means from finitely many reference samples, and their uncertainty summaries are over seeds. The mixture-weight study instead evolves densities with exact conditional expectations and no mixture fit, and it reports deterministic numerical checks.

E.1 Sliced Wasserstein distance for the synthetic tests

For two indexed samples $X = (x_i)_{i=1}^m$ and $Y = (y_i)_{i=1}^m$ in $\mathbb{R}^d$, let $v_1, \dots, v_L$ be fixed unit directions and use parenthesized indices for the sorted projections. The empirical distance and its scalar

normalization are

$$\widehat{\mathrm{SW}}_2(X,Y) = \left\{ \frac{1}{Lm} \sum_{\ell=1}^{L} \sum_{i=1}^{m} \left(\langle v_\ell, X \rangle_{(i)} - \langle v_\ell, Y \rangle_{(i)} \right)^2 \right\}^{1/2}, \qquad \sigma_{\mathrm{tar}} = \sqrt{\frac{\mathrm{tr}(\mathrm{Cov}_{p_0}(X_0))}{d}}. \quad \text{(E.1)}$$

Here $X_0 \sim p_0$ is a single target draw. Tables report $100\widehat{\mathrm{SW}}_2/\sigma_{\mathrm{tar}}$ using (E.1).

E.2 Gaussian score recovery

The anisotropic target in Section 11.1 is $\mathcal{N}((0.6,-0.4), \left(\begin{smallmatrix} 1.6 & 0.5 \\ 0.5 & 0.7 \end{smallmatrix}\right))$. The first reference law is $\mathcal{N}((0.6,-0.4), 4I_2)$. The second is an equal mixture with means $(-2,1)$ and $(2,-1)$, both with covariance $\mathrm{diag}(0.9^2, 0.6^2)$. Each law supplies 4096 reference samples for seeds 42–44. Each run produces 4096 reverse endpoints and is compared with 4096 independent target samples, shared by the two reference laws within the seed. The two runs in a seed also share terminal samples and Brownian increments.

Both runs use the two-dimensional field implementation in Appendix F.3: $T = 5$, $t_{\min} = 10^{-4}$, 240 Heun steps, 65 field times, and 129×129 reference and query grids, with spatial half-width at least 9. Pre-interpolation residuals compare every tabulated score value with the analytic noisy Gaussian score. This analytic score is used only for evaluation and is not an input to the estimated field. The largest absolute residual over nodes, times, and seeds is below 1.3×10^{-10} (Table 2). The maximum relative RMS residuals over seeds are 4.70×10^{-14} and 1.07×10^{-13} for the Gaussian and bimodal reference laws. The largest difference between paired endpoints of the two runs is 7.95×10^{-14}. All endpoints are finite, and no gate safeguard activates. Sample discrepancies use the 64 fixed directions and scalar normalization in (E.1). Their nonzero values include numerical integration and finite-output sampling error, although the cancellation of Section 6 holds to numerical precision.

E.3 Exact-expectation calculation for the mixture-weight study

The mixture family in Section 11.2 is

$$p_{0,\Delta} = \tfrac{1}{2}\mathcal{N}(-\Delta\sigma/2, \sigma^2) + \tfrac{1}{2}\mathcal{N}(\Delta\sigma/2, \sigma^2),$$
$$q_{0,\Delta} = 0.7\mathcal{N}(-\Delta\sigma/2, \sigma^2) + 0.3\mathcal{N}(\Delta\sigma/2, \sigma^2), \qquad \sigma = 0.7.$$

Only the initial component weights differ. Component overlap is $2\Pr\{Z < -\Delta/2\}$, $Z \sim \mathcal{N}(0,1)$. As separation grows, the initial KL tends to $-\frac{1}{2}\log(4 \cdot 0.7 \cdot 0.3)$, while clean Fisher becomes small. The target-weighted Fisher discrepancy F_{mix} in (11.1) has no factor of $1/2$. Its initial value is computed independently by positive-integrand quadrature of the analytic score difference. KL is integrated separately under each target component. At separations 5, 7, and 9, the first-round gains are 0.174, 0.00991, and 0.000199, compared with sensitivities 0.141, 0.00981, and 0.000233. The first-round gains and sensitivities differ by less than 25% relative to the sensitivities at these separations. The two curves in Figure 2(a) use these measures without fitting or rescaling them to agree. After 30 rounds, forward KL is 6.78×10^{-6}, 0.0399, and 0.0852, respectively.

The density solver uses $[-10, 10)$, horizon $T = 8$, terminal density $\mathcal{N}(0,1)$, and endpoint time zero. Numerically integrated current-law OU moments use ZoomFFT with Gaussian Fourier multipliers. Each frozen reverse-SDE density is evolved with a conservative Chang–Cooper finite-volume generator, zero boundary flux, and two-stage L-stable singly diagonally implicit Runge–Kutta integration (SDIRK2). The ascending time grid starts at zero, takes increments $\min\{\delta t, 0.065(t + 0.003)\}$, shortens the last step at T, and is traversed in reverse. The primary

trajectories use 3072 cells and maximum step $\delta t = 0.01$, retaining all 30 rounds for $\Delta = 5, 7, 9$. Other selected first steps use the same grid. No mixture projection or endpoint sampling is performed.

The raw scalar gate is $\alpha_t^2/(\alpha_t^2 + \gamma_t H_t^q)$. A denominator with magnitude below 10^{-8} retains its sign and is set to that magnitude. The gate is capped at 50 and floored at $10^{-3}\alpha_t^2/(\alpha_t^2 + \gamma_t/\sigma^2)$. Below marginal density 2×10^{-13}, the solver uses its inward Gaussian tail continuation. No safeguard activates on retained probability-weighted queries, and no negative mass is repaired in the selected trajectories or numerical controls. Reconstructed KL values from saved densities agree with the recorded trajectories to 7.94×10^{-15} absolute error. The maximum mass drift in normalized saved trajectory densities is 9.14×10^{-14}.

Table 15 records the resolution check for the smallest first-step signal. Medium-to-fine change is 0.3804% of the gain, within our 5% tolerance. For $\Delta = 9$, panel (a) of Figure 2 uses the finest-grid value. Panel (b) uses the base grid for the entire trajectory. Expanding the domain to $[-12.5, 12.5)$ with 3840 cells at the same spacing and maximum step changes the base-grid gain by 1.04×10^{-11} relative. A Gaussian density-solver control gives one-round KL 3.06×10^{-11}. Separate controls driven by exact analytic noisy target scores give KL 2.05×10^{-10} at separation 5 and 1.08×10^{-9} at separation 9. Only the control runs use analytic noisy target scores. The grid and domain checks measure deterministic numerical error.

Table 15: **Refinement of the separation-9 first-step gain.** The initial component weights are 0.7/0.3, the domain is $[-10, 10)$, and g_0 is fractional forward-KL reduction in in (11.2). Successive rows refine the deterministic solver, and the final refinement changes the gain by 0.38%.

Cells	Maximum time step	g_0
3072	0.0100	0.00020303335
6144	0.0050	0.00019989021
12,288	0.0025	0.00019913279

E.4 Exact target for the gate-ablation experiment

The target in Section 11.3 and figure 3 is the following fixed density on $\mathbb{R}^8$:

$$p_0(x) = \frac{|\det A|}{Z_\epsilon^8} \exp\left[-\sum_{j=1}^{8} v_\epsilon((Ax)_j)\right],$$

$$v_\epsilon(u) = \tfrac{1}{2}u^2 + 2\epsilon \log\cosh(u/\epsilon), \qquad \epsilon = 0.04, \qquad Z_\epsilon = \int_{\mathbb{R}} e^{-v_\epsilon(u)}\,\mathrm{d}u.$$

The transformed coordinates $z = Ax$ are independent under this density. the determinant factor returns it to the original x coordinates. All eight coordinates contribute non-Gaussian curvature. The same target is used for every gate and validation seed.

With $z = Ax$, the ridge target's exact derivatives are

$$s_0(x) = -A^\top\{z + 2\tanh(z/\epsilon)\}, \qquad H_0(x) = A^\top \operatorname{diag}\{1 + 50\operatorname{sech}^2(z_j/\epsilon)\}A.$$

The hyperbolic functions act componentwise. The fixed matrix A is constructed using one NumPy `default_rng(20261106)` generator. Draw two successive standard-normal 8×8 arrays in row order, and take their NumPy QR Q factors, first U, then V, without changing the signs of their columns. Set

$$A = U \operatorname{diag}(1, 1, 1.5, 1.5, 2, 2, 3, 3)V^\top.$$

Its Hessian satisfies $I_8 \preceq H_0(x) \preceq 459 I_8$. All eight ridge coordinates are non-Gaussian. Independent exact target samples draw each z_j by rejection from a Laplace law of rate 2, accepting with probability

$$\exp\{-z_j^2/2 - 2\epsilon \log(1 + e^{-2|z_j|/\epsilon})\},$$

and then apply $x = A^{-1}z$. One-dimensional quadrature gives $\mathrm{Var}(z_j) = 0.2543156962$ and target scale $\sigma_{\mathrm{tar}} = 0.3388145209$. Exact target samples, this transform, and normalization information are used for evaluation only.

E.5 Gate definitions and reconstruction settings

For the current reference samples $X_1, \dots, X_N$, let $w_i(y,t)$ be the normalized OU weights in (I.1) and set

$$b_i = \frac{\alpha_t X_i - y}{\gamma_t}, \qquad c_i = \frac{s_0(X_i)}{\alpha_t}, \qquad \delta_i = c_i - b_i, \qquad \widehat{M}_q = \frac{1}{N}\sum_i s_0(X_i) s_0(X_i)^\top.$$

The matrix $\widehat{M}_q$ averages outer products of the target score over the current reference samples. Write $\bar{u} = \sum_i w_i u_i$ for a weighted mean. In particular, $\bar{b} = \widehat{b}^q(y,t)$ and $\bar{\delta} = \widehat{m}_t^q(y)$. With ridge scale η_{ridge}, the weighted covariance, empirical observed information, and ridge are

$$\widehat{C}_{uv} = \sum_i w_i (u_i - \bar{u})(v_i - \bar{v})^\top, \qquad \widehat{H}^q = \sum_i w_i H_0(X_i), \qquad \lambda_{\mathrm{ridge}} = 10^{-8} + \eta_{\mathrm{ridge}}\, \mathrm{tr}(\widehat{C}_{\delta\delta})/8.$$

Each gate G gives the score $\bar{b} + G\bar{\delta}$, computed from the same reference samples as the gate. The gates are

$$G_{\mathrm{Tweedie}} = 0, \qquad G_{\mathrm{TSI}} = I_8,$$

$$G_{\mathrm{uniform\ scalar}} = \frac{\alpha_t^2}{\alpha_t^2 + \gamma_t\, \mathrm{tr}(\widehat{M}_q)/8} I_8, \qquad G_{\mathrm{uniform\ matrix}} = \alpha_t^2 (\alpha_t^2 I_8 + \gamma_t \widehat{M}_q)^{-1},$$

$$G_{\mathrm{local\ scalar}} = \mathrm{clip}_{[0,1]}\left(\frac{-\mathrm{tr}(\widehat{C}_{b\delta})}{\max\{\mathrm{tr}(\widehat{C}_{\delta\delta}), 10^{-20}\}}\right) I_8, \qquad G_{\mathrm{local\ matrix}} = -\widehat{C}_{b\delta}(\widehat{C}_{\delta\delta} + \lambda_{\mathrm{ridge}} I_8)^{-1},$$

$$\widehat{G}^q_{\mathrm{LFGI}} = \alpha_t^2 (\alpha_t^2 I_8 + \gamma_t \widehat{H}^q)^{-1}.$$

The spatially uniform scalar and matrix rows are q-moment analogues of control-variate schedules, labeled q-CV and q-DPSMC in the table [23, 40]. The local scalar row is a conditional analogue of the scalar control-variate coefficient (q-CVSI) [23]. The local matrix row is the centered Matrix Blend regression [14]. In the opposite interpolation convention used by DPSMC, its coefficient is $I - G$. With exact expectations under the target law, matrix regression and LFGI give the same gate under the regularity conditions in [14]. With moments under $q \neq p_0$, the two substitutions can give different gates. We compare their effects on the GAD update.

The matrix ridge uses $\eta_{\mathrm{ridge}} = 0.001$, selected on pilot seed 42 from $\{0.001, 0.01, 0.1\}$ before validation. Covariances are computed by stable two-pass centering with every OU weight retained. The local regression matrix is not symmetrized or spectrally clipped. Its regularized denominator is positive definite. The uniform inverse is positive definite because $\widehat{M}_q \succeq 0$. On this target, $\widehat{H}^q \succ 0$, so q-LFGI uses its raw Hessian resolvent without clipping or an inverse safeguard.

The reported runs start from $\mathcal{N}(0, I_8)$ and use $N = 1024$, four rounds, $T = 5$, $t_{\min} = 10^{-4}$, and 300 stochastic Heun steps uniform in $\log(\mathrm{expm1}(2t))$. The predictor and corrector share the Brownian increment, and no final denoising or annealing is applied. Each round replaces the reference samples by all 1024 endpoints. Within each seed from 43 to 47, all gates share the

initial samples, terminal samples, and Brownian increments. Different seeds are independent. TSI stops on nonfinite states or a coordinate magnitude exceeding 10^6. All five TSI runs fail by round three. Table 3 records the failures in place of a round-three mean.

Metrics use 1024 GAD endpoints, 1024 independent target samples per seed, and 256 fixed unit directions obtained by normalizing Gaussian vectors from `default_rng(85411)`. The target samples are shared across gates and rounds within a run. The table multiplies the normalized SW_2 by 100. A second independent target sample gives the target/target comparison. Display draws are not used in the metrics or as reference samples. The displays use the score fields from seed 43 and the first two coordinates of $z = Ax$, with 10,000 fresh paths per panel. The target display sample is independent of the metric samples and is repeated in the first column of Figure 3. Histograms share axes and logarithmic probability-per-bin colors without smoothing. Bin counts are divided by the full sample count. Panels annotate off-panel percentages of at least 0.05%. Exact-target contours enclose 50% and 90% projected probability.

The round-three reporting cutoff was chosen after inspecting the original four-round results. Table 16 records all finite fourth-round results. q-LFGI improves in all five seeds, with paired mean reduction 0.391 ± 0.432 percentage points of target scale. The alternative uniform and matrix gates improve more between rounds three and four. The local scalar gate also slows, so the comparison does not assert that only q-LFGI has approached a practical plateau. Sensitivity checks on the pilot seed for the number of reference samples, the step count, the matrix ridge, and a 2048-direction metric are kept separate from the validation runs.

Table 16: **Fourth-round gate cutoff check.** Normalized full-dimensional SW_2 in percent, mean $\pm$ sample SD over the same five validation seeds and 1024-point metric samples. Lower is better. TSI fails in all five runs by round three and has no fourth-round result. The target/target scale is 7.87 ± 0.29.

Gate	Round 3	Round 4
Tweedie	199.41 ± 2.18	200.78 ± 3.07
Uniform scalar (q-CV)	71.06 ± 2.13	36.62 ± 0.83
Uniform matrix (q-DPSMC)	63.46 ± 2.53	36.47 ± 2.81
Local scalar (q-CVSI)	12.12 ± 0.75	11.18 ± 0.29
Local Matrix Blend	30.05 ± 3.47	23.48 ± 1.17
q-LFGI	8.56 ± 0.40	8.17 ± 0.46

F Low-dimensional sampler experiment details

This appendix specifies the connected two-dimensional targets, the GAD/MALA/HMC comparison, configuration selection, and the displays. The one-dimensional illustration is described separately in Appendix G.1. Inverse model and metric details remain in Appendices C and D.

F.1 Exact targets for the four sampler benchmarks

The four rows of Figure 4 use the following normalized densities on $\mathbb{R}^2$, in the same top-to-bottom order. Each has the Gaussian-mixture form $p_0(x) = \sum_j w_j \mathcal{N}(x; \mu_j, \Sigma_j)$, with means and covariances in the displayed physical coordinates.

- The Gaussian has $w_1 = 1$, $\mu_1 = (0.6, -0.4)$, and $\Sigma_1 = \left(\begin{smallmatrix} 1.6 & 0.5 \\ 0.5 & 0.7 \end{smallmatrix}\right)$.
- The unequal-lobes target has $w = (0.32, 0.68)$, means $(-1.25, 0)$ and $(1.25, 0.3)$, and both covariances $\mathrm{diag}(0.85^2, 0.70^2)$.

- The ring has 12 equal weights, means $2.1(\cos\phi_j, \sin\phi_j)$, $\phi_j = 2\pi j/12$, $j = 0, \ldots, 11$, and covariances $0.65^2 I_2$.
- The S-shaped chain has 13 equal weights, $u_j = -2.1 + 0.35j$, means $(u_j, 1.35\sin(1.3u_j))$, $j = 0, \ldots, 12$, and covariances $0.47^2 I_2$.

The neighboring components overlap. The three non-Gaussian targets have negative observed-information eigenvalues at some locations, so they are outside Assumption 8.1. For responsibilities $r_j(x) = w_j \mathcal{N}(x; \mu_j, \Sigma_j)/p_0(x)$ and component scores $a_j = \Sigma_j^{-1}(\mu_j - x)$, exact derivatives are

$$s_0(x) = \sum_j r_j a_j, \qquad H_0(x) = \sum_j r_j \Sigma_j^{-1} - \sum_j r_j (a_j - s_0)(a_j - s_0)^\top.$$

These derivatives are evaluated at the current reference samples. No exact noisy mixture scores or target samples enter the score field of Algorithm 1. Exact target sampling is used for evaluation and display only.

Sample errors use (E.1) with $v_\ell = (\cos(\pi\ell/64), \sin(\pi\ell/64))$, $\ell = 0, \ldots, 63$. The scalar σ_{tar} is computed from the analytic mixture covariance. No covariance whitening is applied. Final metrics select 1536 samples from each method and compare them with the same 1536 independent target samples within a seed. The GAD and target sample indices are spread evenly over their saved 4096-point sample sets. MCMC selection retains 384 evenly spaced production states per chain, including rejection repeats. It creates no duplicate observations by resampling, although actual chain rejections can repeat coordinates. The target/target comparison uses another independent target sample. Seeds 42–44 define three complete runs, and the reported SD uses divisor $3 - 1$, rather than being a standard error or a confidence interval.

F.2 Display controls

The displays in this subsection use the preselected seed 42 and the window $[-5, 5]^2$. The main and equal-base displays use 128×128 bins. For the main display, the target panels start with 60,000 independent target draws. The GAD panels use 60,000 independent reverse-SDE draws transported by the third-round score field, which is computed from the saved round-two reference samples. No further round is run. All columns apply the same display procedure. We resample each method's base sample set uniformly and add independent Gaussian jitter of standard deviation 0.10 in physical coordinates, producing 60,000 displayed points. MALA and HMC use their original production states with empirical weights, including rejection repeats. The MCMC panels show resampled chain states. The colors give log probability per bin with the full sample count in the denominator, including points outside the window.

Figure 12 restricts every method to the 1536 samples used in Table 4 and then applies the display procedure of Figure 4. The target and GAD histograms then also become granular, which shows how much of the smoothness in Figure 4 comes from the number of base draws. Figure 13 shows the metric samples in 48×48 bins. Figure 14 shows independent target and GAD draws in 128×128 bins. Both display the original samples without resampling or jitter. These figures separate the cheap drawing of many samples from a fixed GAD field from the error measured on samples of equal size. None of them changes Table 4.

F.3 GAD field, transport, and request counts

The common initial law is $\mathcal{N}((0.6, -0.4), 4I_2)$. GAD uses 4096 reference samples per round and three rounds, each starting fresh terminal samples from $\mathcal{N}(0, I_2)$. Stochastic Heun integration uses 240 steps from $T = 5$ to $t_{\min} = 10^{-4}$, uniform in $u = \log(\text{expm1}(2t) + 0.2)$, with the same

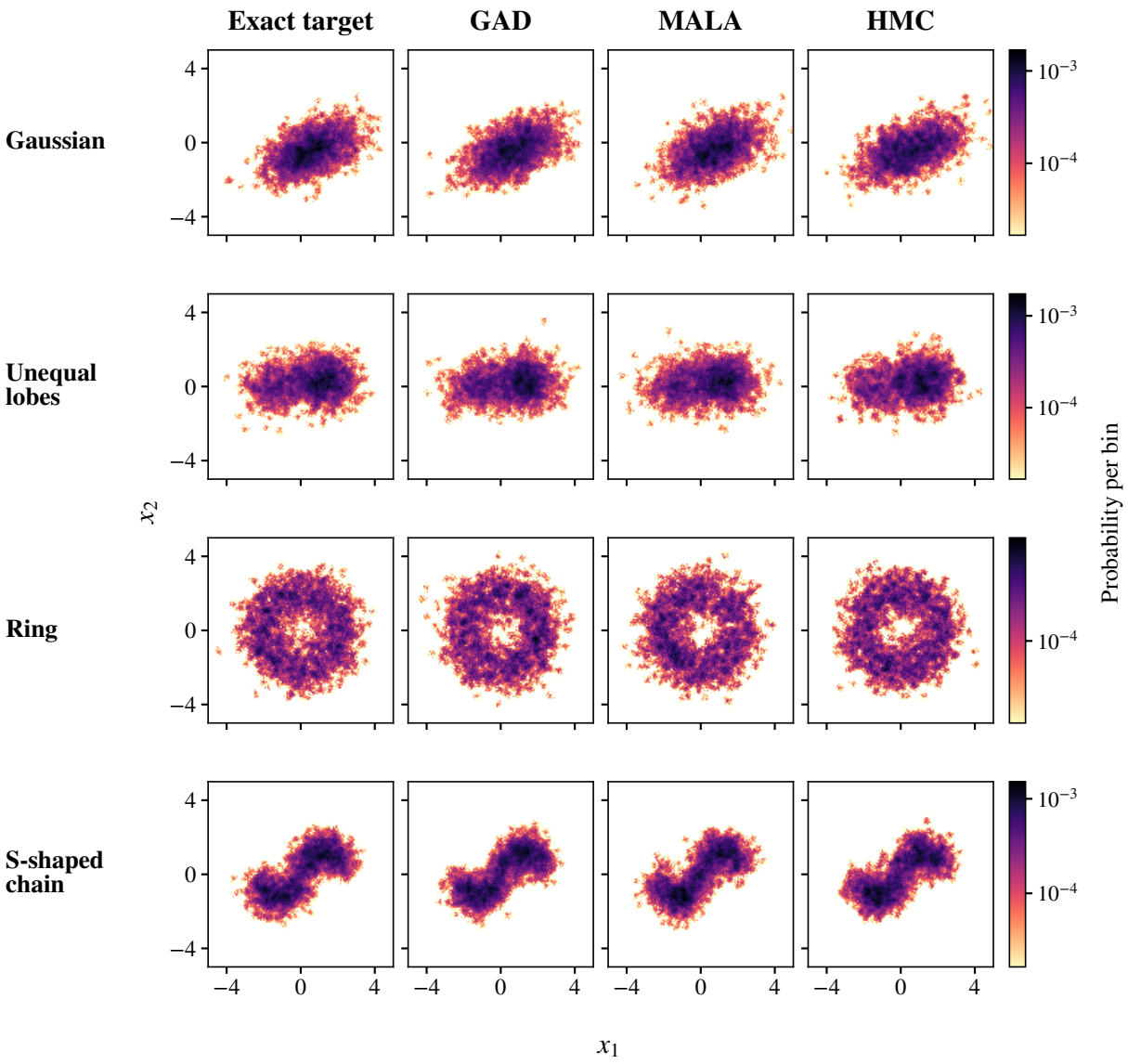


Figure 12: **High-resolution display with equal base samples.** Each target and sampler panel starts with its 1536 metric samples from seed 42 and applies the same resampling, jitter, and color scale as Figure 4. With equal base counts, the target and GAD histograms also become granular. Appendix F.2 gives the display settings.

Brownian increment in predictor and corrector. There is no final denoising or density fit. The frozen score is tabulated at 65 times uniform in u, interpolated linearly in time and bilinearly in space. Beyond the spatial table the nearest boundary score is used. Transported samples can leave the grid.

Current-law moments use a Cartesian reference grid with 129 nodes per coordinate. Its half-width is $B = \max\{9, \max_i \|X_i\|_\infty + 0.1\}$. Each occupied cell retains empirical mass, mean clean position, mean target gradient, and mean target Hessian. The OU kernel is evaluated at the cell node. At time t, the 129×129 query grid has half-width $\alpha_t B + 6\sqrt{\gamma_t}$. Separable Gaussian-kernel products are rescaled before normalization. Near-underflow rows are recomputed by log-sum-exp over occupied cells. The exact same cell moments are retained in this fallback.

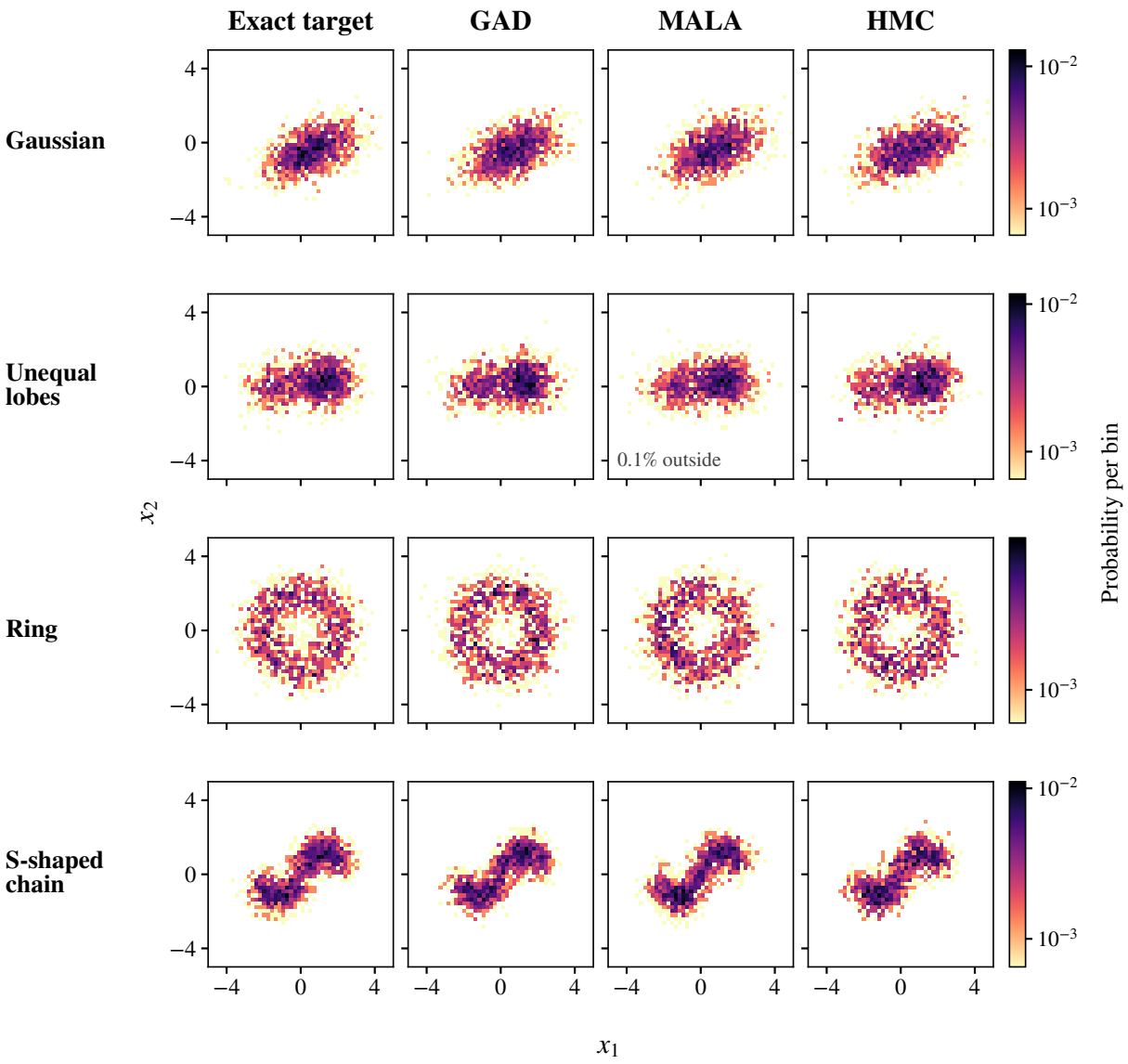


Figure 13: **Samples used by the two-dimensional metrics.** Rows show the four targets and columns show target, GAD, MALA, and HMC samples in the original coordinates. Each seed-42 panel uses the 1536 metric samples directly. Axes and log probability-per-bin colors are shared within each row. Table 4 uses all points, including those outside the displayed window. Appendix F.2 gives the binning and normalization.

For an eigenvalue λ of the conditional mean observed information, the raw gate eigenvalue is $g_{\text{raw}} = \alpha_t^2/(\alpha_t^2 + \gamma_t\lambda)$. Shifted denominators with magnitude below 10^{-8} are replaced by that signed magnitude, with positive sign at zero. Nonpositive raw gates are replaced by 10^{-3}. Positive values are capped at 50, while positive values below 10^{-3} remain unchanged. These safeguards, cell quadrature, interpolation, finite horizon, and time discretization are approximations not covered by Theorem 8.4.

The budget is $W = n_E + n_G + n_H = 24{,}576$, where n_E, n_G, n_H count requested energy, gradient, and Hessian evaluations by derivative order. GAD requests a gradient and Hessian for each reference sample in each round, giving $(n_E, n_G, n_H) = (0, 12{,}288, 12{,}288)$. The count W

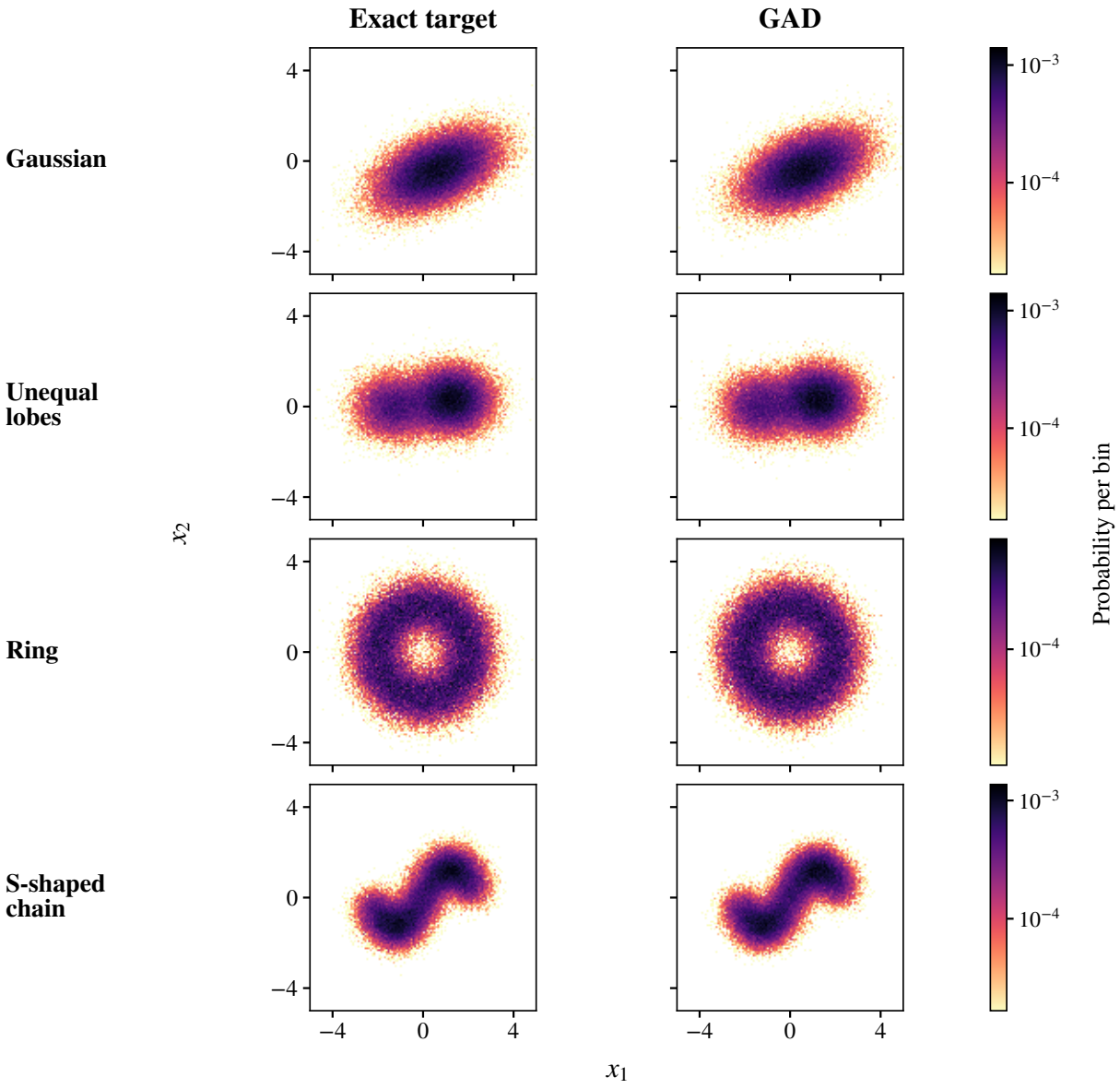


Figure 14: **Independent target and GAD draws.** Each target is paired with output from the third-round GAD field for seed 42. Each panel uses 60,000 independent draws in the original coordinates, without resampling or jitter. Log probability-per-bin colors are shared within each row. The additional GAD draws require no new target derivatives. Appendix F.2 gives the binning and normalization.

counts each requested quantity once. Field tabulation, transport, and display generation reuse the derivatives already evaluated at the reference samples. The quantitative comparison uses the samples drawn for the metric.

F.4 MALA and HMC at the fixed allowance

MALA proposes $x' = x + \varepsilon^2 s_0(x)/2 + \varepsilon\xi$, $\xi \sim \mathcal{N}(0, I_2)$, with the exact Hastings correction. HMC refreshes momentum from $\mathcal{N}(0, I_2)$, takes leapfrog steps with unit mass, and applies the Hamiltonian acceptance test [29]. Both use the same target and four initial states from the saved

initial GAD reference samples, hence the same broad initial law. Neither uses covariance, Hessian, or MAP preconditioning. Both retain rejection repeats.

Warmup uses 20% of the total request allowance, including initial energy and gradient evaluations. A common step across the four chains is adapted by dual averaging with initial $\varepsilon = 0.2$, target mean acceptance 0.574 for MALA and 0.8 for HMC, and constants $\gamma_{\mathrm{DA}} = 0.05$, $t_{\mathrm{DA}} = 10$, $\kappa_{\mathrm{DA}} = 0.75$, and $\mu_{\mathrm{DA}} = \log(10\varepsilon_{\mathrm{initial}})$. Adapted steps are constrained to $[10^{-4}, 5]$. The averaged log step is frozen for production. HMC trajectory lengths are drawn uniformly from $\{5, \ldots, 15\}$, with a final state-independent shortening if needed to fit the remaining allowance. No tuning uses independent target samples.

Each MALA proposal requests one new energy and gradient per chain, reusing the current values. Its final request count is $(12{,}288, 12{,}288, 0)$. HMC requests one gradient per leapfrog step and only the final energy, reusing the current values. The gradient and energy counts depend on the trajectory lengths. The runs spend 24,572–24,576 requests under the common allowance, with at most four requests left unspent and no Hessian requests. MALA produces 9832 production states per run. In seed 42, HMC produces 1764–1808 production states, depending on the target. The 1536 states used in the metric fit within each run's production states, without extending a chain or resampling.

F.5 Configuration selection

We chose 4096 reference samples and three rounds after an earlier study that varied both. In that study, with 4096 samples in the metric, the mean normalized SW_2 of this setting across the four targets is 3.488, compared with 3.693 at round four and 3.604 at round eight. Three rounds gave the smallest average error in that study. Nearby settings gave similar values. We fixed the number of rounds for all targets and seeds. Table 4 reports the separate evaluation with 1536 samples.

G One-dimensional illustration details

The illustration in Section 7.1 fits a mixture to each current law. It differs from the exact-expectation calculation in Appendix E.3 and from the sample-based estimator in Algorithm 1. The following settings describe only that illustration.

G.1 Mixture fit, transport, and display

- **Target.** $p_0 = \sum_{j=1}^{3} w_j \mathcal{N}(\mu_j, \sigma_j^2)$ with $w = (0.40, 0.37, 0.23)$, $\mu = (-3.2, 0, 1.7)$, and $\sigma = (0.72, 0.85, 0.58)$. The score $s_0 = (\log p_0)'$ and the observed information $H_0 = -(\log p_0)''$ are evaluated in closed form from the mixture responsibilities.
- **Initialization.** The analytic mixture $q_0 = 0.34\mathcal{N}(-0.25, 1.45^2) + 0.33\mathcal{N}(0, 1.45^2) + 0.33\mathcal{N}(0.25, 1.45^2)$, with mean -0.0025 and variance 2.144. It is near-Gaussian, not exactly Gaussian, and is not represented by samples.
- **Conditional expectations and projection.** Each q_k is a three-component Gaussian mixture. The OU-posterior moments entering m_t^q and H^q are computed by 16-point Gauss–Hermite quadrature under the mixture conditional, not by the sample-based weights (I.1). The frozen score field is tabulated at 1201 equally spaced points on $y \in [-7, 5.5]$ and 91 times, and linearly interpolated in y during transport. The 5000 endpoints of each round are fitted by unweighted expectation–maximization with three components (at most 100 iterations, tolerance 10^{-7}, variance regularization 10^{-4}). The fitted mixture is q_{k+1}.

- **Gate and indefinite observed information.** The exact H_0 is used, with no positive surrogate. The raw gate is $G_{\text{raw}} = \alpha_t^2/A^q$ with $A^q = \alpha_t^2 + \gamma_t H^q$. Values with $|A^q| < 10^{-8}$ are set to magnitude 10^{-8} with their sign kept, G_{raw} is clipped to $[-50, 50]$, and the transport uses $G = \max(G_{\text{raw}}, 10^{-3})$. In round one, $A^q \le 0$ at 8.42% of the 91×1201 tabulated queries and the full rule activates at 8.52%. In later rounds neither occurs.
- **Reverse diffusion.** Terminal base $\mathcal{N}(0, 1)$, $t_{\max} = 2.5$, $t_{\min} = 5 \times 10^{-3}$, 90 steps on a uniform time grid, the stochastic reverse-OU Heun scheme, and no final denoising step. The terminal mismatch is $\mathrm{KL}(p_{t_{\max}} \| \mathcal{N}(0, 1)) = 0.0028$.
- **Rounds and output.** Five rounds, each drawing fresh terminal samples and Brownian increments for 5000 samples. Rounds 0, 1, 2, 3, and 5 are displayed.
- **Displayed densities.** Histograms with 319 equal bins on $[-6.8, 5.0]$ (width 0.0370), smoothed by a Gaussian filter with standard deviation 0.9 bins and renormalized by trapezoidal integration. The target curve uses 400,000 draws from p_0, and each q_k curve uses 300,000 draws from the fitted mixture q_k. These draws affect only the display and the KL estimate, not the iteration.
- **Forward-KL estimator.** With x_i the bin centres and $\widehat{q}_k$ the smoothed q_k histogram,

$$\widehat{\mathrm{KL}}(p_0 \| q_k) = \sum_i{}^{\text{trap}} p_0(x_i)\{\log[p_0(x_i) + 10^{-12}] - \log[\widehat{q}_k(x_i) + 10^{-12}]\},$$

  where $\sum^{\text{trap}}$ is trapezoidal quadrature and p_0 is evaluated exactly. Rounds 0 to 5 give 0.5724, 0.0761, 0.0254, 0.0091, 0.0042, and 0.0035. The KL of the fitted mixtures by direct quadrature is 0.5691, 0.0756, 0.0255, 0.0086, 0.0041, and 0.0035.
- **Estimator floor.** With a 300,000-draw target sample in place of q_k, the plotted estimator gives $(2.06 \pm 0.38) \times 10^{-4}$ over 30 seeds. A nonparametric estimate from a single 5000-sample output, which this experiment does not use, would have a floor of 0.0113 ± 0.0044 over 100 seed pairs (5–95% range 0.0054–0.0192). The iteration itself carries the sampling noise of fitting a three-component mixture to 5000 endpoints.
- **Seeds.** The transport producing q_{k+1}, $k = 0, \ldots, 4$, uses seed $41 + k$ for its terminal sample and Brownian path. The target display sample uses seed 999, and the q_k display samples use seed $2000 + k$.

H Allen–Cahn and Navier–Stokes inverse problems

These two initial-condition inversions qualify the application results in Section 13. Both forward models evolve a nonlinear partial differential equation (PDE). The question is whether the resulting posterior departs enough from the Gaussian based on its MAP and local curvature to justify the GAD rounds. We examine coefficient distributions and spatial field moments against NUTS, then compare the improvements under a common request budget.

H.1 Forward models and observations

Both problems use the posterior in (13.1) with $\alpha \sim \mathcal{N}(0, I_d)$, a periodic unit square, a 32×32 grid, and one synthetic dataset generated with seed 42. In the Allen–Cahn problem the order parameter satisfies

$$\partial_t u = 0.045^2 \Delta u + u - u^3, \qquad u(0) = B\alpha.$$

The columns of B are the leading 32 eigenvectors of the grid covariance $\exp(-\|x - x'\|/0.1)$, scaled by the square roots of their eigenvalues. The Fourier solver uses implicit diffusion and explicit

reaction with step 0.0005. We observe u at 40 fixed grid sites at time 0.33 with independent Gaussian noise of standard deviation 0.01.

In the Navier–Stokes problem the vorticity satisfies

$$\partial_t \omega + v \cdot \nabla \omega = 7.5 \times 10^{-4} \Delta \omega + f, \qquad v = (\partial_y \psi, -\partial_x \psi), \qquad \Delta \psi = \omega.$$

In this vorticity equation, $\omega = \partial_y v_x - \partial_x v_y$. The initial field is a mean-zero Fourier expansion with 36 coefficients. The basis uses a cosine and a sine for each of the first 18 nonzero integer wave vectors k on the half-plane $k_x > 0$ or $k_x = 0,\ k_y > 0$, ordered by $(\|k\|^2, \|k\|_1, k_x, k_y)$. Each mode has unit grid Euclidean norm before weighting by $1.2 \exp[-((2\pi\, 0.075 k_x)^2 + (2\pi\, 0.13 k_y)^2)/2]$. The forcing on the grid is

$$\begin{aligned} f(x,y) = {} & 0.14 \sin(2\pi(1.5x + 0.35y)) + 0.08 \cos(2\pi(0.25x + 2.5y)) \\ & + 0.04 \sin(2\pi(3x - 0.75y)). \end{aligned}$$

The solver uses Crank–Nicolson diffusion, explicit advection and forcing, and step 0.05. It keeps Fourier modes with $|k_x|, |k_y| \le 32/3$ for dealiasing and zeros the vorticity mean after each step. At time 10.5, we apply the Fourier multiplier $\exp[-((k_x/8)^2 + (k_y/4)^2)/2]$ and observe the filtered vorticity at 80 fixed grid sites with independent Gaussian noise of standard deviation 0.005. Appendix I.6 identifies the files containing the frozen sensor indices, observations, and basis arrays for both problems.

H.2 Sampling and evaluation

Both GN-GAD and MAP-pMALA start from MAP–Laplace. MAP-pMALA runs eight chains with 64 warmup steps per chain and initial step size $h = 0.25$ in the proposal of Section 10. GN-GAD uses 1000 reference samples per round, Gauss–Newton curvature, and 300 flow steps on a geometric grid from 10 to 10^{-7}. We report the first three rounds of the archived five-round Allen–Cahn and four-round Navier–Stokes runs. The five paired sampler seeds are 43–47. At each round, the MAP-pMALA checkpoint is the last complete chain batch within the cumulative $E + G + H$ allowance of (13.2), including common MAP setup and MAP-pMALA warmup. MAP–Laplace is evaluated once per target at setup cost.

Each target has a fixed NUTS reference with four chains, each with 3000 warmup and 5000 retained draws. Target acceptance is 0.95 and maximum tree depth in sampling is 12. Neither run records a divergence. Maximum rank-normalized $\widehat{R}$ is 1.00099 for Allen–Cahn and 1.00100 for Navier–Stokes. We whiten coefficients using the mean and covariance of the 20,000 reference draws. In those coordinates, SW_2 is the root mean square of the one-dimensional Wasserstein-2 distances over 128 fixed unit directions. Each candidate contributes at most 1000 states to this metric and the reference contributes all 20000. The deterministic stratified MAP-pMALA subsample changes with the cumulative checkpoint. Covariance error is the relative Frobenius discrepancy in the whitened coordinates. Initial-field mean and pointwise standard deviation errors are relative Euclidean discrepancies from the corresponding NUTS fields. Moment estimates use all saved states, including MAP-pMALA rejection repeats. At the third checkpoint the moment estimates use 10000 Gaussian draws, 1000 GN-GAD samples, and 3976 cumulative MAP-pMALA states.

H.3 Posterior shape and field uncertainty

The NUTS projections in Figure 15 show one roughly elliptical region in each view. MAP–Laplace and GN-GAD recover the broad shapes, including the strongly tilted Allen–Cahn projection in the prior-whitened coefficients.

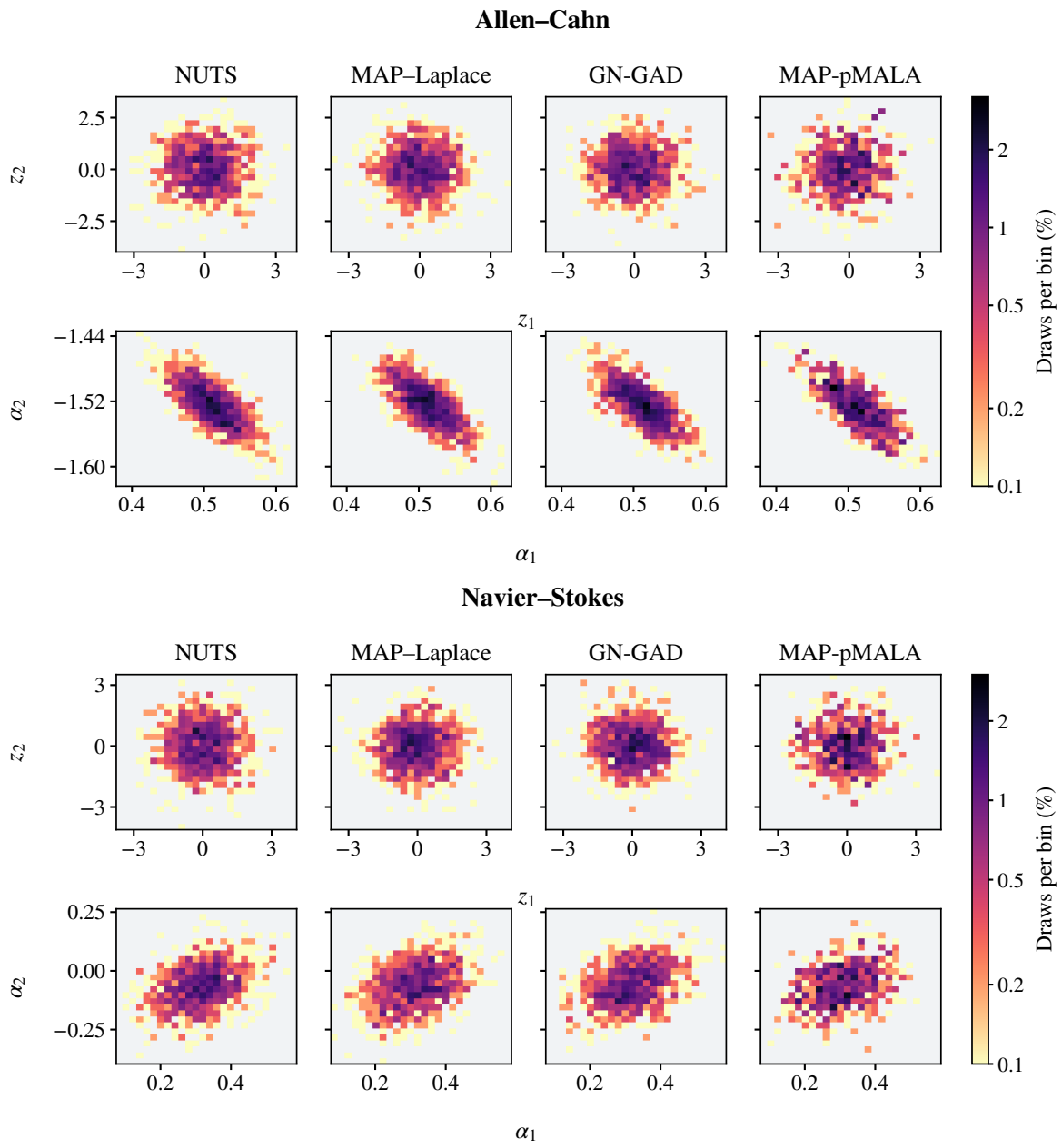


Figure 15: **Coefficient distributions for the two initial-condition inversions.** Allen–Cahn is above Navier–Stokes. Within each seed-43 block, the upper row gives coordinates (z_1, z_2) in the two lowest-curvature MAP Hessian directions. They are centered at the MAP and divided by Laplace standard deviations. The lower row gives the first two prior-whitened coordinates (α_1, α_2). GN-GAD and MAP-pMALA use the round-three request checkpoint. Each panel has 1000 states, with deterministic subsampling for NUTS, MAP–Laplace, and MAP-pMALA. Each row shares limits and 26×26 bins. Color shows the percentage of draws per bin on a common log scale within each problem. Gray bins are empty.

MAP–Laplace and both samplers reproduce the main features of the NUTS mean and the regions of larger pointwise uncertainty in Figure 16. The projections and field maps use one sampler seed and probe selected features of the posterior.

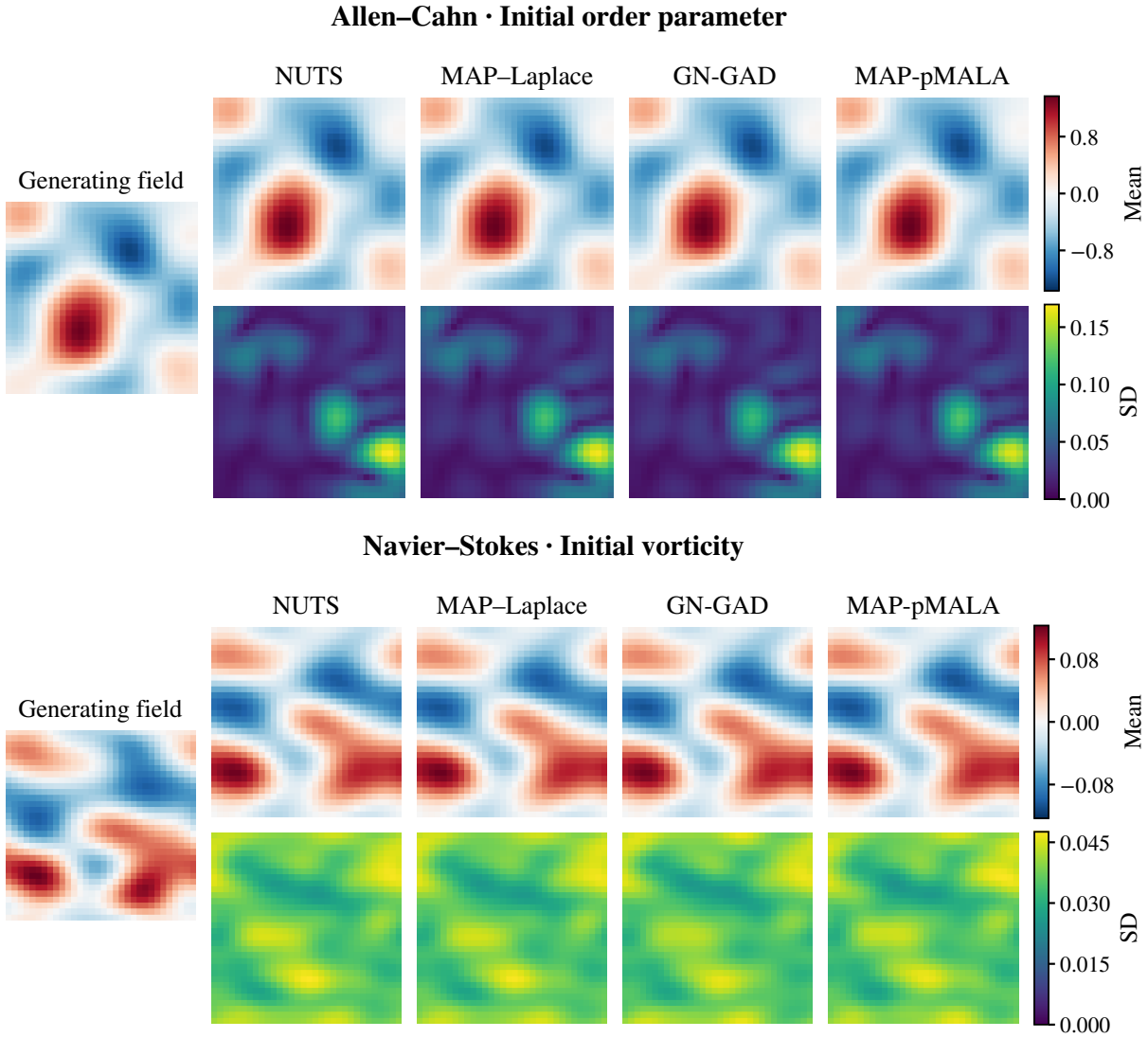


Figure 16: **Reconstructed initial fields and pointwise uncertainty.** The initial order parameter (Allen–Cahn) and vorticity (Navier–Stokes) are shown on the 32×32 spatial grid defined in Appendix H.1. Within each problem, synthetic truth and posterior means share a color scale, and posterior standard deviations share a separate scale. GN-GAD and MAP-pMALA are shown at the round-three request checkpoint for seed 43. Appendix H.2 gives the samples used for the maps.

H.4 Accuracy at matched request allowances

At round three, GN-GAD has lower SW_2 than MAP-pMALA in all five pairs on each problem (Figure 17). Mean SW_2 for MAP–Laplace, GN-GAD, and MAP-pMALA is 0.0651, 0.0624, 0.1014 for Allen–Cahn and 0.0895, 0.0834, 0.1052 for Navier–Stokes. The gain over MAP–Laplace is about 0.003 and 0.006, respectively. In Allen–Cahn the MAP–Laplace and GN-GAD errors

are close to the NUTS split discrepancy (0.0595), a descriptive finite-sample scale for SW_2. MAP–Laplace gives the smallest observed covariance and initial-field moment errors on both targets.

The MAP–Laplace relative error in pointwise field standard deviation is about 1.2% for each problem, compared with 2.2% and 2.4% for GN-GAD on Allen–Cahn and Navier–Stokes, respectively. Mean recorded times on the NVIDIA T4 GPU through the third checkpoint are 12.9 and 5.9 minutes for GN-GAD and MAP-pMALA on Allen–Cahn, and 52.9 and 7.7 minutes on Navier–Stokes, including common MAP setup and MAP-pMALA warmup.

For Navier–Stokes seed 47 at round two, the 32-draw coupled check between 300 and 600 flow steps gave a MAP-whitened RMS discrepancy of 0.0518 against a tolerance of 0.05. We retain the 300-step samples, which initialized round three. All round-three local checks passed, and the round-three SW_2 ordering against MAP-pMALA holds in each of the four other pairs.

The two field inversions condition low-dimensional initial fields on spatial measurements of the evolved order parameter or filtered vorticity. The resulting uncertainty in the displayed coefficients is compatible with fluctuations around a single reconstructed field. In acoustic inversion, each range difference confines the source near a curved locus at one pulse time, and the temporal prior couples those loci into a family of plausible routes. In earthquake location, the sparse picks allow lateral position and depth to trade off as origin times and shared velocities vary. The NUTS views in Figures 6 and 8 illustrate this curved uncertainty. The ellipsoidal contours of the MAP Gaussian keep the same principal directions throughout, so they capture only the local orientation of such a tradeoff. The annealed GN-GAD runs have the lowest mean primary sliced Wasserstein error in those two comparisons (Tables 5 and 6). In the two field inversions, the gain in sliced Wasserstein distance over MAP–Laplace is small. MAP–Laplace already agrees closely with NUTS on the reported field summaries.

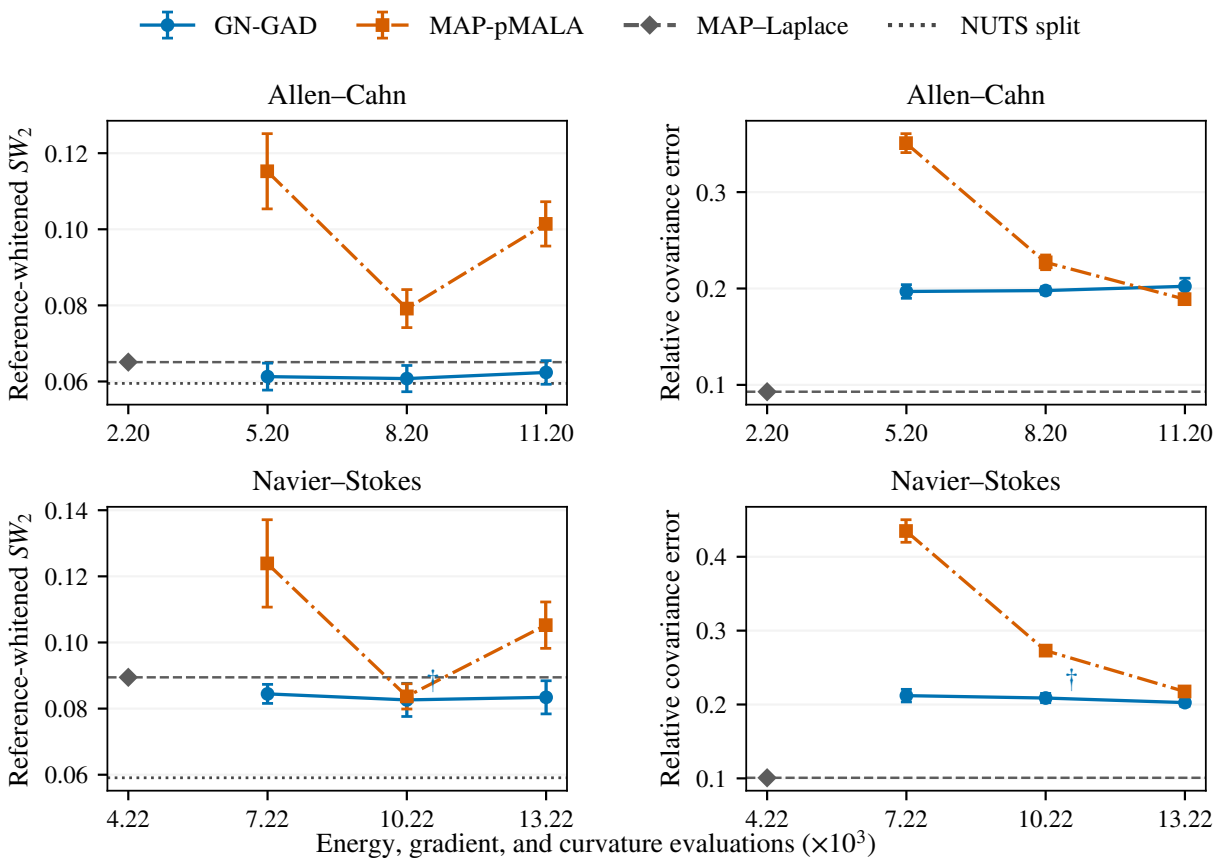


Figure 17: **Request-matched initial-condition inversions.** Reference-whitened SW_2 (left) and relative Frobenius covariance error in the same coordinates (right) for Allen–Cahn and Navier–Stokes. Lower values indicate closer agreement with NUTS. The horizontal axis counts pointwise energy, gradient, and curvature requests, including common MAP setup and MAP-pMALA warmup. The three sampler checkpoints correspond to GN-GAD rounds one through three. Markers and bars show means and one sample SD over five paired seeds. The MAP–Laplace diamond is at its setup cost, and the dashed line shows its error. The dotted line gives the mean SW_2 between a 1000-draw NUTS subsample and its disjoint remainder over 100 splits. Appendix H.2 defines the metrics and sample sets. The dagger marks Navier–Stokes GN-GAD round two, where seed 47 exceeded the coupled integration tolerance. The mean includes all five seeds.

I Implementation and comparison details

I.1 The sample-based estimator

At GAD round k, let

$$X_{1:N}^{(k)} = (X_1^{(k)}, \dots, X_N^{(k)})$$

be the current unweighted reference samples. For a query (y, t), define the normalized OU weights

$$w_i(y,t) := \frac{K_t^{\mathrm{OU}}(y \mid X_i^{(k)})}{\sum_{j=1}^N K_t^{\mathrm{OU}}(y \mid X_j^{(k)})}. \tag{I.1}$$

The implementation evaluates these weights through equivalent Gaussian-kernel logits in log space. With unweighted reference samples, averages using the weights in (I.1) are self-normalized importance-sampling (SNIS) averages with the OU kernel as the weight function. The effective sample size (ESS) of these weights at (y,t) is

$$N_{\mathrm{eff}}(y,t) := \left(\sum_i w_i(y,t)^2\right)^{-1}.$$

For each reference sample, the inverse-problem implementations cache the exact clean score at the current likelihood power β and the Gauss–Newton approximation

$$H^{\mathrm{GN}}_{i,\beta} := H^{\mathrm{GN}}_{0,\beta}(X_i^{(k)}) = I_d + \beta DF(X_i^{(k)})^\top \Gamma^{-1} DF(X_i^{(k)}),$$

where DF is the Jacobian of the unwhitened forward map. Substituting $J = \Gamma^{-1/2}DF$ in (9.4) gives the cached matrix. The flow reuses the cached score and curvature at each integration stage. Study-specific safeguards are described in Appendices C and D.

Using the weights (I.1), the sample-based conditional average of the Gauss–Newton matrices and its shifted form are

$$\widehat{H}^q_{\mathrm{GN}}(y,t) := \sum_i w_i(y,t) H^{\mathrm{GN}}_{i,\beta}, \qquad \widehat{A}^q_{\mathrm{GN}}(y,t) := \alpha_t^2 I_d + \gamma_t \widehat{H}^q_{\mathrm{GN}}(y,t).$$

The empirical Gauss–Newton gate is

$$\boxed{\widehat{G}^q_{\mathrm{GN}}(y,t) = \Psi_t(\widehat{H}^q_{\mathrm{GN}}(y,t)) = \alpha_t^2 (\widehat{A}^q_{\mathrm{GN}}(y,t))^{-1}.}$$

The symmetric resolvent uses linear solves or an equivalent eigendecomposition, with the numerical guards given in Appendices C and D. The sample-based score field is

$$\widehat{s}_t^{a,q}(y) = \widehat{b}^q(y,t) + \widehat{G}^q_{\mathrm{GN}}(y,t)\widehat{m}_t^q(y), \qquad \widehat{m}_t^q(y) := \widehat{c}^q(y,t) - \widehat{b}^q(y,t), \tag{I.2}$$

where $\widehat{b}^q$, $\widehat{c}^q$, and $\widehat{H}^q_{\mathrm{GN}}$ use the same reference samples and the same query-wise OU weights in the reported GAD implementation.

The acoustic and earthquake experiments integrate the flow (10.1) with the score estimate (I.2). We use the classical fourth-order Runge–Kutta method on logarithmic OU-time grids. The synthetic studies use the reverse-SDE integrators stated in their appendices. Finite terminal time, nonzero endpoint time, conditional estimation from finitely many samples, and time discretization are numerical approximations. Theorem 8.4, which concerns the exact-expectation reverse SDE, does not cover them.

I.2 GN consistency and likelihood changes

With the current OU posterior, define

$$H^q_{\mathrm{GN}}(y,t) = \mathbb{E}_{\rho^q_{y,t}}[H^{\mathrm{GN}}_0(X_0)],$$
$$A^q_{\mathrm{GN}}(y,t) = \alpha_t^2 I_d + \gamma_t H^q_{\mathrm{GN}}(y,t), \qquad G^q_{\mathrm{GN}}(y,t) = \alpha_t^2 (A^q_{\mathrm{GN}}(y,t))^{-1}.$$

Here $A^q_{\mathrm{GN}} \succ 0$ by (9.1). Define $e^q_{t,\mathrm{GN}} = d_t - G^q_{\mathrm{GN}} m^q_t$. Where the Poisson moments are defined, (5.2) gives

$$\boxed{e^q_{t,\mathrm{GN}} = \left(I_d - (A^q_{\mathrm{GN}})^{-1} A^q\right) d_t - \alpha_t (A^q_{\mathrm{GN}})^{-1} r_H.} \tag{I.3}$$

Table 17: **Inverse-problem GN-GAD settings.** Direct and annealed GN-GAD use the same number of samples per round, number of rounds, initial law, and allowance. The acoustic allowance includes 2941 MAP setup requests. The earthquake runs start from the prior and require no MAP setup.

Problem	Dimension	Samples / rounds	Final allowance	Initial law
Acoustic	16	2048 / 4	27517	Equal mixture of prior and MAP–Laplace
Earthquakes	23	1500 / 6	27000	Prior $\mathcal{N}(0, I_{23})$

The first term in (I.3) measures the effect of replacing A^q by A^q_{GN}. The second retains the flux covariance r_H from (5.1), now multiplied by the Gauss–Newton inverse.

Fix the likelihood power β. Write A^q_β for the shifted conditional mean of the exact observed information and $A^q_{\mathrm{GN},\beta}$ for its Gauss–Newton counterpart. The exact observed information is $H_{0,\beta} = H^{\mathrm{GN}}_{0,\beta} + \beta R_{\mathrm{nl}}$, with R_{nl} as in (9.2). Wherever the conditional moments and Poisson representation are defined, the extra factor in (I.3) satisfies

$$I_d - (A^q_{\mathrm{GN},\beta})^{-1} A^q_\beta = -\gamma_t (A^q_{\mathrm{GN},\beta})^{-1} \mathbb{E}_{\rho^q_{y,t}}[\beta R_{\mathrm{nl}}(X_0)].$$

The matrix $A^q_{\mathrm{GN},\beta}$ is positive definite. Bounding the curvature term in (I.3) also requires a bound on the conditional mean of βR_{nl}. The experiments assess the curvature approximation numerically.

When the likelihood power changes, the previous endpoints supply reference samples for a new target. The score and Gauss–Newton matrix must both use the new power. The inverse-problem experiments measure the combined error across rounds for the chosen schedule. The Gaussian tests check terminal time and grid settings on the surrogate. The comparison with NUTS assesses the overall sampling error on the nonlinear target for both direct and annealed runs.

I.3 Inverse-problem comparison settings

Each problem uses one synthetic dataset chosen in an exploratory study. The MCMC settings are selected by their final primary sliced Wasserstein error on seed 42 and are fixed for seeds 43–47. The evaluation seeds share the dataset and NUTS samples. The physical histograms in Figures 6 and 8 use seed 43.

Direct and annealed GN-GAD share their initial samples and terminal Gaussian draws within each seed. Direct transport uses full likelihood power at every round. The annealed powers are $(0.07, 0.45, 1, 1)$ for acoustic inversion and $(0.02, 0.07, 0.2, 0.45, 1, 1)$ for earthquake location. The settings in Table 17 are shared by the two GAD variants within each problem. MAP-pMALA instead starts from MAP–Laplace. Other MCMC methods use the corresponding GAD initial law, except where explicitly identified as MAP starts or continuations.

The count W includes setup, initialization, warmup or pilot runs, and rejected proposals. It excludes the NUTS runs, metric evaluation, and display draws, and it does not include the work of computing and integrating the GAD score field from cached derivatives. The first allowances in Figure 9 are $9085 = 2941 + 6144$ requests for acoustic inversion, including MAP setup and three requests per initial reference sample, and 4500 for earthquake location. A snapshot can use fewer requests than its allowance when the next complete step would exceed it. The per-problem records in Appendices C and D give the actual expenditures.

I.4 MCMC kernels, calibration, and retained laws

For $U_\beta(\alpha) = \|\alpha\|^2/2 + \beta U_{\text{like}}(\alpha)$, fixed-metric MALA, run as a population of parallel chains, proposes

$$\alpha' = \alpha - \frac{\epsilon^2}{2} C \nabla U_\beta(\alpha) + \epsilon C^{1/2} \xi, \qquad \xi \sim \mathcal{N}(0, I_d),$$

with the full Hastings correction. Here C is identity, the inverse exact full-posterior MAP Hessian, or a separately learned pilot covariance. GN-smMALA instead uses $C(\alpha) = (I_d + \beta J(\alpha)^\top J(\alpha))^{-1}$ and includes both forward and reverse state-dependent Gaussian densities, including log determinants, in its MH correction [17]. It does not add a metric-derivative drift. HMC uses fixed inverse mass C, momentum $p \sim \mathcal{N}(0, C^{-1})$, kinetic energy $p^\top C p/2$, position update $\alpha \leftarrow \alpha + \epsilon C p$, and L leapfrog steps followed by a Hamiltonian MH correction. For MAP-pMALA, the step h in Section 10 satisfies $h = \epsilon^2/2$.

Each population method has a fixed production step and metric. Rejections produce repeated states. Invalid proposals are rejected. Annealed families change likelihood power at the GAD checkpoints specified in Appendix I.3. At each checkpoint, the reported empirical posterior uses the latter half of the $\beta = 1$ production states. Direct families retain the latter half of their production states. MAP-pMALA retains all post-warmup production states. The acoustic sensitivity check is in Table 10. Intermediate-power snapshots in Figure 9 are progress diagnostics, not posterior-production endpoints.

For the acoustic problem we evaluate 48 settings on seed 42, using the same dataset as the GAD runs. The population size is $N_{\text{pop}} \in \{64, 128\}$. Annealed identity MALA uses $\epsilon \in \{0.003, 0.01, 0.03\}$, annealed MAP-pMALA uses $\{0.03, 0.1, 0.3\}$, direct and annealed GN-smMALA use $\{0.1, 0.3, 0.8\}$, and annealed HMC combines identity/MAP inverse masses with the corresponding fixed-metric step ranges and $L \in \{8, 24\}$. For each method we select the setting with the lowest final joint SW_2 among those that produce full-power samples. The acoustic configuration search uses 1251696 requests in total, reported separately from each selected run. The four acoustic likelihood powers are the six earthquake powers with 0.02 and 0.2 removed.

For the earthquake problem we evaluate 240 configurations per seed in addition to direct GN-GAD, annealed GN-GAD, and MAP-pMALA. Direct, annealed, and GAD-continuation MCMC families use the same ranges of population size and step, with an additional independently learned pilot mass for MALA. MAP-start HMC uses $N_{\text{pop}} \in \{4, 8\}$ and $L \in \{8, 24, 64\}$. Replica exchange uses four independent ladders with 8 or 16 geometrically spaced powers from 0.01 to one, and adjacent swaps. Only the cold chain contributes posterior states. SMC uses conditional ESS fractions 0.5 or 0.8, adaptive powers, systematic resampling at each stage, and three MALA/GN-smMALA moves or two HMC moves per stage. The pilot covariance is estimated from a separate set of 32 states, each moved ten times by GN-smMALA at step 0.3. Its covariance has 5% diagonal shrinkage and a positive trace-scaled ridge. The request count includes the pilot. Evaluation uses only the production states. GAD-continuation runs start from the annealed GN-GAD samples after round four in the same seed, count its 18000 requests, and then run at $\beta = 1$. SMC retains all of its full-target production states. The other population methods retain the latter half of their full-target production states.

Final E2–E5 block SW_2 on seed 42 selects a single earthquake configuration per family/kernel, including its available mass choices. We also select an annealed MAP-pMALA configuration from the settings that use the MAP preconditioner. The isotropic configuration has the lowest calibration error among the annealed fixed-metric MALA settings for both problems. Table 18 lists the selected settings. The NUTS samples used for selection never enter the transition kernels, initial states, or within-run adaptation. If the remaining allowance cannot cover one

Table 18: **Selected MCMC settings.** Chosen by final primary SW_2 on calibration seed 42, then reused at every checkpoint and seeds 43–47. N_{pop} is population size, ϵ is the stated proposal/leapfrog step, and L is HMC trajectory length in leapfrog steps. MAP-pMALA adapts only during its warmup, which counts toward the allowance. A dash means not applicable. SMC additionally lists conditional ESS fraction c. Replica exchange lists the number of temperatures n_{temp}.

Method	Metric / inverse mass	N_{pop}	ϵ	L or extra
Acoustic				
Annealed HMC	Identity	128	0.01	8
Annealed MALA	Identity	128	0.03	–
Annealed MAP-pMALA	MAP	128	0.03	–
Annealed GN-smMALA	Local GN	64	0.8	–
GN-smMALA	Local GN	64	0.8	–
Earthquakes				
Direct MALA	Identity	128	0.03	–
Direct HMC	Identity	128	0.003	24
Annealed MALA	Identity	128	0.03	–
Annealed MAP-pMALA	MAP	128	0.03	–
GN-smMALA	Local GN	128	0.8	–
Annealed GN-smMALA	Local GN	128	0.8	–
Annealed HMC	Identity	64	0.03	8
GAD then MAP-pMALA	MAP	64	0.1	–
GAD then GN-smMALA	Local GN	64	0.3	–
GAD then HMC	Identity	64	0.03	24
MAP-start HMC	MAP	8	0.1	24
Replica-exchange HMC	MAP	32	0.1	24, $n_{temp} = 8$
Replica-exchange GN-smMALA	Local GN	64	0.8	–, $n_{temp} = 16$
SMC–MALA	Pilot	64	0.1	–, $c = 0.8$
SMC–GN-smMALA	Local GN	64	0.3	–, $c = 0.8$

more complete step of all chains, or one more GAD round, a run stops and we record the number of requests actually used. The allowance applies to each selected run. We report the cost of the configuration search separately.

I.5 Numerical error and work between target requests

A dense observed-information matrix takes $O(d^2)$ storage per reference sample. Factoring the shifted dense matrix takes $O(d^3)$ work per integration query. Constructing and integrating the field from cached derivatives adds work beyond the target requests in (13.2).

The iteration in Theorem 8.4 uses exact conditional expectations and observed information in the reverse SDE (3.1). In the score estimate (I.2), weighted averages using (I.1) introduce sampling error. The ESS measures weight concentration at a query. The first term of (I.3) gives the error from substituting Gauss–Newton curvature in the exact-expectation gate. A numerical implementation also incurs shifted linear-solve error and transport discretization error, stops at positive t_{min}, and computes each new score field from finitely many endpoints. The inverse problems use (10.1). In annealed runs, the target also changes with the likelihood power. The numerical results measure the combined effect of these choices.

I.6 Reproducibility records

The acoustic record combines the original `arrival120` case of the four-round, five-seed study with `acoustic120_extended_run_RESULTS.zip`. It preserves exact observations and covariance basis, the shared MAP and reference, original GAD and MAP-pMALA histories, the annealed GAD reference samples and MCMC trajectories, frozen calibration choices, per-budget metrics, output-window checks, request counts, and separate display records. The reference and the direct GAD and MAP-pMALA runs are shared with that four-round study.

The earthquake record is `Quake_Integration_Handoff.zip`. It contains the 23-dimensional model, the fixed data and source, the settings for six rounds, the NUTS diagnostics, all 7596 metric rows, and the configurations selected on seed 42. The record includes selected physical projections for one seed, all-seed figures, and data for the figures regenerated from the record. Reanalysis of the full trajectories requires the original campaign outputs.

Both studies use float64 sampling and differentiation. Their generating environments record JAX/jaxlib 0.6.2, NumPyro 0.19.0, and NumPy 2.2.6, together with hardware and run metadata. The acoustic reference and selected-run timing records originate from different stages.

The Allen–Cahn and Navier–Stokes records are `GAD_ALLEN_CAHN_FIVE_SEED_RESULTS.zip` and `GAD_NAVIER_STOKES_R4_S300_FIVE_SEED_RESULTS.zip`. They preserve the fixed data, including the training and holdout sensor indices, the forward solvers, the NUTS draws, and the sampler checkpoints. For each target, the `target_arrays.npz` and `target_constants.npz` files in its directory under `code/frozen/` contain the arrays used by the solver. The three-round report uses the saved checkpoints.

The generating parameters are used to simulate the data. The starting distributions and preconditioners are obtained from the prior, MAP fit, or separately generated pilot samples as specified for each method in Section 10 and appendix I.4. NUTS reference draws are used for evaluation and the configuration selection criterion. The request totals follow the saved run records. Configuration search and extra display draws are recorded separately.

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